ETAS ISOLAR-A/B v12.7.0

ISOLAR-A

ISOLAR-A ERROR DESCRIPTION

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1. About this Manual

This document provides list of validations error. For each error message a description of the cause and possible solutions are included.

2. Dext Validations

This section deals with the error description of Dextvalidations. DextValidations are specific to Dext template in autosar.

2.1 DEXT4x_constr_1324 [Constr_1324]

Error Description

Duplicate VIN (Vehicle IdentificationNumber) for DiagnosticDataIdentifier.

How to Solve

Remove duplicate value.

2.2 DEXT4x_constr_1325 [Constr_1325]

Error Description

If Service Datamapping refers to DataElement then correspondingDataElementshouldnot have the attribute arraySizeSemantics set to VariableSize..

QuickFix

Remove the attributes whichare notpermitted.

2.3 DEXT4x_constr_1326 [Constr_1326]

Error Description

The allowedattributes ofSwDataDefProps for the aggregationinthe role DiagnosticDataElement.swDataDefProps are definedas Follows: baseType.baseTypeDefinition.baseTypeEncoding,baseType.baseTypeDefinition.baseTypeSize, baseType.baseTypeDefinition.byteOrder, compuMethod,dataConstr,displayFormatandunit.

QuickFix

Change array SizeSemantics to fixed size.

2.4 DEXT4x_constr_1327 [Constr_1327]

Error Description

If the enclosingDiagnosticContributionSetis ofcategory DIAGNOSTICS_ECU_EXTRACT, thenDiagnosticContributionSetshall refer tomaximumone DiagnosticEcuInstanceProps andthe referredDiagnosticEcuInstanceProps shall refer tomaximum1EcuInstance.

QuickFix

a. Remove additional EcuInstance fromDiagnosticEcuInstanceProps
b. Remove additional DiagnosticEcuInstanceProps fromDiagnosticContributionSet.ECUInstance.

2.5 DEXT4x_constr_1328 [Constr_1328]

Error Description

The ecuinstance referredinthe ServiceTable under the contributionset-shall alsobe referredinthe DiagnosticEcuInstanceProps.

QuickFix

a. Change EcuInstance referredinDiagnsoticServiceTable
b. Change EcuInstance referredinDiagnosticEcuInstanceProps

2.6 DEXT4x_constr_1331 [Constr_1331]

Error Description

The attribute DiagnosticEcuReset.customSubFunctionNumber shall only existifthe value ofDiagnosticEcuReset.category is outside the standardizedsetofvalues as definedby [TPS_DEXT_01056]:

a. HARD_RESET
b. KEY_OFF_ON_RESET
c. SOFT_RESET
d. ENABLE_RAPID_POWER_SHUT_DOWN
e. DISABLE_RAPID_POWER_SHUT_DOWN

QuickFix

a. Remove the customSubFunctionNumber

2.7 DEXT4x_constr_1332 [Constr_1332]

Error Description

The allowedvalue for DiagnosticEcuReset.customSubfunction-Number shall always be withinthe closedinterval 0x40.. 0x7E. (64-126).

QuickFix

a. Enter validvalue (Between64to126)

2.8 DEXT4x_constr_1333 [Constr_1333]

Error Description

MemoryIdentifiersreferredbyReqDownloadandReqUploadser-vicesshallnothaveHighAddressandLowAd-dressvaluesset.

QuickFix

a. Removethevalueforanyoneoftheattributes
b. RemoveMemoryidentifierreference

2.9 DEXT4x_constr_1334 [Constr_1334]

Error Description

The attribute DiagnosticComControl.customSubFunctionNumber shall only existifthe value ofDiagnosticComControl.category is out-side the standardizedsetofvalues.

_ENABLE_RX_AND_TX

_DISABLE_RX_AND_TX

_ENABLE_RX_AND_DISABLE_TX

_DISABLE_RX_AND_ENABLE_TX

_ENABLE_RX_AND_DISABLE_TX_WITH_ENHANCED_ADDRESS_INFOR-MATION

_ENABLE_RX_AND_TX_WITH_ENHANCED_ADDRESS_INFORMATION

QuickFix

a. Remove Category

2.10 DEXT4x constr_1335 [Constr_1335]

Error Description

Giventhe fulfilmentofconstr_1334, the value ofattribute Diagnos-ticComControl.customSubFunctionNumber shall be inthe range 0x40-0x5F.

QuickFix

a. Remove Category

2.11 DEXT4x_constr_1336 [Constr_1336]

Error Description

The value ofattribute DiagnosticComControlSpecificChannel.subnetNumber shall be withinthe closedinterval 114.

QuickFix

a. Enter the value between1to14

2.12 DEXT4x_constr_1337 [Constr_1337]

Error Description

The value ofattribute DiagnosticComControl-SubNodeChannel.subNodeNumber shall notexceedthe closedinterval 0-65535.

QuickFix

a. Enter the number inrange 0-65535

2.13 DEXT4x_constr_1338 [Constr_1338]

Error Description

The number ofaggregatedperiodicRate withinthe contextofone DiagnosticReadDataByPeriodicIDClass shall be withinthe closedinterval 1-3.

How to Solve

1.If periodicRate is not defined, then one periodicRate element should created under DiagnosticReadDataByPeriodicIDClass

2.If there are more than 3 periodicRate defined, then remove the additional periodicRate elements.

2.14 DEXT4x_constr_1339 [Constr_1339]

Error Description

Ina complete DiagnosticExtract,the attribute DiagnosticRoutine.startshall always existfor any givenDiagnosticRoutine.

How to Solve

Addstart(DiagnosticStartRoutine) toDiagnosticRoutine.

2.15 DEXT4x_constr_1340 [Constr_1340]

Error Description

ADiagnosticServiceSwMappingthatreferencesDiagnosticRoutineControl,withDiagnosticStartRoutineintherollstart,shouldbeconfigure dwithSwcS gnosticRoutineNeedswiththeattributediagRoutineTypesettoSynchronous.

How to Solve

ChangethevalueofdiagRoutineTypetosynchronous.

2.16 DEXT4x_constr_1341 [Constr_1341]

Error Description

ADiagnosticServiceSwMappingthatreferencesDiagnosticRoutineControl,withDiagnosticStopRoutineintherollstop,shouldbeconfiguredwithSwcSe gnosticRoutineNeedswiththeattributediagRoutineTypesettoSynchronous.

How to Solve

ChangethevalueofdiagRoutineTypetoAsynchronous.

2.17 DEXT4x_constr_1342 [Constr_1342]

Error Description

AllowedvaluesofSecurityAccess-->RequestSeedId.

How to Solve

ProvideanoptiontousertoenterthevaluewithinRange.

2.18 DEXT4x_constr_1344 [Constr_1344]

Error Description

ApplicabilityofDataTypetypedbyDiagnosticDataElement-mappedtotheDia-gnosticServiceDataMapping.

How to Solve

1.Provide option for user to select the applicationprimitive datatype with category VALUE

2.Provide an option for user to select an ApplicationArrayDataType whose Array Element category is VALUE.

2.19 DEXT4x_constr_1346 [Constr_1346]

Error Description

DiagnosticServiceSwMapping.serviceInstancecontainsanyoneoftheval-ues:Dia-gnosticRoutine,DiagnosticSecurityAccess,DiagnosticReadDataBy-Identifier,Dia-gnosticWriteDataByIdentifier,DiagnosticIOControl.

How to Solve

Popup a dialog which shows the available valid subclasses such as Diag-nosticRoutineControl,DiagnosticSecurityAccess,DiagnosticReadDataBy-Identifier, DiagnosticWriteDataByIdentifier,DiagnosticIOControl, so that user can choose.

2.20 DEXT4x_constr_1347 [Constr_1347]

Error Description

ForanygivenDiagnosticServiceSwMapping,oneandonlyoneofthefollow-ingreferencesshallexist: DiagnosticServiceSwMap-ping.mappedFlatSwcServiceDependency DiagnosticServiceSwMap-ping.mappedSwcServiceDependencyInSystem DiagnosticServiceSwMap-ping.mappedBswServiceDependency.

How to Solve

Provide an option for user to choose to retain the reference.

2.21 DEXT4x_constr_1349 [Constr_1349]

Error Description

The value ofudsDtcValue shall be unique toany other DTC andDTC groupvalue.

How to Solve

Change the value ofudsDtcValue inDiagnosticTroubleCodeUds.

2.22 DEXT4x_constr_1350 [Constr_1350]

Error Description

The value ofDiagnosticTroubleCodeGroup.groupNumber shall be unique toany other DTC andDTC groupvalue.

How to Solve

Change the value ofGroupNumber inDiagnosticTroubleCodeGroups.

2.23 DEXT4x_constr_1352 [Constr_1352]

Error Description

If the attribute DiagnosticTrouble Code Props.maxNumber FreezeFrameRecords exists than the attribute DiagnosticTrouble-CodeProps.freezeFrame shall notexistor vice versa.

How to Solve

1. Removevalueforattributefreezeframe.
2. Remove value for attribute maxNumberFreezeFrameRecords Allow theuser tochoose the e lementtobe re-moved(freezeframeormaxNumberFreezeFrameRecords)

2.24 DEXT4x_constr_1353 [Constr_1353]

Error Description

ForallDTCPropsmappedtoaContributionSetatleastoneofMaxNofF-FRecordsorFreezeFrameshallexistfor eachofDTCProps

How to Solve

SetdefaultvalueformaxNumberFreezeFrameRecordsSettingfreezeframe mightneedcreationofelementifitdoesnotexist.Henceitisnotconsidered-forquickfix.

2.25 DEXT4x_constr_1354 [Constr_1354]

Error Description

Ifone ofthe attributes DiagnosticTrouble-CodeProps.maxNumberFreezeFrameRecordsor Dia-gnosticTrouble-CodeProps.freezeFrame exists thenthe attribute DiagnosticTrouble-CodeProps.snapshotRecordContentshall exist

2.26 DEXT4x_constr_1359 [Constr_1359]

Error Description

If DiagDebounceAlgorithmProps aggregates DiagEventDebounceCoun-terBased, only thenthe attribute DebounceCounterStorage canbe con-figured.

How to Solve

Remove the value ofattribute DebounceCounterStorage.

2.27 DEXT4x_constr_1360 [Constr_1360]

Error Description

AggregationofDiagEventDebounceMonitorInternal withinDi-agDebounceAlgorithmProps is notallowed.

How to Solve

Remove DiagEventDebounceMonitorInternal.

2.28 DEXT4x_constr_1361 [Constr_1361]

Error Description

Only one DiagnosticEventToEnableConditionGroupMappingcanrefer a DiagnosticEvent.

How to Solve

Remove other references for DiagnosticEvent.

2.29 DEXT4x_constr_1362 [Constr_1362]

Error Description

Only one DiagnosticEventToStorageConditionGroupMappingcanrefer a DiagnosticEvent.

How to Solve

Remove other references for DiagnosticEvent.

2.30 DEXT4x_constr_1365 [Constr_1365]

Error Description

Number ofEvents configuredinside a ResponseOnEventservice should-notmore than255

2.31 DEXT4x_constr_1366 [Constr_1366]

Error Description

WhenDaignosticResponseOnEventdefines multiple events thatrefer to-dataIdentifier, thenthe attribute idofthe dataIdentifiers shouldbe unique

2.32 DEXT4x_constr_1376 [Constr_1376]

Error Description

The reference inthe role DiagnosticTroubleCodeProps.memoryDestina-tionshall notexceedthe upper multiplicity 2.

2.33 DEXT4x_constr_1377 [Constr_1377]

Error Description

Number ofreferences inDiagnosticTroubleCodeProps.memoryDestina-tionshouldnotbe more than2ifone ofthe references is a Diagnos-ticMemoryDestinationMirror.

How to Solve

Optiontoremove references (whichone todelete tobe chosenby user).

2.34 DEXT4x_constr_1378 [Constr_1378]

Error Description

The value ofthe attribute DiagnosticMemoryDestinationUserDefined.memoryIdshall be unique withinany given DiagnosticExtract.

2.35 DEXT4x_constr_1379 [Constr_1379]

Error Description

Withinthe scope ofone DiagnosticContributionSetonly one DiagnosticMemoryDestinationPrimary shall exist.

How to Solve

WhichDiagnosticMemoryDestinationPrimary toretaintobe chosenby user.

2.36 DEXT4x_constr_1380 [Constr_1380]

Error Description

Withinthe scope ofone DiagnosticContributionSetonly one DiagnosticMemoryDestinationPrimary shall exist.

How to Solve

Optiontoremove values ofDiagnosticMemoryDestinationMirror (whichone todelete tobe chosenby user).

2.37 DEXT4x_constr_1394 [Constr_1394]

Error Description

If the attribute maxNumberofElements exists, the value shouldbe greater thanzero.

How to Solve

Enter value greater than0.

2.38 DEXT4x_constr_1405 [Constr_1405]

Error Description.

If the reference DiagnosticProtocol.serviceTable exists thenthe value ofDiagnosticProtocol.protocolKindshall be identical tothe value ofDiagnosticServiceTable.protocolKind.

QuickFix

a. Modify the attribute DiagnosticServiceTable.protocolkindtomatchthe DiagnosticProtocol. protocolkind.

2.39 DEXT4x_constr_1406 [Constr_1406]

Error Description

All diagnosticconnections inservice table referredina protocol shall alsobe referredinDiagnosticProtocol.diagnosticConnection.

How to Solve

Refer the diagnosticConnectioninDiagnosticProtocol alsowhichare referredinDiagnosticServiceTable.

2.40 DEXT4x_constr_1411 [Constr_1411]

Error Description.

Atmostone ofthe attributes inthe followinglistshall exist:_DiagnosticMemoryIdentifier.memoryHighAddressLabel _DiagnosticMemoryIdentifier.memoryHighAddress.

QuickFix

a. Remove memoryHighAddressLabel
b. Remove memoryHighAddress

2.41 DEXT4x_constr_1412 [Constr_1412]

Error Description

Atmostone ofthe attributes inthe followinglistshall exist:_DiagnosticMemoryIdentifier.memoryLowAddressLabel _DiagnosticMemoryIdentifier.memoryLowAddress.

QuickFix

a. Remove memoryLowAddressLabel
b. Remove memoryLowAddress.

2.42 DEXT4x_constr_1419 [Constr_1419]

Error Description

Ifattribute DiagnosticSecurityLevel.accessDataRecordSize exists, the value is greater thanzero.

How to Solve

Enter value whichis greater than0.

2.43 DEXT4x_constr_1421 [Constr_1421]

Error Description

ThevaluesofDiagnosticDynamicallyDefine-DataIdentifierClass.subfunctionshallnotrepeat,i.e.everyvalueof DiagnosticDynamicallyDefineDataIdentifierSubfunctionEnum(clearDynamicallyDefineDataIdentifier, defineByIdentifier,defineByMemoryAddress)shallatmostappearonceinthesubfunctionattribute.

How to Solve

Removetherepeatedsubfunctions.

2.44 DEXT4x_constr_1447 [Constr_1447]

Error Description

The values 0x00, 0x20, 0x40, 0x60, 0x80, 0xA0, 0xC0, and0xE0are not allowedtoappear inthe value ofDiagnosticParameterIdentifier.id.

QuickFix

Enter value indecimal format.

2.45 DEXT4x_constr_1448 [Constr_1448]

Error Description

The allowedinterval for values ofDiagnosticParameterIdentifier.idshall notexceed[0- 255].

QuickFix

Enter value indecimal format.

2.46 DEXT4x_constr_1449 [Constr_1449]

Error Description

The value ofDiagnosticParameterIdentifier.dataElement.dataEle-ment.arraySizeSemantics shall notbe settovariable Size.

QuickFix

Set the value of DiagnosticParameterIdentifier.dataElement.dataEle-ment.arraySizeSemantics to fixed size Shall not be set to variableSize.

2.47 DEXT4x_constr_1450 [Constr_1450]

Error Description

If a DiagnosticServiceSwMappingor DiagnosticService-DataMappingre-fers toa DiagnosticRequestCurrentPowertrainData anda DiagnosticDa-taElementthenone ofthe followingshall exists:

a. SwcServiceDependency referencedby the same Diagnos-ticServiceSwMappingshall aggregate anObdPidServiceNeeds inthe role serviceNeeds.
b. The BswServiceDependencyIdentreferencedby the same Diag-nostic-ServiceSwMappingshall aggregatan ObdPidServiceNeeds inthe role serviceNeeds.DiagnosticServiceDataMapping
c. If DiagnosticServiceDataMappingrefers toDiagnosticDataEle-mentthatis aggregatedbyDiagnosticParameterIdentifier thenone oftwoalternative model configurations shall exist:
d. SwcServiceDependency referencedby the same Diagnos-ticServiceSwMappingshall aggregate anObdPidServiceNeeds inthe role serviceNeeds.
e. The BswServiceDependencyIdentreferencedby the same Diag-nosticServiceSwMappingshall aggregate an ObdPidServiceNeeds inthe role serviceNeeds.

How to Solve

Configure ObdPidServiceNeeds toSwcServiceDependency whichis re-ferredinDiagnosticServiceSwMappingand whichhas instance type Diag-nosticRequestCurrentPowertrainData .

2.48 DEXT4x_constr_1451 [Constr_1451]

Error Description

a. Ifa DiagnosticServiceSwMappingrefers toa DiagnosticRequestVehicleInfoanda thenone ofthe followingshall exists:
b. SwcServiceDependency referencedby the same DiagnosticServiceSwMappingshall aggregate anObdInfoServiceNeeds inthe role serviceNeeds.
c. The BswServiceDependencyIdentreferencedby the same Diagnostic-
d. ServiceSwMappingshall aggregate anObdInfoServiceNeeds inthe role serviceNeeds.

How to Solve

a. 1.Create an element ObdInfoServiceNeeds within the SwcServiceDependency referenced by the same DiagnosticServiceSwMapping
b. 2.Create an element ObdInfoServiceNeeds within the BswServiceDependencyIdent referenced by the same.

2.49 DEXT4x_constr_1452 [Constr_1452]

Error Description

If the reference DiagnosticProtocol.serviceTable exists then the value of DiagnosticProtocol.protocolKind shall be identical to the value of DiagnosticServiceTable.protocolKind.

How to Solve

A DiagnosticServiceSwMapping refers to a DiagnosticRequestControlOfOnBoardDevice then create a ObdControlServiceNeeds.

2.50 DEXT4x_constr_1455 [Constr_1455]

Error Description

Only one DiagnosticJ1939Node shouldrefer a J1939NmNode.

How to Solve

Allow the user tochoose towhichDiagnosticJ1939Nodes toretain(other reference will have tobe removed).

2.51 DEXT4x_constr_1456 [Constr_1456]

Error Description

The value ofthe attribute DiagnosticTroubleCodeJ1939.fmi shouldbe inthe interval 0to31.

How to Solve

Requestthe user enter a validvalue withinthe range.

2.52 DEXT4x_constr_1457 [Constr_1457]

Error Description

All DiagnosticTroubleCodeJ1939withattribute kindsettoserviceOnly andrefer the same DiagnosticJ1939Node shouldreference same memory destination.

How to Solve

Popup a dialog to display all the available Memory destination, so that user is able to choose one DiagnosticMemoryDestination to be reffered in DiagnosticTroubleCodeJ1939.

2.53 DEXT4x_constr_1458 [Constr_1458]

Error Description

Ifa DiagnosticTroubleCodeJ1939withattribute kindsettoserviceOnly refers a memory destination, thenthe same memory destinationshouldnotbe referredinanyother DiagnosticTroubleCodeJ1939whose attribute kindis notsettoserviceOnly.

How to Solve

Quick Ambiguous incase ifthere are multiple DiagnosticTroubleCodeJ1939; hence notpossible.

2.54 DEXT4x_constr_1458 [Constr_1458]

Error Description

Ifa DiagnosticTroubleCodeJ1939withattribute kindsettoserviceOnly refers a memory destination, thenthe same memory destinationshouldnotbe referredinanyother DiagnosticTroubleCodeJ1939whose attribute kindis notsettoserviceOnly.

How to Solve

Quick Ambiguous incase ifthere are multiple DiagnosticTroubleCodeJ1939; hence notpossible.

2.55 DEXT4x_constr_1460 [Constr_1460]

Error Description

DiagnosticInfoType.idcannothave the followingvalues, 0x20, 0x40, 0x60, 0x80, 0xA0, 0xC0, and0xE0.

How to Solve

Provide option for user to enter correct value other than 0,32, 64, 96, 128, 160, 192, and 224.

2.56 DEXT4x_constr_1461 [Constr_1461]

Error Description

DiagnosticTestRoutineIdentifier.idcannothave the followingvalues, 0x20, 0x40, 0x60, 0x80, 0xA0, 0xC0, and 0xE0.

How to Solve

Provide option for user to enter correct value other than 0x00, 0x20, 0x40, 0x60, 0x80, 0xA0, 0xC0, and 0xE0.

2.57 DEXT4x_constr_1462 [Constr_1462]

Error Description

DiagnosticTestResult.testIdentifier.idcannothave the followingvalues, 0,32, 64, 96, 128, 160, 192, and224.

How to Solve

Provide option for user to enter correct value other than 0,32, 64, 96, 128, 160, 192, and 224.

2.58 DEXT4x_constr_1464 [Constr_1464]

Error Description

DiagnosticEnvironmentalConditioncontains DiagnosticEnvConditionFormula whose attribute NrcValue shouldbe inthe interval 2to254.

How to Solve

Provide optionfor user tocorrectthe value.

2.59 DEXT4x_constr_1465 [Constr_1465]

Error Description

These are the allowedvalues for CompareType withinDiagnosticEnvDataCondition.

How to Solve

Provide optionfor user tocorrectthe value.

2.60 DEXT4x_constr_1466 [Constr_1466]

Error Description

These are the allowedvalues for CompareType withinDiagnosticEnvDataCondition: isEqual andisNotEqual.

How to Solve

Allow the user to select any one of the values from isEqual or isNotEqual.

2.61 DEXT4x_constr_1467 [Constr_1467]

Error Description

Mode Elementreferredinthe DiagnosticEnvModeConditionshouldbe withinDiagnosticEnvironmentalCondition whichDiagnosticEnvModeConditionis a part.

How to Solve

Allow the user tochoose the mode element.

2.62 DEXT4x_constr_1470 [Constr_1470]

Error Description

The value ofDiagnosticParameter.bitOffsetshall only be settoa multiple of8.

How to Solve

Enter value inrange 0to255.

2.63 DEXT4x_constr_1472 [Constr_1472]

Error Description

The attribute DiagnosticDataIdentifier.supportInfoByte shall notexistifthe value ofDiagnosticDataIdentifier.idis outside the range 0xF400-0xF4FF.

QuickFix

Remove SupportInfoByte attribute

2.64 DEXT4x_constr_1509 [Constr_1509]

Error Description

For all DiagnosticTroubleCodeProps thatrefer toDiagnosticMemoryDestinationPrimary inthe role memoryDestinationthere shouldnotbe notwoextendedDataRecord.recordNumber withthe same value.

2.65 DEXT4x_constr_1510 [Constr_1510]

Error Description

For all DiagnosticTroubleCodeProps thatrefer toDiagnosticMemoryDestinationMirror inthe role memoryDestinationthere shouldnotbe notwoextendedDataRecord.recordNumber withthe same value.

2.66 DEXT4x_constr_1511 [Constr_1511]

Error Description

For all DiagnosticTroubleCodeProps thatrefer toDiagnosticMemoryDestinationUserDefinedinthe role memoryDestinationthere shouldnotbe notwoextendedDataRecord.recordNumber withthe same value.

2.67 DEXT4x_constr_1512 [Constr_1512]

Error Description

For all DiagnosticTroubleCodeProps thatrefer toDiagnosticMemoryDestinationPrimary inthe role memoryDestinationthere shouldnotbe notwoFreezeframe.recordNumber withthe same value.

2.68 DEXT4x_constr_1513 [Constr_1513]

Error Description

For all DiagnosticTroubleCodeProps thatrefer toDiagnosticMemoryDestinationMirror inthe role memoryDestinationthere shouldnotbe notwoFreezeframe.recordNumber withthe same value.

2.69 DEXT4x_constr_1514 [Constr_1514]

Error Description

For all DiagnosticTroubleCodeProps thatrefer toDiagnosticMemoryDestinationUserdefinedinthe role memoryDestinationthere shouldnotbe notwofreezeframe.recordNumber withthe same value.

2.70 DEXT4x_constr_1584 [Constr_1584]

Error Description

DiagnosticDataElementshouldnotbe usedby more thanone DiagnosticServiceDataMappingthatrefers to DataPrototype utilizedby PortPrototype whose RoleBasedPortAssignmenthas the attribute role settoIOControlRequest.

How to Solve

Remove the reference ofDiagnosticDataElementfromDiagnosticServiceDataMappingifithas beenreferredmore thanonce.

2.71 DEXT4x_constr_1590 [Constr_1590]

Error Description

Any givenDiagnosticEventshall atmostonce be referencedfroma DiagnosticMasterToSlaveEventMapping.

How to Solve

Remove the reference ofDiagnosticEventeither fromMasterEventor fromSlaveEventifithas beenreferenced more thanonce.

2.72 DEXT4x_constr_1612 [Constr_1612]

Error Description

The reference fromDiagnosticRoutineControl (via its abstractbase class DiagnosticServiceInstance) inthe role accessPermissiontoDiagnosticAccessPermissionshall notbe used.

QuickFix

Remove the reference for accessPermissionfromServiceInstances

2.73 DEXT4x_constr_1616 [Constr_1616]

Error Description

The attribute DiagnosticExtendedDataRecord.customshall only existifthe attribute DiagnosticExtendedDataRecord.trigger is settothe value DiagnosticRecordTriggerEnum.custom.

QuickFix

Setthe attribute Trigger toCUSTOM.

2.74 DEXT4x_constr_1617 [Constr_1617]

Error Description

The attribute DiagnosticFreezeFrame.customTrigger shall only existifthe attribute DiagnosticFreezeFrame.trigger is settothe value DiagnosticRecordTriggerEnum.custom.

QuickFix

Setthe attribute Trigger toCUSTOM

2.75 DEXT4x_constr_1633 [Constr_1633]

Error Description

DiagnosticResponseOnEvent.eventandDiagnosticResponseOnEvent.responseOnEventActioncannotbe defined together.

QuickFix

a. Remove all Change Trigger
b. Remove ResponseOnEventAction

2.76 DEXT4x_constr_1711 [Constr_1711]

Error Description

For DiagnosticMemoryDestinationMirror attribute typeOf-FreezeFrameRecordNumerationshouldnotbe set.

QuickFix

Remove the value for typeOfFreezeFrameRecordNumeration

2.77 DEXT4x_constr_1721 [Constr_1721]

Error Description

InDiagnosticIOControl, the value ofthe attribute BitNumber under the elementcontrolEnableMaskBitshouldbe unique.

QuickFix

Provide anoptionfor user tochange the value.

2.78 DEXT4x_constr_1722 [Constr_1722]

Error Description

DiagnosticResponseOnEvent.eventandDiagnosticResponseOnEvent.re-sponseOnEventActioncannotbe defined together.

QuickFix

Remove the DiagnosticDataElements.

2.79 DEXT4x_constr_1725 [Constr_1725]

Error Description

For DiagnosticMemoryDestinationMirror attribute dtcStatusAvailabil-ityMask shouldnotbe set.

QuickFix

Remove the value for dtcStatusAvailabilityMask.

2.80 DEXT4x_constr_1762 [Constr_1762]

Error Description

For each of the DiagnosticEventPortMapping, the references to ServiceDependency can only be set through one of the following role: BswServiceDependency or swcFlatServiceDependency or SwcServiceDependencyInSystem

QuickFix

Configure only one of the references BswServiceDependency or swcFlatServiceDependency or SwcServiceDependencyInSystem for DiagnosticEventPortMapping

Remove the value for dtcStatusAvailabilityMask.

2.81 DEXT4x_constr_1759 [Constr_1759]

Error Description

For each of the DiagnosticOperationCyclePortMapping, the references to SwcServiceDependency can only be set through one of the following role: swcFlatServiceDependency or SwcServiceDependencyInSystem

QuickFix

Configure only one of the references swcFlatServiceDependency or SwcServiceDependencyInSystem for DiagnosticEventPortMapping

2.82 DEXT4x_constr_1752 [Constr_1752]

Error Description

For each of the DiagnosticEnableConditionPortMapping, the references to SwcServiceDependency can only be set through one of the following role: swcFlatServiceDependency or SwcServiceDependencyInSystem

QuickFix

Configure only one of the references swcFlatServiceDependency or SwcServiceDependencyInSystem for DiagnosticEnableCondition-PortMapping

2.83 DEXT4x_constr_1753 [Constr_1753]

Error Description

For each of the DiagnosticStorageConditionPortMapping, the references to SwcServiceDependency can only be set through one of the following role: swcFlatServiceDependency or SwcServiceDependencyInSystem

QuickFix

Configure only one of the references swcFlatServiceDependency or SwcServiceDependencyInSystem for DiagnosticStorageCondition-PortMapping

2.84 DEXT4x_constr_1782 [Constr_1782]

Error Description

DiagnosticDataElement that is aggregated by the DiagnosticExtended-DataRecord should be referred in the role DataElement within Diagnos-ticDemProvidedDataMapping

QuickFix

Configure DataElement with DiagnosticDataElement which are configured under the DiagnosticParameter in the role “recordElement” under DiagnosticExtendedDataRecord

2.85 DEXT4x_constr_1435 [Constr_1435]

Error Description

If same diagnosticEvent is referred in DiagnosticEventPortMapping.diagnosticEvent and DiagnosticEventToDebounceAlgorithmMapping then the concrete subclass of respective debounceAlgorithms ie. Subclass of debounceAlgorithms of DiagnosticEventToDebounceAlgorithmMapping and DiagnosticEventPortMapping should be identical

QuickFix

Configure the same type of DebounceAlgorithm in DiagnosticEventToDebounceAlgorithmMapping similar to the one configured within the DiagnosticEventPortMapping

2.86 DEXT4x_constr_1843 [Constr_1843]

Error Description

For each of the DiagnosticEventPortMapping, the reference diagnostic Event is mandatory

QuickFix

Configure DiagnosticEvent for DiagnosticEventPortMapping

2.87 DEXT4x_constr_1844 [Constr_1844]

Error Description

For each of the DiagnosticOperationCyclePortMapping, the reference operationCycle is mandatory

QuickFix

Configure operationCycle for DiagnosticOperationCyclePortMapping

2.88 DEXT4x_constr_1845 [Constr_1845]

Error Description

For each of the DiagnosticEnableConditionPortMapping, the reference enableCondition is mandatory

QuickFix

Configure enableCondition for DiagnosticEnableConditionPortMapping

2.89 DEXT4x_constr_1846 [Constr_1846]

Error Description

For each of the DiagnosticStorageConditionPortMapping, the reference diagnosticStorageCondition is mandatory

QuickFix

Configure the reference diagnosticStorageCondition

2.90 DEXT4x_constr_1847 [Constr_1847]

Error Description

For each of the DiagnosticDemProvidedDataMapping, the reference dataElement is mandatory

QuickFix

Configure DataElement for DiagnosticDemProvidedDataMapping

2.91 DEXT4x_constr_1823 [Constr_1823]

Error Description

For DiagnosticTestRoutineIdentifier configuring the value for attributes id is mandatory

QuickFix

Configuring the Value for the attribute id

3. General Validations

This section deals with the error description of general validations. This is applicable for all the autosar elements.

3.1 ASW4X_GEN_0001

Error Description

All the Identifiable (i.e. Containers which has Shortname) should have unique absolute qualified path.

How to Solve

a. Delete one of the Identifiable.
b. Update the short name of the element or its parent to have unique qualified path within the project.

Error Description

Two elements have same Short Label configured in Variation Point.

How to Solve

Update the short name of the element or its parent to have unique qualified path within the project

3.2 ASW4X_GEN_0002_A

Error Description

For an Identifiable configuring Shortname is mandatory.

How to Solve

Pop up a dialog with autogenerated Shortname.

3.3 ASW4X_GEN_0002_B

Error Description

The length of the Shortname should not be more than 256 characters.

How to Solve

Truncate the Shortname to 128 characters in length.

3.4 ASW4X_GEN_0002_C

Error Description

ShortName must match regular expression "[a-zA-Z]([a-zA-Z0-9]|_[a-zA-Z0-9])*_?"

How to Solve

Configure a shortname adhering to the regular expression“[a-zA-Z]([a-zA-Z0-9]|_[a-zA-Z0-9])*_?”.

3.5 ASW4X_GEN_0003

Error Description

All the mandatory references and attributes must be configured.

How to Solve

Configure all the mandatory references and attributes.

3.6 ASW4X_GEN_0004

Error Description

Two different Identifiables should not have the same UUID. It should be unique to ensure inter operability between tools.

How to Solve

Update the UUID attribute to be unique.

3.7 ASW4X_GEN_0005

Error Description

ARElement should not have any proxy references.

How to Solve

a. Delete the Identifiable.
b. Configure the reference with a valid element or remove if not mandatory / required

3.8 ASW4X_GEN_0006

Error Description

SwDataDefPropsConditional should configure variationpoint when the Multiplicity of SwDataDefPropsConditional is greater than one.

How to Solve

a. Delete the Identifiable.
b. Multiplicity of SwDataDefPropsConditional is 1 but it is allowed to create multiple due to variant nature. So bind SwDataDefPropsConditionals to different variants or remove invalid SwDataDefPropsConditionals

3.9 ASW4X_GEN_2521

Error Description

The shortLabel in variation point shall be unique, it must be unique within the next enclosing Identifiable.

How to Solve

Configure unique shortLabel for each variation point within the next enclosing Identifiable.

3.10 ASW4X_GEN_0007

Error Description

Split configuration present in the same file

HowToSolve

Merge the split file configuration to the existing file

3.11 ASW4X_GEN_2504

Error Description

In Variation Point for an element, the latest BindingTime is set to other than preCompileTime

HowToSolve

In Varition Point for the element, Set the latest BindingTime to PRE-COMPILE-TIME

3.12 ASW4X_GEN_2514

Error Description

The combination of attribute ShortLabel of VariationPoint and short-Name within the enclosing identifiable should be unique.

HowToSolve

Re-configure the shortLable of VariationPoint or the shortName within the enclosing identifiable, so that the combination is unique.

3.13 ASW4X_GEN_2519

Error Description

Same SystemConstant is referred in more than one SwSystemCon-stValue, with different value attributes

HowToSolve

Configure same value attributes to SwSystemConstValue

3.14 ASW4X_GEN_2594

Error Description

cyclic dependency - The value of the element swSystemconstvalue re-fers to the same system constant which causes a cyclic dependency.

HowToSolve

Reconfigure the value of swSystemconstvalue to have a valid integer value

4. Software Validations

This sectiondeals withthe error descriptionofsoftware validations. Software Validations are specifictosoftware template inautosar.

1. ComponentValidations
2. CompositionValidations
3. Data Type Validations
4. Internal Behavior Validations
5. Other Validations

Component Validations

This section deals with the error description of software component validations. Software Component Validations are specific to software template in autosar.

4.1 ASW4X_CPT_0001 [Constr_1092]

Error Description

A ParameterSwComponentType shouldhave PPortPrototypes configuredwhichare referringtoa ParameterInterface.

How to Solve

a. Delete the PPortPrototype.
b. Reconfigure the provider interface ofPPortPrototype withParameterInterface.

4.2 ASW4X_CPT_0002 [Constr_1102]

Error Description

Ina SwComponentType, iftwoor more PortPrototypes configuredthroughClientServerInterface shouldnothave ApplicationError withsame Shortname butwithdifferentErrorCode.

How to Solve

a. Delete the PortPrototype.
b. Delete the configuredPortInterface.
c. Delete the ApplicationError elements configuredinthe Portinterface.
d. Reconfigure the ApplicationError elements withdifferentShortnames.

4.3 ASW4X_CPT_0003

Error Description

Ina CompositionSwComponentType, ifAssemblySwConnector is connectingtwoPortPrototypes whichare configuredthroughModeSwitchInterface andthe same are available intwodifferentSwComponentType, thenboth the SwComponentType shouldbe mappedtosame ECU Instances via SwcToEcuMappinginsystem.

How to Solve

a. Reconfigure the EcuInstance inthe SwcEcuMappingelementofthe System.
b. Reconfigure the SwComponentPrototype.
c. Reconfigure the Provider Interface/ RequiredInterface ofthe PortPrototype toother thanModeSwitchInterface.
d. Delete the ports configured.

4.4 ASW4X_CPT_0004 [Constr_4004]

Error Description

A PortPrototype whichhas SenderRecieverAnnotations configuredinitshouldonly refer toSenderRecieverInterface

How to Solve

a. Delete the SenderRecieverAnnotations.
b. Reconfigure the PortPrototype withSenderRecieverInterface.

4.5 ASW4X_CPT_0005 [Constr_4005]

Error Description

A PortPrototype whichhas ClientServerAnnotations configuredinitshouldonly refer toClientServerInterface.

How to Solve

a. Delete the ClientServerAnnotations.
b. Reconfigure the PortPrototype withClientServerInterface.

4.6 ASW4X_CPT_0006 [Constr_4006]

Error Description

A PortPrototype whichhas ParameterPortAnnotations configuredinitshouldonly refer toParameterInterface.

How to Solve

a. Delete the ParameterPortAnnotations
b. Reconfigure the PortPrototype withParameterInterface.

4.7 ASW4X_CPT_0007 [Constr_4005]

Error Description

A PortPrototype whichhas ModePortAnnotations configuredinitshouldonly refer toModeSwitchInterface.

How to Solve

a. Delete the ModePortAnnotations
b. Reconfigure the PortPrototype withModeSwitchInterface.

4.8 ASW4X_CPT_0008 [Constr_4008]

Error Description

A PortPrototype whichhas TriggerAnnotations configuredinitshouldonly refer toTriggerInterface.

How to Solve

a. Delete the TriggerAnnotations
b. Re configure the PortPrototype withTriggerInterface.

4.9 ASW4X_CPT_0009 [Constr_4009]

Error Description

A PortPrototype whichhas NvDataAnnotations configuredinitshouldonly refer toNvDataInterface.

How to Solve

a. Delete the NvDataAnnotations
b. Re configure the PortPrototype withNvDataInterface.

4.10 ASW4X_CPT_0010 [Constr_4010]

Error Description

A PortPrototype which has DelegatedPortAnnotation configured in it should be only available within a CompositionSwComponentType as a DelegationPort.

How to Solve

a. Delete the DelegatedPortAnnotation.
b. Delete the PortPrototype.
c. Re configure the PortPrototype withinCompositionSwComponentType.

4.11 ASW4X_CPT_0011 [Constr_1043]

Error Description

If a PortPrototype is configuredwitha particular PortInterface thenthe respective ComSpecs thatcanbe configuredis as per the below table:

	SenderRecieverInterface	ClientServerInterface	ParameterInterface	ModeSwitchInterface	NvDataInterface	TriggerInterface
PPortPrototype	QueuedSenderComSpec, NonqueuedSenderComSpec	ServerComSpec	ParameterProvideComSpec	ModeSwitchSenderComSpec	NA	NA
RPortPrototype	QueuedReceiverComSpec, NonqueuedReceiverComSpec	ClientComSpec	ParameterRequireComSpec	ModeSwitchReceiverComSpec	NvRequireComSpec	NA

How to Solve

a. Delete the PortPrototype.
b. Delete the erroneous ProvidedComSpecs/RequiredComSpecs.
c. Re configure the PortPrototype withrespective ComSpecs.

4.12 ASW4X_CPT_0012 [Constr_1103]

Error Description

In an AtomicSwComponentType, any RPortPrototype configured with SenderRecieverInterface which has NonQueuedComSpecwhichhas the attribute "EnableUpdate"is settoTRUEandthe attribute "DataElement"is configuredwithVariableDataPrototype, then the configuredDataElement shouldnotbe referredby any DataReadAccess in any of the Runnable entity available in the SwcInternalBehavior which belongs to the SwComponentType inwhichthe RPortPrototype resides.

How to Solve

a. Delete the RPortPrototype.
b. Re configure the PortInterface ofRPortPrototype withother thanSenderRecieverInterface.
c. Re configure the DataElementofthe NonQueuedComSpecwitha differentone whichis notconfiguredinany RunnableEntity ofthe parentSwComponentType.
d. Delete the DataReadAccess element.
e. Delete the NonQueuedComSpecelementofthe RPortPrototype.

4.13 ASW4X_CPT_0013 [Constr_1100]

Error Description

In a CompositionSwComponentType, if any SwComponentPrototype referring to the SwComponentType which has unconnected RPortPrototype, then this RPortPrototype should either have RequiredComSpecs configured with "Init Value" or the attribute "maybeUnconnected" configured with value "true".

How to Solve

a. Delete the RPortPrototype.
b. Configure the RequiredComSpecs for the unconnectedRPortPrototype withits InitValue.

4.14 ASW4X_CPT_0014 [Constr_1137]

Error Description

A PPortPrototype which is configured with ParameterInterface should be available in ParameterSwComponentType only.

How to Solve

a. Delelte the PportPrototype.
b. Re configure the ProvidedInterface ofPPortPrototype witha differentInterface other thanParameterInterface.

4.15 ASW4X_CPT_0015 [Constr_1045]

Error Description

In a SwComponentType, if PortPrototype referring to the PortInterface contains ParameterDataPrototype which refers to the ApplicationDataType, then the mapped ImplementationDataType (via DataTypeMap) configured to the SwBaseType whichhas attribute “BaseTypeEncoding”shouldbe settoone ofthe following: { "2C", "IEEE754", "ISO-8859-1", "ISO-8859-2", "WINDOWS-1252", "UTF-8", "UCS-2", "NONE", "BOOLEAN"}.

How to Solve

a. Delete the ParameterDataPrototype.
b. Re configure the BaseTypeEncodingattribute ofthe SwBaseType toone among{ "2C", "IEEE75- 4", "ISO-8859-
c. 1", "ISO-8859-2", "WINDOWS-1252", "UTF-8", "UCS-2", "NONE", "BOOLEAN"}

4.16 ASW4X_CPT_0016 [Constr_1033]

Error Description

In a CompositionSwComponentType which contains AssemblySwConnector, in which the PortProto-type is a Sender/Reciever shouldnotbe involvedinm:ncommunication.

How to Solve

a. Delete the Assembly Connector.
b. Configure the PortPrototypes suchthatitis notinvolvedinm:ncommunication.

4.17 ASW4X_CPT_0017 [Constr_1037]

Error Description

Ina SwComponentPrototype, which is typedby a SwComponentType will have RPortPrototypes configured, if this RPortPrototype is configured with ClientServerInterface and this RPortPrototype is used in an AssemblySwConnector where inwhichthe PPortPrototype is alsoconnectedwithClientServerInterface thenitis an error.

How to Solve

a. Delete the RPortPrototype whichis configuredwithClientServerInterface.
b. Reconfigure the RPortPrototype witha PortInterface other thanClientServerInterface.

4.18 ASW4X_CPT_0018 [Constr_1169]

Error Description

In a Trigger Element, the only allowed values for the attribute Trigger.swImplPolicy are either STANDARD (in which case the Trigger processing does not use a queue) or QUEUED (in which case the processing of Triggers positively uses a queue).

How to Solve

a. Delete the Trigger Element.
b. Reconfigure the "SwImplPolicy"attribute ofthe Trigger elementtoone ofthe {"STANDARD", "QUEUED"}.

4.19 ASW4X_CPT_0019 [Constr_2049]

Error Description

Ina SwComponentType, twoor more PortPrototypes shall notrefer todifferentModeSwitchInterfaces whichcontains ModeDeclarationGroupPrototype referringtothe ModeDeclarationGroupwithsame Shortname butcontains ModeDeclarationwithdifferentShortnames.

How to Solve

a. Delete the PortPrototypes.
b. Re configure the ModeDeclarations ofthe ModeDeclarationGroups tosame.
c. Re configure the Shortnames ofModeDeclarations tosame.

4.20 ASW4X_CPT_0020 [Constr_1172]

Error Description

Ina ModeDeclarationGroupPrototype Element, the value ofattribute "SwCalibrationAccess"should be one among the following: {"NOT-ACCESSIBLE", "READ-ONLY"}.

How to Solve

Delete the ModeDeclarationGroupPrototype Element.

Reconfigure the "SwCalibrationAccess" attribute of the Trigger element to one of the {"NOT-ACCESSIBLE", "READ-ONLY"}.

4.21 ASW4X_CPT_0021 [Constr_1166]

Error Description

In a PortPrototype which is typed by ModeSwitchInterface contain ModeDeclarationGroupPrototype referring to ModeDeclarationGroup should have corresponding ModeRequestTypeMap configured in the SwInternalBehavior of the SwComponentType where the PortPrototype is created. This ModeRequestTypeMap should be configured withthe same ModeDeclarationGroupofthe ModeSwitchInterface andanImplementationDataType.

How to Solve

a. Delete the PortPrototype.
b. Reconfigure the ModeRequestTypeMapwithcorrespondingModeDeclarationGroupandImplementationDataType.

4.22 ASW4X_CPT_0022 [Constr_1167]

Error Description

In a ModeRequestTypeMap configured under DataTypeMappingSet, the referred ImplementationDataType should have category attribute set to either 'VALUE' or 'TYPE_REFERENCE' that in turn refers to an ImplementationDataType of category VALUE. And if the referred ImplementationDataType has SwBaseType configured through the attribute "BaseType". The value of the attribute "BaseTypeEncoding" of the SwBaseType configuredthroughBaseTypeDirectDefinitionshouldbe "NONE".

How to Solve

a. Delete ModeRequestTypeMap.
b. Re configure the ImplementationDataType whose category is either 'VALUE' or 'TYPE_REFERE- NCE'.
c. Re configure the BaseTypeEncodingattribute ofSwBaseType to'NONE'.
d. Delete the SwBaseType configured.

4.23 ASW4X_CPT_0023 [Constr_1144]

Error Description

a. Ina ComplexDeviceDriverSwComponentType or inanEcuAbstractionSwComponentType, the configuredattribute "HardwareElements"shouldbe oftype "HwType".
b. Ina SensorActuatorSwComponentType, the configuredattribute "SensorActuator"shouldbe oftype "HwType".

How to Solve

a. Delete the SwComponentType.
b. Reconfigure the HardwareElements/SensorActuator attribute toHwType.

4.24 ASW4X_CPT_0024 [Constr_2017]

Error Description

a. In a ServiceProxySwComponentType, the value of the "IsService" attribute of the PortInterface configured in the PPortPrototype shouldbe settoTRUE.
b. Ina ServiceProxySwComponentType, the value ofthe "IsService"attribute ofthe SenderRecieverInterface configuredinthe RPortPrototype shouldbe settoTRUE.

How to Solve

a. Delete the PortPrototype.

b. Reconfigure the "IsService"attribute ofthe PortInterface toTRUE.

4.25 ASW4X_CPT_0025 [Constr_2009]

Error Description

InanNvBlockSwComponentType, any PortPrototype shall only refer to NvDataInterface or ClientServerInterface only.

How to Solve

a. Delete the PortPrototype.
b. Re configure the PortInterface ofPortPrototype toone ofthe NvDataInterface or ClientServer- Interface.

4.26 ASW4X_CPT_0027A [Constr_1148]

Error Description

Ina NvBlockSwComponentType, the value ofthe "IsService"attribute ofthe PortInterface configu- redinthe PortPrototype shouldbe settoFALSE.

How to Solve

a. Delelte the PortPrototype.
b. Delete the configuredPortInterface.
c. Re configure the "IsService"attribute ofthe PortInterface toFALSE.

4.27 ASW4X_CPT_0029 [Constr_2012]

Error Description

In a NvBlockDescriptor, the configured RAM and ROM blocks should be configured with compatible ImplementationDataTypes

Note: TwoImplementationDataTypes are compatible when

a. Bothare notnull andnotproxy.
b. Bothare configuredwithsame category.
c. All the subelements ofbothare ofsame size.
d. All the subelements ofbothare identical.

How to Solve

a. Delete the RAM& ROMblocks configured.
b. Re configure the "Type"attribute ofthe RAM& ROMblocks withcompatible ImplementaionDataTypes.

4.28 ASW4X_CPT_0030 [Constr_2014]

Error Description

InanNvBlockSwComponentType, the "Role"attribute configuredinthe ClientServerPortofthe NvBlockDescriptor shouldbe one amongthe following: {"NvMService", "NvMNotifyJobFinished", "NvMNotifyInitBlock", "NvMAdmin", "NvMirror"}.

How to Solve

a. Delelte the configuredClientServerPort.
b. Re configure the "Role"attribute ofthe ClientServerPorttoone amongthe {"NvMService", "NvMNotifyJobFinished", "NvMNotifyInitBlock", "NvMAdmin", "NvMirror"}.

4.29 ASW4X_CPT_0031 [Constr_2015]

Error Description

a. In NvBlockSwComponentType, the RTE Event defined SwcInternalBehavior should be only OperationInvokedEvent.
b. In NvBlockSwComponentType, the RunnableEntity defined in SwcInternalBehavior should be configured with mandatory attributes {Symbol, CanBeInvokedConcurrently}.
c. In NvBlockSwComponentType, the RunnableEntity defined in SwcInternalBehavior should be configured with OperationInvokedEvent.
d. In NvBlockSwComponentType, the PortAPIOption defined in SwcInternalBehavior should be configured with PortArgValues.

How to Solve

a. Delete the RTEEvent.
b. Delete the RunnableEntity.
c. Delete the PortAPIOption.
d. Re configure the RTEEventinSwcInternalBehavior withOperationInvokedEvent.
e. Configure the mandatory attributes {Symbol, CanBeInvokedConcurrently}ofthe RunnableEntity.
f. Re configure the StartOnEventattribute ofthe RunnableEntity withOperationInvokedEvent.
g. Configure the PortArgValues inthe PortAPIOptionelement.

4.30 ASW4X_CPT_0032 [Constr_1129, 1130]

Error Description

a. In a NonqueuedReceiverComSpec, the SwImplPolicy attribute definedinthe DataElementshouldbe one among the following: {"CONST","FIXED","MEASUREMENT-POINT","STANDARD"}
b. Ina QueuedReceiverComSpec, the SwImplPolicy attribute definedinthe DataElementshouldbe setto "QUEUED".

How to Solve

a. Delete the DataElement.
b. Reconfigure the SwImplPolicy attribute ofNonqueuedReceiverComSpec to one among the {"CONST","FIXED", "MEASUREMENT-POINT", "STANDARD"}
c. Reconfigure the SwImplPolicy attribute ofQueuedReceiverComSpectoQUEUED.

4.31 ASW4X_CPT_0033 [Constr_1188, 1189]

Error Description

Ina QueuedReceiverComSpec, the attribute "HandleOutOfRange"shouldbe setto"EXTERNAL_REPLACEMENT"if andonly ifthe reference "ExternalReplacement"is configuredeither withVariableDataPrototype or ParameterDataPrototype.

How to Solve

a. Delete the QueuedReceiverComSpec.
b. Re configure the "HandleOutOfRange" attribute to "EXTERNAL_ REPLACEMENT" when the "ExternalReplacement"attribute is configured.
c. Configure the "ExternalReplacement"attribute witheither VariableDataPrototype or Paramet- erDataPrototype when"HandleOutOfRange"attribute is setto"EXTERNAL_REPLACEMENT"

4.32 ASW4X_CPT_0034 [Constr_1131, 1132]

Error Description

a. Ina NonQueuedSenderComSpec, the SwImplPolicy attribute definedinthe
DataEle mentshouldbe settoone amongthe following: {“CONST”, “FIXED”,
“MEASUREMENT-POINT”, “STANDARD”}
b. In a QueuedSenderComSpec, the SwImplPolicy attribute definedinthe
DataElementshouldbe setto "QUEUED".
a. How to Solve
c. Delete the DataElement.

d. Reconfigure the SwImplPolicy attribute of NonqueuedSenderComSpec to one among the {"CONST","FIXED","MEASUREMENT-POINT","STANDARD"}

e. Reconfigure the SwImplPolicy attribute ofQueuedSenderComSpec-toQUEUED.

4.33 ASW4X_CPT_0035 [Constr_1044]

Error Description

Ina DataFilter, the referredDataType shouldhave the BaseType as Integer.

How to Solve

Configure the BaseType of the DataType to an Integer.

4.34 ASW4X_CPT_0036 [Constr_1128]

Error Description

If a Runnable Entity is referred in more than one OperationInvokedEvent of a SwcInternalBehavior, then all the PortPrototype mapped to the operation (via POperationInAtomicSwcInstanceRef inside OperationInvokedEvent) shouldhave the identical QueueLengthconfiguredinServerComSpecwhichis inside PortPrototype.

How to Solve

a. Delete the extra OperationInvokedEvents.
b. Delete the extra ServerComSpecelements ofthe PPortPrototype.
c. Re configure the QueueLengthattribute ofthe ServerComSpecelements tobe same.
d. Delete the PPortPrototype.

4.35 ASW4X_CPT_0037 [Constr_1147]

Error Description

InanAtomicSwComponentType, all the PortGroupavailable initshall have category attribute settoone among the following: {"MODE_MANAGEMENT”, "PARTIAL_NETWORKING"}.

How to Solve

a. Delete the PortGroup.
b. Reconfigure the category ofPortGroupwitheither "MODE_MANAGEMENT"or "PARTIAL_NETWORKING".

Composition Validations

This section deals with the error description of software composition validations. Software Composition Validations are specifictosoftware template inautosar.

4.36 ASW422_CPT_0031[Constr-2015]

Error Description

The SwcInternalBehavior ofa NvBlockSwComponentType is only permittedtodefine the following:

a. OperationInvokedEvents
b. RunnableEntitys triggeredby OperationInvokedEvents (serverRunnableEntitys)
c. RunnableEntitys whichdefines only the mandatory attributes symbol andcanBeInvokedConcurrently
d. PortAPIOptions definingPortDefinedArgumentValues
e. TimingEvents (whichmay include references toModeDeclarations inthe role disabledMode)
f. DataReceivedEvents (whichmay include references toModeDeclarations inthe role disabledMode)
g. SwcModeSwitchEvents
h. RunnableEntitys triggeredby TimingEvents
i. RunnableEntitys triggeredby DataReceivedEvents
j. RunnableEntitys triggeredby SwcModeSwitchEvents"

How to Solve

Remove the elements whichare not

4.37 ASW4X_COMP_0001 [Constr_1032]

Error Description

Ina DelegationSwConnector, ifOuterPortis connectedtoProviderPort/RequiredPortthenInnerPortalsoshould be connectedtoProviderPort/RequiredPortrespectively.

How to Solve

a. Delete the configuredInnerPort.
b. Reconfigure the InnerPortaccordingtothe OuterPort.
c. Reconfigure the OuterPortaccordingtothe InnerPort.

4.38 ASW4X_COMP_0002 [Constr_1035]

Error Description

In a CompositionSwComponentType, a SwComponentPrototype should not refer to a CompositionSwComponentType whichhas cyclicdependency.

How to Solve

a. Delete the SwComponentPrototype.
b. Reconfigure the Type attribute ofSwComponentPrototype toany other CompositionSwComponentType which does nothave cyclicdependency.

4.39 ASW4X_COMP_0003 [Constr_1086]

Error Description

In a CompositionSwComponentType, two SwConnectors (Delegatio/Assembly) should not have same configuration.

How to Solve

a. Delete the duplicate SwConnector.
b. Reconfigure the attributes ofthe SwConnectors todifferentconfiguration.

4.40 ASW4X_COMP_0004_A [Constr_1087]

Error Description

Ina CompositionSwComponentType, the ContextComponentattribute of the Provider and Requestor Ports of the AssemblySwConnector shouldbe configured.

How to Solve

a. Delete the erroneous Provider/Requestor ports.
b. Configure the ContextComponentattribute ofthe Provider/Requestor port.

4.41 ASW4X_COMP_0004_B [Constr_1087]

Error Description

In a CompositionSwComponentType, the TargetPortattribute ofthe Provider andRequestor Ports ofthe AssemblySwConnector shouldbe configured.

How to Solve

a. Delete the erroneous Provider/Requestor ports.
b. Configure the TargetPortattribute ofthe Provider/Requestor port.

4.42 ASW4X_COMP_0004_C [Constr_1087]

Error Description

In a CompositionSwComponentType, the Provider and Requestor Ports of the AssemblySwConnector should be configured.

How to Solve

a. Delete the erroneous Provider/Requestor ports.
b. Configure the Provider/Requestor port.

4.43 ASW4X_COMP_0004_D [Constr_1087]

Error Description

In a CompositionSwComponentType, the SwComponentPrototype configured in the PPortInCompositionInstanceRef through "ContextComponent" attribute of the AssemblySwConnector, should be from the same CompositionSwComponentType inwhichthe SwConnector is configured.

How to Solve

a. Delete the erroneous Provider/Requestor ports.
b. Configuretthe "ContextComponent"attribute ofPPortInCompositionInstanceReftothe SwComp-onentPrototype whichbelongs tothe same CompositionSwComponentType inwhichthe SwConnector is configured.

4.44 ASW4X_COMP_0004_E [Constr_1087]

Error Description

In a CompositionSwComponentType, the parentofthe "TargetPort"ofthe Provider andRequestor Portofthe AssemblySwConnector shouldnotbelongtoServiceProxySwComponentType.

How to Solve

a. Delete the Provider/Requestor ports.
b. Reconfigure the TargetPortofProvider/Requestor PortofAssemblySwConnector todifferentSwComponentType other thanServiceProxySwComponentType.

4.45 ASW4X_COMP_0004_G [Constr_2016]

Error Description

Ina CompositionSwComponentType, ifthe type of the "ContextComponent"ofthe Provided/RequiredPortis NvBlockSwComponentType thenthe PortInterface ofthe "TargetPort"configuredinthe Provided/RequiredPortshould be either "NvDataInterface"or "SenderReceiverInterface".

How to Solve

a. Delete the Provider/Requestor ports.
b. Reconfigure the TargetPortofProvider/Requestor PortofAssemblySwConnector todifferentother thanNvBlockSwComponentType.
c. Reconfigure the PortInterface ofthe TargetPortofthe Provided/RequiredPorttoeither "NvDataInterface"or "SenderReceiverInterface".

4.46 ASW4X_COMP_0005_A [Constr_1088]

Error Description

In a CompositionSwComponentType, the Provider and Requestor Ports of the DelegationSwConnector should be configured.

How to Solve

Provide option for user to select PPOrtInCompositionInstanceRef or RPOrtInCompositionInstanceRef within DelegationConnector.

4.47 ASW4X_COMP_0005_B [Constr_1088]

Error Description

Ina CompositionSwComponentType, the ContextComponentattribute of the Provider and Requestor Ports of the DelegationSwConnector shouldbe configured.

How to Solve

Delete the erroneous Provider/Requestor ports.

Configure the ContextComponentattribute ofthe Provider/Requestor port.

4.48 ASW4X_COMP_0005_C [Constr_1088]

Error Description

In a CompositionSwComponentType, the SwComponentPrototype configured in the PPortInCompositionInstanceRef through "ContextComponent"

attribute of the DelegationSwConnector, should be from the same CompositionSwComponentType inwhichthe SwConnector is configured.

How to Solve

a. Delete the erroneous Provider/Requestor ports.
b. Configure the "ContextComponent"attribute ofPPortInCompositionInstanceReftothe SwComponentPrototype whichbelongs tothe same CompositionSwComponentType inwhichthe SwConnector is configured.

4.49 ASW4X_COMP_0005_D [Constr_1088]

Error Description

a. In a CompositionSwComponentType, the TargetPort attribute of the Provider and Requestor Ports of the AssemblySwConnector shouldbe configured.
b. In a CompositionSwComponentType, the TargetPort attribute of the Provider and Requestor Ports of the AssemblySwConnector should belong to the referred SwComponentPrototype which is inturn configured through"ContextComponent"attribute ofthe Provider/Requestor Port.

How to Solve

a. Delete the erroneous Provider/Requestor ports.
b. Configure the TargetPortattribute ofthe Provider/Requestor port.

4.50 ASW4X_COMP_0005_E [Constr_1088]

Error Description

a. Ina CompositionSwComponentType, the OuterPortofthe DelegationSwConnector shouldbe configured.
b. Ina CompositionSwComponentType, the OuterPortofthe DelegationSwConnector configuredshouldbelongto the CompositionSwComponentType inwhichthe SwConnector is configured.

How to Solve

Setattribute isService totrue [Change the attribute isService ofthe portinterface ofthe ports involvedinthe SwConnector totrue]

4.51 ASW4X_COMP_0007 [Constr_1174]

Error Description

InSwConnector , one or bothports involvedare service ports ofa CompositionSwComponentType. Service ports are notallowedinthe contextofCompositionSwComponentType.

How to Solve

Setthe attribute isService totrue (Change the attribute isService ofthe portinterface ofthe ports involvedinthe SwConnector totrue).

4.52 ASW4X_COMP_0008 [Constr_1174]

Error Description

A CompositionSwComponentType cannotrefer toa PortPrototype whichis configuredwitha PortI- nterface whose "isService"attribute is settoTRUE.

How to Solve

a. Delete the PortPrototype.
b. Delete the configuredPortInterface.
c. Set"isService"attribute ofthe PortInterface toFALSE. Data Type Validations
d. This sectiondeals withthe error descriptionofsoftware data type validations. Software Data Type Validations are specifictosoftware template inautosar.

4.53 ASW4X_COMP_0013 [Constr_1197]

Error Description

If a CompositeNetworkRepresentation is defined then it should cover the entire set of leaf elements of corresponding ApplicationCompositeDataType. For each leaf exactly one CompositeNetworkRepresentation should be defined.

How to Solve

a. Configure the RequiredComSpecs for the unconnected RPortPrototype with its InitValue.
b. Delete the RPortPrototype.

4.54 ASW4X_DT_0003 [Constr_1010]

Error Description

In a SwBaseType, if "NativeDeclaration" attribute of BaseTypeDefinition is not defined, then Shortname of SwBaseType shouldbe one ofthe platformor standardtypes definedby AUTOSAR. {i.e StandardTypes: Boolean, uint8, uint16, uint32, sint8, sint16, sint32, uint8_least, uint16_least, uint32_least, sint8_least, sint16_least, sint32_ least, float32, float64}.

How to Solve

a. Configure the "NativeDeclaration"attribute ofthe BaseTypeDefinition.
b. Re configure the Shortname of the SwBaseType to one of the Standard types {i.e Standard Types: Boolean, uint8, uint16, uint32, sint8, sint16, sint32, uint8_least, uint16_least, uint32_least, sint8_least, sint16_least, sint32_least, float32, float64}.

4.55 ASW4X_DT_0004 [Constr_1011, 1012, 1013]

Error Description

a. In a SwBaseType <Short Name1>, the "Category" attribute shouldbe configured with either FIXED_LENGTH or VARIABLE_LENGTH.
b. Ina SwBaseType, if the "Category"attribute is configuredwith-FIXED_LENGTH thenthe "BaseTypeSize"of the BaseTypeDirectDefinitionshouldbe configured.
c. Ina SwBaseType, the "Category"attribute is configuredwithVARIABLE_LENGTH thenthe "MaxBaseTypeSize" ofthe BaseTypeDirectDefinitionshouldbe configured.

How to Solve

a. Delete the BaseTypeDirectDefinition.
b. Configure the BaseTypeSize attribute ifthe category ofSwBaseType is FIXED_LENGTH.
c. Configure the MaxBaseTypeSize attribute ifthe category ofSwBaseType is VARIABLE_LENGTH.

4.56 ASW4X_DT_0004_411 [Constr_1011, 1012, 1013]

Error Description

a. In a SwBaseType, the "Category" attribute is configured with FIXED_LENGTH but the "BaseTypeSize" of the BaseTypeDirectDefinitionshouldbe configured.
b. In a SwBaseType, the "Category" attribute is configured with FIXED_LENGTH but the "MaxBaseTypeSize" of the BaseTypeDirectDefinitionshouldnotbe configured.
c. In a SwBaseType, the "Category" attribute is configured withVARIABLE_LENGTH but the "MaxBaseTypeSize" ofthe BaseTypeDirectDefinitionshouldbe configured.
d. In a SwBaseType, the "Category" attribute is configured with VARIABLE_LENGTH but the "BaseTypeSize" of the BaseTypeDirectDefinitionshouldnotconfigured.

How to Solve

a. Delete the SwBaseType/BaseTypeDirectDefinition.
b. Re configure the SwBaseType basedonthe Category attribute as below
IfCategory = FIXED_LENGTH thenconfigure BaseTypeSize of-BaseTypeDirectDefinition.

IfCategory = FIXED_LENGTH thendon'tconfigure MaxBaseTypeSize of-BaseTypeDirectDefinit- ion.

IfCategory = VARIABLE_LENGTH thenconfigure MaxBaseTypeSize of-BaseTypeDirectDefiniti-on.

IfCategory = VARIABLE_LENGTH thendonotconfigure BaseTypeSize of-BaseTypeDirectDefini- tion.

4.57 ASW4X_DT_0006 [Constr_1143]

Error Description

a. In ApplicationDataType, the value of "Category" attribute should be one among the {VALUE, VAL_BLK, STRUCTURE, ARRAY, STRING, BOOLEAN, CURVE, MAP, COM_AXIS, RES_AXIS}.
b. In ApplicationRecordElement, the "Category" attribute should not be configured with one among the {COM_ AXIS, RES_AXIS}.
c. In ApplicationRecordElement, the value of "Category" attribute should be one among the {VALUE, VAL_BLK, STRUCTURE, ARRAY, STRING, BOOLEAN, CURVE, MAP}.

d. In ImplementationDataType, the value of "Category" attribute should be one among the {VALUE, DATA_ REFERENCE, FUNCTION_REFERENCE, TYPE_REFERENCE, STRUCTURE, UNION, ARRAY}.
e. In ApplicationArrayElement, the value of "Category" attribute should be one among the {VALUE, VAL_BLK, STRUCTURE, ARRAY, STRING, BOOLEAN, CURVE, MAP, COM_AXIS, RES_AXIS}.

How to Solve

a. Delete the DataType.
b. Re configure the Category attribute ofrespective DataTye withits respective value.

4.58 ASW4X_DT_0007 [Constr_1008]

Error Description

InanApplicationPrimitiveDataType, the "Category"attribute shouldnotbe configuredwitheither STRUCTUREor ARRAY.

How to Solve

a. Delete the DataType.
b. Re configure the Category attribute ofApplicationPrimitiveDataTye withother thanSTRUCTUREor ARRAY.

4.59 ASW4X_DT_0008 [Constr_1022]

Error Description

In a CompuMethod, Both limits(UpperLimit and LowerLimit) of PhysicalToInternal and InternalToPhysical CompuScales shouldbe specified.

How to Solve

a. Allow the user to configure the range for upper Limit
b. Allow the user to configure the range for Lower Limit
c. If upper and lower Limit containers are not available then this needs to be created.

4.60 ASW4X_DT_0009 [Constr_1025]

Error Description

Ina CompuMethod, the value ofCompuNominatorDenominator inany CompuScale shouldnotbe zero.

How to Solve

Allow the user to input the value of CompuDenominator attribute to value other than 0.

4.61 ASW4X_DT_0010 [Constr_1004]

Error Description

Ina SwComponentType, the SwcInternalBehavior will have DataTypeMappingSets referredwhichinturnhas DataTypeMaps. This DataTypeMaps shouldnothave reference tomore thanone ImplementationDataType for one ApplicationDataType.

How to Solve

a. Delete the DataTypeMappingSet.
b. Re configure the DataTypeMaps ofDataTypeMappingSetsuchthatithas only one Implementa- tionDataType for every ApplicationDataType.

4.62 ASW4X_DT_0011 [Constr_1133]

Error Description

In a CompuMethod, the Vtvalues ofthe CompuScaleContents/CompuInverseValue definedinany CompuScale shouldbe unique.

How to Solve

a. Delete the CompuScale.
b. Configure unique values ofVtinCompuScaleContents/CompuInverseValue.

4.63 ASW4X_DT_0012 [Constr_1146]

Error Description

In a CompuMethod, when the 'Symbol' attribute of any CompuScale within the CompuMethod is configured then the category of the CompuMethod

should be one among the {SCALE_LINEAR_AND_TEXTTABLE, SCALE_ RATIONAL_AND_TEXTTABLE, TEXTTABLE, TAB_NOINTP, BITFIELD_TEXTTABLE}.

How to Solve

a. Delete CompuMethod.
b. Delete the CompuScale for whichSymbol attribute is configured.
c. Re configure the category ofCompuMethodwithone amongthe {SCALE_LINEAR_AND_TEXTTA- BLE, SCALE_ RATIONAL_AND_TEXTTABLE, TEXTTABLE, TAB_NOINTP, BITFIELD_TEXTTABLE}.

4.64 ASW4X_DT_0013 [Constr_1026]

Error Description

In a VariableDataPrototype, the Unit values defined in the SwDataDefPropsConditional and CompuMethod shouldbe compatible.

Ina AutosarDataType the Unitvalues definedinthe SwDataDefPropsConditional andCompuMe- thodshouldbe compatible.

How to Solve

a. Delete the CompuMethod/SwDataDefPropsConditional.
b. Configure the Units ofCompuMethodandSwDataDefPropsConditional suchthatbothare compatible.

4.65 ASW4X_DT_0014 [Constr_1093]

Error Description

a. Ina ApplicationPrimitiveDataType, ifthe "Category"attribute is configuredwithSTRING thenthe SwTextProps shouldbe configuredinthe SwDataDefPropsConditional.
b. In a ApplicationPrimitiveDataType, if the "Category" attribute is configured with STRING then the ArraySizeSemantics ofthe SwTextProps shouldbe configured.
c. Ina ApplicationPrimitiveDataType, ifthe "Category"attribute is configuredwithSTRING thenthe SwMaxTextSize ofthe SwTextProps shouldbe configured.

How to Solve

a. Delete the SwDataDefPropsConditional.
b. Configure the SwTextProps.
c. Configure the ArraySizeSemantics/ SwMaxTextSize inthe SwTextProps

4.66 ASW4X_DT_0016 [Constr_2051]

Error Description

InApplicationValueSpecification, the SwValueContshouldbe configured-withSwArraysize.

How to Solve

a. Delete the ApplicationValueSpecification.
b. Configure the SwArraySize ofthe SwValueCont

4.67 ASW4X_DT_0017 [Constr_1177]

Error Description

a. In SwPointerTargetProps, the "TargetCategory" should be configured with one among the {FUN-CTION_ REFERENCE, DATA_REFERENCE, TYPE_REFER-ENCE}.
b. InSwPointerTargetProps, ifthe "TargetCategory"is configuredwithVALUE-thenthe SwBaseTy-pe withNativeDeclarationsetto'VOID' shouldbe config-ured.

How to Solve

a. Delete the SwPointerTargetProps.
b. Configure the SwPointerTargetProps withSwBaseType whose NativeDeclara-tionis configured.

4.68 ASW4X_DT_0018 [Constr_1178]

Error Description

In an ImplementationDataType, the SwDataDefPropsConditional should be configured with only one attribute amongthe [BaseType, Implementa-tionDataType, SwPointerTargetProps]

How to Solve

a. Delete the SwDataDefPropsConditional.
b. Configure the SwDataDefPropsConditional withonly one attribute amongthe [BaseType, ImplementationDataType, SwPointerTargetProps]

4.69 ASW4X_DT_0019 [Constr_1017]

Error Description

a. Ina SwDataDefPropsConditional, whenthe "SwImplPolicy"attribute is configuredwithCONST thenthe "SwCalibrationAccess"attribute shouldbe configuredwithvalue READ-WRITE.
b. Ina SwDataDefPropsConditional, whenthe "SwImplPolicy"attribute is configuredwithFIXED th- enthe "SwCalibrationAccess"attribute shouldbe configuredwithvalue NOT-ACCESSIBLE.
c. In a SwDataDefPropsConditional, when the "SwImplPolicy" attribute is configured with MEASUR- EMENTPOINT thenthe "SwCalibrationAccess"attribute shouldbe configuredwithvalue READ -ONLY.
d. In a SwDataDefPropsConditional, when the "SwImplPolicy" attribute is configured with QUEUED then the "SwCalibrationAccess"attribute shouldbe configuredwithvalue NOT-ACCESSIBLE.

How to Solve

a. Delete the SwDataDefPropsConditional.
b. Re configure the SwCalibrationAccess attribute accordingtothe SwImplPolicy attribute.

4.70 ASW4X_DT_0020 [Constr_2533]

Error Description

Ina SwDataDefPropsConditional, whenthe value ofattribute "SwRecordLayoutGroupIndex"is settoZEROthen the Category attribute ofthe ApplicationDataType shouldbe settoVALUEor VAL_B- LK.

How to Solve

a. Delete the SwDataDefPropsConditional.
b. Re configure the Category attribute ofApplicationDataType toeither VALUEor VAL_BLK.

4.71 ASW4X_DT_0021 [Constr_1142]

Error Description

InCompuMethod, the "Category" attribute should be configured withvalue one among the [IDENTICAL, LINEAR,

SCALE_LINEAR, SCALE_LINEAR_AND_TEXTTABLE, RAT_FUNC,SCALE_RAT_FUNC, SCALE_RATIONAL_AND_ TEXTTABLE, TEXTTABLE, TAB_NOINTP, BITFIELD_TEXTTABLE]

How to Solve

a. Delete the CompuMethod.
b. Configure the Category attribute ofCompuMethodtoone amongthe [IDENTICAL, LINEAR, SCA -LE_LINEAR,
c. SCALE_LINEAR_AND_TEXTTABLE, RAT_FUNC, SCALE_RAT_FUNC, SCALE_RATIONAL_AND_TEXTTABLE, TEXTTABLE, TAB_NOINTP, BIT-FIELD_TEXTTABLE]

4.72 ASW4X_DT_0022 [Constr_2245]

Error Description

InanAutosarDataType, the "InvalidValue"attribute configuredinthe SwDataDefPropsConditional shouldbe with inthe limits definedinthe Data-ConstrRule.

How to Solve

Option to set the InvalidValue range upper and lower limit within the range specified in ImplementationDatatype.

4.73 ASW4X_DT_0023 [Constr_2550]

Error Description

a. In an AutosarDataType, the "Unit" attribute configured in the SwDataDe-fPropsConditional and ApplicationPrimitiveDataType shouldbe same.
b. InanAutosarDataType, the "Unit"attribute configuredinthe Com-puMethodandApplicationPrimitiveDataType shouldbe same.

How to Solve

a. Delete the SwDataDefPropsConditional.
b. Reconfigure the Unitvalue attribute ofSwDataDefPropsConditional/Com-puMethodandApplicationPrimitiveDataType tobe same.

4.74 ASW4X_DT_0024 [Constr_2551]

Error Description

In an ApplicationDataType, the "BaseType" attribute of the SwCalprmAxis de-fined in the SwDataDefPropsConditional shouldnotbe configured.

How to Solve

a. Delete the SwCalprmAxis.
b. Delete the configuredBaseType.

4.75 ASW4X_DT_0025 [Constr_1134]

Error Description

a. Ina CompuMethod, the "Category"attribute is configuredwithTEXTTABLE-butthe CompuPhys -ToInternal is notconfigured.
b. Ina CompuMethod, the "Category"attribute is configuredwithTEXTTABLE-butthe CompuInter- nalToPhys is notconfigured.
c. Ina CompuMethod, the "Category"attribute is configuredwithTEXTTABLE-butthe Upper/Low-er Limitofthe CompuScale definedinthe CompuInter-nalToPhys is notset.

How to Solve

a. Option to remove the attribute Unit or PhyConstr if configured
b. Allow Configuring CompuInternalToPhys with CompuScale defining Upper/ Lower Limit

4.76 ASW4X_DT_0026 [Constr_2548]

Error Description

In an AutosarDataType, the "DataConstr" attribute configured in the SwDataDefPropsConditional and ApplicationPrimitiveDataType shouldbe same.

How to Solve

a. Delete the DataType.
b. Re configure the DataConstr ofSwDataDefPropsConditional andApplica-tionPrimitiveDataType tobe same.

4.77 ASW4X_DT_0027 [Constr_1016]

Error Description

InanAutosarDataType, the "InvalidValue"attribute configuredinthe SwDataDe-fPropsConditional shouldbe one amongthe [NumericalValueSpecification, TextValueSpecification, CostantRference].

How to Solve

a. Delete the SwDataDefPropsConditional.
b. Configure the InvalidValue attribute of the SwDataDefPropsConditional to be one among the [NumericalValueSpecification, TextValueSpecification,Con-stantReference].

4.78 ASW4X_DT_0028 [Constr_2549]

Error Description

a. InanAutosarDataType, the "Unit"attribute configuredinthe SwAxisIndividual andCompuMethodshouldbe same.
b. InanAutosarDataType, the "Unit"attribute configuredinthe SwAxisIndividual andApplicationPrimitiveDataType shouldbe same.
c. InanAutosarDataType, the "Unit"attribute configuredinthe SwAxisIndividual andSwVariableRefProxy should be same.

How to Solve

a. Delete the SwAxisIndividual.
b. Re configure the Unitvalues tobe same.

4.79 ASW4X_DT_0029 [Constr_1014]

Error Description

Ina SwBaseType, the attribute "BaseTypeEncoding"Configuredshouldbe one ofthe followingvalues: 1C, 2C, BCD-P, BCD-UP, DSP-FRACTIONAL, SM, IEEE754, ISO-8859-1, ISO-8859-2, WINDOWS-1252, UTF-8,UTF-16, UCS-2, NONE, VOID, BOOLEAN.

How to Solve

Configure the BaseTypeEncodingattribute tobe one amongthe "1C, 2C, BCD-P, BCD-UP, DSP-FRACTIONAL, SM,

IEEE754, ISO-8859-1, ISO-8859-2, WINDOWS-1252, UTF-8,UTF-16, UCS-2, NONE, VOID, BOOLEAN"

4.80 ASW4X_DT_0030 [Constr_2536]

Error Description

Ina CompuMethodofcategory "BITFIELD_TEXTTABLE", the Vtvalues ofthe CompuScaleConstantContents/CompuConst/CompuConstTextCon-tentdefinedshouldnotcontain"|"

How to Solve

a. Delete the "|"inVtvalue ofCompuConstTextContent.
b. Reconfigure the Category ofthe CompuMentodwithValue other thanBIT-FILED_TEXTTABLE.

4.81 ASW4X_DT_0031 [Constr_2561]

Error Description

a. In a DataContr, the value ofthe ConstrLevel configuredunder the Data-ConstrRule is limitedtothe following:0: This represents socalled""hardlim-its"". They shall always be specified.
b. 1: This represents socalled""softlimits"". Softlimits may be violatedafter confirmationby the user ofanMCDSystem. Other values may exist, butthe semantics is outside ofthe AUTOSARscope.

How to Solve

a. Delete the ConstrLevel configuration.
b. Setthe ConstrLevel value toeither 0or 1.

4.82 ASW4X_DT_0032 [Constr_2544]

a) Error Description

InanApplicationDataType, the limits definedinthe DataConstr shouldbe equal toor withinthe limits definedin CompuMethod.

How to Solve

a. Delete the limits definedinDataConstr
b. Configure the limits ofDataConstr ofApplicationDatatype tobe equal or withinthe limits ofcompuMethod.

b) Error Description

InanApplicationDataType, the limits definedinthe DataConstr shouldbe equal toor withinthe limits definedin ImplementationDataType.

How to Solve

1. Delete the limits definedinDataConstr.
2. Configure the limits ofDataConstr ofApplicationDatatype tobe equal or withinthe limits ofImplementationDataType.

c) Error Description

InanImplementationDataType, the limits definedinthe DataConstr shouldbe equal toor withinthe limits defined inBaseType..

How to Solve

a. Delete the limits definedinDataConstr.
b. Configure the limits ofDataConstr ofApplicationDatatype tobe equal or withinthe limits ofBaseType.

4.83 ASW4X_DT_0033 [Constr_1007]

a) Error Description

Allowedattribute ofswDataDefProps for ApplicationDatatypes andtheir allowedmultiplies.

4.84 ASW4X_DT_0034 [Constr_1009]

a) Error Description

Allowedattribute ofswDataDefProps for ImplementationDatatypes andtheir allowedmultiplies.

4.85 ASW4X_DT_0035 [Constr_rte_sws_3851]

Error Description

The CompuMethod with category "TEXTTABLE", "SCALE_LINEAR_AND_TEXTTABLE", or "SCALE_RATIONAL_

AND_TEXTTABLE"thatcontains a CompuScale witha pointrange

a. The attribute symbol ofthe CompuScale shouldbe available andnotempty
b. The stringspecifiedinthe VT elementofthe CompuConstofthe CompuScale is a validC identifier
c. The attribute shortLabel ofCompuScale is available andnotempty.

How to Solve

a. Symbol shouldbe available andnotempty.
b. Value ofvariable vtshouldbe a validC identifier
c. ShortLabel shouldbe available andnotempty.

4.86 ASW4X_DT_0036 [Constr_1006]

a) Error Description

The Categories allowed for SwPointerTargetProps are VALUE, DATA_REFERENCE, FUNCTION_REFERENCE, TYPE_REFERENCE, STRUCTURE, UNION, ARRAY.

How to Solve

Reconfigure the value TargetCategory of SwPointerTargetProps to have any of the following VALUE, DATA_ REFERENCE, FUNCTION_REFERENCE, TYPE_REFERENCE, STRUCTURE, UNION, ARRAY.

b) Error Description

The validCategories allowedfor the SystemConstantis - VALUE.

How to Solve

Reconfigure the value ofthe attribute "Category"ofSystemConstanttoVALUE.

c) Error Description

The valid Categories allowed for Returntype_SwServiceArg and - Arguments_SwServiceArg are DATA_ REFERENCE, FUNCTION_REFERENCE,TYPE_REFERENCE.

How to Solve

Reconfigure the value Category of SwServiceArg with any of the following DATA_REFERENCE, FUNCTION_ REFERENCE,TYPE_REFERENCE.

d) Error Description

The validCatergories allowedfor McDataInstance - VALUE, VAL_BLK, STRUCTURE, UNION, ARRAY, BIT, HOST, STRING, BOOLEAN, COM_AXIS, RES_AXIS, CURVE, MAP.

How to Solve

Reconfigure the value Category of McParameterInstance or McVariableInstance to contain any of the following VALUE, VAL_BLK, STRUCTURE, UNION, ARRAY, BIT, HOST, STRING, BOOLEAN, COM_AXIS, RES_AXIS, CURVE, MAP.

4.87 ASW4X_DT_0038 [Constr_1020]

Error Description

The DataType referred in the SharedAxisType of the SwAxisGrouped is not compatible with the DataType referredby the ParameterDataPrototype inthe AutosarParameterRef.

How to Solve

Reconfigure ShareAxisType andTargetDataPrototype tobe ofthe same Data type.

4.88 ASW4X_DT_0065 [Constr_1288]

Error Description

A complete list of the SwDataDefProps and other attributes and their multiplicities whichare allowedfor a given category.

How to Solve

Remove value of attribute {2}for the SwDataDefProps of implementationDataType.

4.89 ASW4X_DT_0039 [Constr_1241]

Error Description

The Application Primitive Data Type with Category other than STRING shall not contain InvalidValue.

How to Solve

Provide an option to remove invalidValue within the the CompoundPrimitiveDataType with Category other than STRING.

4.90 ASW421_DT_0057[Constr-1322]

Error Description

The ImplementationDatatype of category ARRAY should not contain ImplementationDatatypeElement with the category as TYPE_REFERENCE.

How to Solve

a. Change the category to STRUCTURE
b. Setthe value for Dynamic array size profile
c. SetBoth

4.91 ASW421_DT_Constr_1289 [Constr-1289]

Error Description

The Dataprototype typedby ApplicationDataType withmentionedcategory havingSwDataDefProps shouldnotbe configuredwiththe attributes, mentionedinthe table 5.31ofsoftware specification

How to Solve

Delete the givenattribute for the SwDataDefProps ApplicationDataType

4.92 ASW422_DT_0016[Constr-2051]

Error Description

IfApplicationValueSpecificationis configuredwithone ofthe followingcategories - CURVE, MAP, CUBOID, CUBE_ 4, CUBE_5, COM_AXIS,RES_AXIS, CURVE_AXIS, or VAL_BLK, andSwValueContis notconfiguredwithSwArraysize, thenitis anerror

How to Solve

Configure the SwArraySize ofthe SwValueCont

4.93 ASW422_DT_0047[Constr-1389]

Error Description

InVariationPointProxy the category ofthe attribute ImplementationDatatype shouldnotbe any ofthe following: STRUCTURE, ARRAY, UNION, FUNCTION_REFERENCE, andDATA_REFERENCE.

Butitshouldbe ofcategory VALUEor TYPE_REFERENCE.After all the TYPE-REFERENCEis resolved,the implementationshouldhave category as VALUE

How to Solve

a. Change the category ofthe attribute ImplementationDatatype toTYPE_REFERENCEor VALUE
b. Change the category ofthe attribute ImplementationDatatype toVALUE
c. Ifthe Category is ImplementationDatatype is TYPE_REFERNCE, thencreate another ImplementationDatatype1 withCategory value andassignittoImplementationDatatype1.

4.94 ASW422_DT_0048[Constr-1397]

Error Description

The TransformerHardErrorEvent should contain either of the attributes Operation or trigger.The attributes operationandtrigger shall be mutually exclusive..

How to Solve

a. Create any one ofthe attribute Operationor trigger for the Transformer-HardErrorEvent
b. Delete any one ofthe attribute Operationor trigger for the Transformer-HardErrorEvent

4.95 ASW422_DT_0049 [Constr_1398]

Error Description

InVariationPointProxy attribute category ofImplementationDatatype shouldnotbe any ofthe following:
STRUCTURE, ARRAY, UNION, FUNCTION_REFERENCE, andDATA_REFERENCE.Butitshouldbe ofcategory VALUEor TYPE_REFERENCE. The allowedcategory are "VALUE"and"TYPE_REFERENCE"andifitis type reference itshouldboil downtoVALUE.

How to Solve

Setcategory toVALUE.

4.96 ASW422_DT_0050 [Constr_1390]

Error Description

For all the RuleBasedValueContelement, the attribute Unitshouldbe assignedwitha value

How to Solve

Provide an option for user to enter a value (for the invalidValue) that is not inbetween the lower and the upper limit of the compuMethod.

4.97 ASW422_DT_0051 [Constr_1393]

Error Description

For all the RuleBasedValueContelement, the attribute Unitshouldbe assignedwitha value.

How to Solve

Configure the elementUnitandassigna value tothe attribute Unit.

4.98 ASW422_DT_0052[Constr-1396]

Error Description

The ImplementationDatatype ofcategory ARRAY shouldnotcontainImplementationDatatypeElementwiththe category as TYPE_REFERENCE.

How to Solve

a. Change the Category value ofImplementationDataTypeElementfromTYPE_REFERENCEtoVALUE
b. Create a new ImplementationdataType ofcategory VALUEandthenassignittothe firstimlementationdata type ofTYPE_REFERENCE

4.99 ASW422_DT_0054[Constr-2058]

Error Description

IfAutosarDataype is inany ofthe followingcategories {CURVE, MAP, CUBOID, CUBE_4, CUBE_5, COM_AXIS, RES_AXIS, CURVE_AXIS, VAL_BLK or ARRAY}, andIfthe attribute swValueContis definedfor anApplicationRuleBasedValueSpecification+RuleBasedValueCont, thenRuleBasedValueContshall always define the attribute Arraysize.

How to Solve

Define the attribute swArraysize for SwValueCount.

4.100 ASW422_DT_0057[Constr-1318]

Error Description

If the value of attribute ImplementationDataType.dynamicArraySizeProfile is set to VSA_LINEAR, the ImplementationDataType shall aggregate a VSA Payload ImplementationDataTypeElement that fulfills all of the followingconditions:

a. The attribute ImplementationDataTypeElement.arraySizeSemantics shall notbe defined
b. The attribute ImplementationDataTypeElement.category shall be settoARRAY.
c. The attribute ImplementationDataTypeElement.arraySize shall notbe defined.
d. The attribute ImplementationDataTypeElement.arraySizeHandlingshall notbe defined.

The VSA Payload ImplementationDataTypeElement shall immediately aggregate another ImplementationDataTypeElementthatshall fulfill all ofthe followingconditions:

a. The attribute ImplementationDataTypeElement.arraySizeSemantics shall be settothe value variableSize.
b. The attribute ImplementationDataTypeElement.arraySize shall be defined.

c. The attribute ImplementationDataTypeElement.arraySizeHandling shall be set to the value allIndicesSameArraySize.

How to Solve

a. Configure VSA PayloadImplementationDataTypeElementwithinthe ImplementationDatatype
b. Delete the value for arraySizeSemantics
c. Configure Category toARRAY
d. Delete the definedarraySize
e. Delete the value for arraySizeHandling
f. Delete the value for arraySizeSemantics
g. Delete the value for arraySize
h. Delete the value for arraySizeHandling

4.101 ASW422_DT_0058[Constr-1315]

Error Description

a. InApplicationArrayDataType the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_ SQUARE which contains an ApplicationArrayElement then the attribute ApplicationArrayElement.arraySizeSemantics shouldbe settoVARIABLE-SIZE.
b. InApplicationArrayDataType the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_ SQUARE which contains an ApplicationArrayElement but the attribute ApplicationArrayElement.maxNumberOfElements is definedwhichis invalid.
c. In ApplicationArrayDataType if the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_ SQUARE which contains an ApplicationArrayElement, then the attribute Applic-
d. ationArrayElement.arraySizeHandlingshouldbe settoinheritedFromArrayElementTypeSize"
e. In ApplicationArrayDataType if the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_SQUARE which contains an ApplicationArrayElement, then the attribute ApplicationArrayElement.type shall notrefer toanApplicationArrayDataType whose dynamicArraySizeProfile is set
f. InApplicationArrayDataType the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_SQUARE which contains an ApplicationArrayElement with the attribute ApplicationArrayElement.type refers to an ApplicationArrayDataType which is last and has to satisfy following conditions for array element withinit
g. The attribute ApplicationArrayElement.arraySizeSemantics shall be settoVARIABLE-SIZE

h. The attribute ApplicationArrayElement.maxNumberOfElements shall be defined
i. The attribute ApplicationArrayElement.arraySizeHandlingsettothe value allIndicesSameArraySize.

j. InApplicationArrayDataType the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_SQUARE which contains an ApplicationArrayElement with the attribute ApplicationArrayElement.type refers to an ApplicationArrayDataType which is last and has to satisfy following conditions for array element withinit

The attribute ApplicationArrayElement.arraySizeSemantics shall be settoVARIABLE-SIZE

The attribute ApplicationArrayElement.maxNumberOfElements shall notbe defined

The attribute ApplicationArrayElement.arraySizeHandlingsettothe value alIndiclesSameArraySize.

The ApplicationArrayElementshall be typedby anApplicationArrayDataType.

How to Solve

a. Setthe arraySizeSemantics value tovariableSize
b. Delete the value for maxNumberOfElements
c. Setthe arraySizeHandlingtoinheritedFromArrayElementTypeSize
d. Delete the value for arraySizeSemantics
e. Setthe value for arraySizeSemantics as VARIABLE-SIZE
f. Setthe value ofmaxNumberOfElements
g. Setthe arraySizeHandlingtoallIndicesSameArraySize
h. Setthe value for arraySizeSemantics as VARIABLE-SIZE
i. Setthe value ofmaxNumberOfElements
j. Setthe arraySizeHandlingtoallIndicesSameArraySize
k. Setthe type toApplicationArrayDataType whose dynamicArraySizeProfile is notset

4.102 ASW422_DT_0059[Constr-1316]

Error Description

a. InApplicationArrayDataType the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_ RECTANGULAR which

contains an ApplicationArrayElement then the attribute ApplicationArrayElement.arraySizeSemantics shouldbe settoVARIABLE-SIZE."

b. InApplicationArrayDataType the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to
c. _ RECTANGULAR which contains an ApplicationArrayElement but the attribute ApplicationArrayElement.maxNumberOfElements shouldbe defined."
d. In ApplicationArrayDataType if the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_ RECTANGULAR which contains an ApplicationArrayElement, then the attribute ApplicationArrayElement.arraySizeHandlingshouldbe settoallIndicesSameArraySize"
e. In ApplicationArrayDataType if the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_ RECTANGULAR which contains an ApplicationArrayElement, then the attribute ApplicationArrayElement.type shall notrefer toanApplicationArrayDataType whose dynamicArraySizeProfile is set
f. InApplicationArrayDataType the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_ RECTANGULAR which contains an ApplicationArrayElement with the attribute ApplicationArrayElement.type refers to an ApplicationArrayDataType which is last and has to satisfy following conditions for array elementwithinit
g. The attribute ApplicationArrayElement.arraySizeSemantics shall be settoVARIABLE-SIZE
h. The attribute ApplicationArrayElement.maxNumberOfElements shall be defined
i. The attribute ApplicationArrayElement.arraySizeHandlingsettothe value allIndicesSameArraySize.
j. InApplicationArrayDataType the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to VSA_ RECTANGULAR which contains an ApplicationArrayElement with the attribute ApplicationArrayElement.type refers to an ApplicationArrayDataType which is last and has to satisfy following conditions for array elementwithinit

   The attribute ApplicationArrayElement.arraySizeSemantics shall be settoVARIABLE-SIZE

   The attribute ApplicationArrayElement.maxNumberOfElements shall notbe defined

   The attribute ApplicationArrayElement.arraySizeHandlingsettothe value allIndicesSameArraySize.

The ApplicationArrayElementshall be typedby anApplicationArray-DataType.

How to Solve

a. Setthe arraySizeSemantics value tovariableSize.
b. configure the value for maxNumberOfElements.
c. Setthe arraySizeHandlingtoallIndicesSameArraySize.
d. Delete the value for arraySizeSemantics.
e. Setthe value for arraySizeSemantics as VARIABLE-SIZE.
f. Setthe value ofmaxNumberOfElements.
g. Setthe arraySizeHandlingtoallIndicesSameArraySize.
h. Setthe value for arraySizeSemantics as VARIABLE-SIZE.
i. Setthe value ofmaxNumberOfElements.
j. Setthe arraySizeHandlingtoallIndicesSameArraySize.
k. Setthe type toApplicationArrayDataType whose dynamicArraySizeProfile is notset.

4.103 ASW422_DT_0060[Constr-1317]

Error Description

a. InApplicationArrayDataType the value ofattribute ApplicationArray-DataType.dynamicArraySizeProfile is set to VSA_ FULLY_ FLEXIBLE which contains an ApplicationArrayElement then the attribute ApplicationArrayElement.arraySizeSemantics shouldbe settoVARIABLE-SIZE."
b. InApplicationArrayDataType the value ofattribute ApplicationArray-DataType.dynamicArraySizeProfile is set to VSA_ FULLY_ FLEXIBLE which contains an ApplicationArrayElement but the attribute ApplicationArrayElement.maxNumberOfElements shouldbe defined."
c. In ApplicationArrayDataType if the value ofattribute ApplicationArray-DataType.dynamicArraySizeProfile is set to VSA_ FULLY_ FLEXIBLE which contains an ApplicationArrayElement, then the attribute ApplicationArrayElement.arraySizeHandlingshouldbe settoallIndicesDifferentArraySize"
d. In ApplicationArrayDataType if the value ofattribute ApplicationArray-DataType.dynamicArraySizeProfile is set to VSA_ FULLY_ FLEXIBLE which contains an ApplicationArrayElement, then the attribute ApplicationArrayElement.type shall notrefer toanApplicationArrayDataType whose dynamicArraySizeProfile is set
e. InApplicationArrayDataType the value ofattribute ApplicationArray-DataType.dynamicArraySizeProfile is set to VSA_ FULLY_ FLEXIBLE which

contains an ApplicationArrayElement with the attribute ApplicationArrayElement.type refers to an ApplicationArrayDataType which is last and has to satisfy following conditions for array elementwithinit

The attribute ApplicationArrayElement.arraySizeSemantics shall be settoVARIABLE-SIZE

The attribute ApplicationArrayElement.maxNumberOfElements shall be defined

The attribute ApplicationArrayElement.arraySizeHandlingsettothe value allIndicesSameArraySize.

f. InApplicationArrayDataType the value ofattribute ApplicationArrayDataType.dynamicArraySizeProfile is set to

VSA_ FULLY_ FLEXIBLE which contains an ApplicationArrayElement with the attribute ApplicationArrayElement.type refers to an ApplicationArrayDataType which is last and has to satisfy following conditions for array elementwithinit

The attribute ApplicationArrayElement.arraySizeSemantics shall be settoVARIABLE-SIZE

The attribute ApplicationArrayElement.maxNumberOfElements shall notbe defined

The attribute ApplicationArrayElement.arraySizeHandlingsettothe value allIndicesDifferentArraySize.

g. The ApplicationArrayElementshall be typedby anApplicationArrayDataType.

How to Solve

a. Setthe arraySizeSemantics value tovariableSize
b. Configure the value for maxNumberOfElements
c. Setthe arraySizeHandlingtoallIndicesDifferentArraySize
d. Delete the value for arraySizeSemantics
e. Setthe value for arraySizeSemantics as VARIABLE-SIZE
f. Setthe value ofmaxNumberOfElements
g. Setthe arraySizeHandlingtoallIndicesSameArraySize
h. Setthe value for arraySizeSemantics as VARIABLE-SIZE
i. Setthe value ofmaxNumberOfElements
j. Setthe arraySizeHandlingtoallIndicesDifferentArraySize

k. Setthe type toApplicationArrayDataType whose dynamicArraySizeProfile is notset 12.

4.104 ASW4X_DT_constr_1694[Constr-1694]

Error Description

The attribute implementationdatatype withinthe SwDataDefProps shouldrefer toonly toanImplementationDatatype

SwDataDefPropsConditional inthis case canappear withinImplementationDataType, withinImplementationDataTypeElementandalsowithinSwPointerTargetProps Note: Category has tobe Type_Reference.

How to Solve

Allow the user tochoose animplementationDataType Quick fix notpossible ifnoImplementationDataType is present

4.105 ASW4X_DT_constr_1712[Constr-1712]

Error Description

Attribute IntendedPartialInitializationCountwithinarrayValueSpecificationshouldbe configuredifApplicationArrayDataType has arraysizesemantics settovariableSize.

How to Solve

May Notpossible

4.106 ASW4X_DT_1718[Constr_1718]

Error Description

a. Applicable toApplicationDataType andImplentationDataType ofCategory array.
b. SwDataDefProps ofApplicationArrayDataType or has the reference value setfor DataConstr thenSwDataDefProps ofApplicationArrayElementcannothave reference value setfor DataConstr.

How to Solve

Remove value ofthe DataConstr reference field(SwDataDefProps ofarrayelementor subelement).

4.107 ASW4X_DT_1719[Constr-1719]

Error Description

SwDataDefProps ofApplicationDataType has the attribute value setfor displayFormatthenSwDataDefProps of ApplicationArrayElementcannothave attribute value setfor displayFormat.

How to Solve

Remove value ofdisplayFormat(SwDataDefProps ofarrayelementor subelement).

4.108 ASW4X_DT_1720[Constr-1720]

Error Description

SwDataDefProps ofApplicationDataType has the attribute value setfor stepsize thenSwDataDefProps ofApplicationArrayElementcannothave attribute value setfor stepsize.

How to Solve

Remove value ofthe stepSize (SwDataDefProps ofarrayelementor subelement).

4.109 ASW4X_DT_0027_411[Constr-1016]

Error Description

If the InvalidValue configured in the ImplementationDataType or ImplementationDataTypeElement is not NumericalValueSpecification, TextValueSpecification or ConstaneReference.

How toSolve

Configure the InvalidValue attribute of the SwDataDefPropsConditional to be one among the [NumericalValueSpecification,

TextValueSpecification,ConstantReference].

4.110 ASW4X_DT_0048_20_11 [constr_1397]

Error Description

The TransformerHardErrorEvent should contain either of the attributes Operation or trigger.The attributes operation and trigger shall be mutually exclusive

How to Solve

Configure any one of the attribute Operation or trigger, for the TransformerHardErrorEvent

4.111 ASW4X_DT_0054_20_11 [constr_2058]

Error Description

If AutosarDataype is in any of the following categories {CURVE, MAP, CUBOID, CUBE_4, CUBE_5, COM_AXIS, RES_AXIS, CURVE_AXIS, VAL_BLK}, and If the attribute swValueCont is defined for an ApplicationRuleBasedValueSpecification+RuleBasedValueCont, then RuleBasedValueCont shall always define the attribute swArraysize

How to Solve

a. Define the attribute swArraysize for SwValueCount.
b. Internal Behaviour Validations
c. This section deals with the error description of software internal behaviour validations. Software Internal Behaviour Validations are specificto-software template inautosar.

4.112 ASW4X_IB_0002 [Constr_1094]

Error Description

Itis notallowedthatseveral RunnableEntity(s)share the same value ofthe attribute symbol.

How to Solve

Reconfigure the value ofthe attribute Symbol ofall the runnable entity tohave unique values.

4.113 3.55ASW4X_IB_0041 [Constr_1228]

Error Description

The RTEEvent that is referenced through Trigger by any WaitPoint, then this RTEEvent should not have

the activationReasonRepresentation configured..

How to Solve

Allow the user to remove the reference value of ActivationReasonRepresentation configured within.

4.114 ASW421_IF_0001 [Constr_1304]

Error Description

The Value of Application error code should not exceed the close interval

How to Solve

Set application error code value between 1 to 63

4.115 ASW421_IF_0010 [Constr_1304]

Error Description

In SenderReceiverInterface, the Invalidation policy can be configured for only the DataElements that has the SwImplPolicy value other than ï¿½QUEUEDï¿½

How to Solve

By default, set the the SwImplPolicy as STANDARD.

4.116 ASW4X_IF_008B [Constr_1270]

Error Description

In a ClientSerevrOperationMapping,Each argument should be referred only once in either firstDataPrototype or seconDataPrototype

How to Solve

In a particular ClientServerOperationMapping, if any argumentPrototype is referred more than once, in either firstDataPrototype or seconDataPrototype, then provide an option for user to choose different value for SecondDataPrototype.

4.117 ASW4X_DT_0041 [Constr_1243]

Error Description

The NumericalOrText in ApplicationValueSpecification should be configured with any one of the two attributes vf and vt.

How to Solve

Option to remove the one of the fields Vf or Vt. By default remove other value by retaining value for Vt which is the first field that can be configured.

4.118 ASW4X_DT_0043 [Constr_1255]

Error Description

If the ApplicationPrimitiveDataType with Category BOOLEAN or STRING has unit configured directly inSwDataDefProps or inside the PhysConstrs, then the unit should refer a physicalDimension with all the exp values either not configured or configured as ï¿½0ï¿½.

How to Solve

The referenced physicalDimension should define exponent value 0.

4.119 ASW421_IF_008A [Constr_1269]

Error Description

In a ClientSerevrOperationMapping, The number of ArgumentPrototype defined in firstOperation and secondOperation should be same.

How to Solve

If the arguments in firstoperation and secondOperation is mismatch, then provide an option for the user to add arguments from the Operation that has less number of arguments so that both will be equal.

4.120 ASW421_IF_0009 [Constr_1304]

Error Description

IfVariableAndParameterInterfaceMappingcontains a firstDataPrototype thatdoes notrefer toa compuMethodof category BITFIELD_TEXTTABLE, thenifbitfieldTextTableMaskFirstattribute is definedinthe TextTableMapping inDataPrototypeMappingthenitis anerror

How to Solve

Change the category ofcompuMethodtoBITFIELD_TEXTTABLE

4.121 ASW421_IF_0010 [Constr_1305]

Error Description

IfVariableAndParameterInterfaceMappingcontains a secondDataPrototype thatdoes notrefer toa compuMethod ofcategory BITFIELD_TEXTTABLE, thenifbitfieldTextTableMaskSecondattribute is definedinthe TextTable-MappinginDataPrototypeMappingthenitis anerror

How to Solve

Change the category ofcompuMethodtoBITFIELD_TEXTTABLE

4.122 ASW421_IF_0011 [Constr_1306]

Error Description

Ina ClientSerevrOperationMapping/VariableAndParameterInterface-Mapping,firstDataPrototype and secondDataPrototype referingtoa com-puMethodofcategory BITFIELD_TEXTTABLE, andTex-tTableMapping.valuePair is configuredthenthe value ofattribute TextTable-Mapping.identicalMappingshouldbe true

How to Solve

Setthe value ofthe TexttableMapping.identicalMappingtofalse

4.123 ASW421_IF_0012 [Constr_1308]

Error Description

The attribute NvBlockNeeds.cyclicWritingPeriodshall existifandonly ifthe at-tribute NvBlockNeeds.storeCyclic exists andits value is settotrue.

How to Solve

1.Remove the value for cyclicWritingPeriod
2.Setthe value ofthe attribute storeCyclictoTRUE

4.124 ASW421_IF_0013 [Constr_1312]

Error Description

IfPRPortPrototype is typedby a parameterInterface, thenitis anerror

How to Solve

Change the ProvidedRequiredInterface ofPRPortPrototype

4.125 ASW422_IF_0015_VariableAndParameterInterfaceMapping [Constr_ 1307]

Error Description

Ifa TextTableMappingelementdefines bitfieldTextTable-MaskFirstor bitfieldTextTableMaskSecondthenall containedText-TableMapping.valuePair.firstValues andTextTableMapping.valuePair.secondValues shouldnotspecify a value, thatwouldbe ruledoutbasedonvalue ofTextTableMapping.mappingDirection

How to Solve

a. Configure TextTableValuePair.firstValues andbitfieldTextTableMaskFirsttobe equal
b. Configure FirstDataPrototype

4.126 ASW422_IF_0015_ClientServerInterfaceMapping [Constr_1307]

Error Description

Ifa TextTableMappingelementdefines bitfieldTextTable-MaskFirstor bitfieldTextTableMaskSecondthenall containedText-TableMapping.valuePair.firstValues andTextTableMapping.valuePair.secondValues shouldnotspecify a value, thatwouldbe ruledoutbasedonvalue ofTextTableMapping.mappingDirection

How to Solve

a. Configure TextTableValuePair.firstValues andbitfieldTextTableMaskFirsttobe equal
b. Configure SecondDataPrototype 3.

4.127 ASW4X_IF_0044[Constr_1635]

Error Description

For structureddata type (ApplictionRecordDataType,ImplementationRecordDataType)

Ifa subElementMappingis configuredfor the RecordElement, thenthe value ofattribute isOptional i. shouldbe same for bothrecordElements referredinfirstElementandsecondElementor ii. shouldnothave value definedfor bothrecordElements referredfor bothfirstElementandsecondElement.

How to Solve

Remove tothe value ofisOptional onbothe recordelements.

4.128 ASW4X_IF_0045[Constr_1726]

Error Description

Whenthere are multiple metaDataItems, metadataItemwithlargestvalue ofattribute lengthwill be infirstpositionandmetaDataItemwithsmallestvalue ofattribute lengthwill atthe lastposition. How toSolve

1. Quick fix is notpossible inthis validation.

4.129 ASW4X_IB_0004 [Constr_1164]

a) Error Description

In SwcInternalBehavior, the runnable entity arguments under the Runnable entity, can be configured only if the Runnable Entity is referedatleastinone ofthe operationInvokedEventavailable insame InternalBehavior.

How to Solve

Reconfigure the Runnable entity ref-

eredinthe OperationInvokedEvent

b) Error Description

In the SwcInternalBehavior Containing the Runnable entity, the number of runnable entity arguments should be equal to the sum of number of PortDefinedArguments and the number of ArgumentDataPrototypes in the ClientServerOperation

How to Solve

Reconfigure the Number of RunnbaleEntityArgs to match the PortDefinedArgument of the PortAPIOption + number ofArgumentDataPrototypes inClientServerOperation.

4.130 ASW4X_IB_0005 [Constr_1165]

Error Description

The existence of a RunnableEntityArgumentis limitedtoRunnableEntitys triggeredby a ClientServerOperation.

How to Solve

a. Delete the RunnableEntityArgument.
b. Configure TargetProvidedOperationinthe OperationInvokedEventtoClientServerOperation.

4.131 ASW4X_IB_0006 [Constr_2031]

Error Description

The value ofthe attribute periodofTimingEventshall be greater than0.

How to Solve

Reconfigure the value ofPeriodtobe greater than0.

4.132 ASW4X_IB_0007 [Constr_1090]

Error Description

a. A RunnableEntity referredby TimingEvents shouldnotbe configuredwith-more thanone WaitPointifitis scheduledonly once inthe EcucReference-Values.
b. In a RunnableEntity there should notbe more than one WaitPoint configuredif it is scheduled only once within the EcucReferenceValues.

How to Solve

a. Delete the extra WaitPoints.
b. Re-schedule the RunnableEntity more thanonce.

4.133 ASW4X_IB_0008A [Constr_1091]

Error Description

The Waitpoint can be triggered only by DataReceivedEvent, DatasendCompletedEvent, modeSiwtchAckEvent, AsynchronousServerCallReturnsEvent.

How to Solve

Reconfigure the WaitPoint to be triggered by DataReceivedEvent, DatasendCompletedEvent, modeSiwtchAckEvent, AsynchronousServerCallReturnsEvent.

4.134 ASW4X_IB_0008B [Constr_1096]

Error Description

A runnableEntity thathas a waitpointshall notbe referencedby a SwcModeSiwtchEvent

How to Solve

a. Delete the waitpoint.
b. Reconfigure the ModeSiwtchEventtorefer a differentRunnableEntity whichdoes nothave a waitpoint.

4.135 ASW4X_IB_0009 [Constr_1097]

Error Description

A RunnableEntity that has a WaitPoint shall not be referenced by a RTEEventthat has a reference in the role disabledMode.

How to Solve

a. Delete the waitpoint.
b. Reconfigure the OperationInvokedEventtorefer a differentRunnableEntity whichdoes nothave a waitpoint.

4.136 ASW4X_IB_0010 [Constr_2026]

Error Description

In a SwcInternalBehavior, the RunnableEntity and the variableDataPrototype configured in the LocalVariable inside the AutosarVariableRef of the WriteLocalVariable or ReadLocalVariable does not belong to same SwcInternalBehavior.

How to Solve

Configure the VariableDataPrototype inthe VariableAccessRefandthe Runnable entity tobelongtosame SwcInternalBehavior.

Error Description

In a SwcInternalBehavior contaning a RunnableEntity, the VariableDataPrototype configured in the reference "LocalVariable" of the VariableAccessRef of the WriteLocalVariable, has to be defined in the role implicitInterRunnableVariableor explicitInterRunnableVariable.

How to Solve

Configure the VariableAccessRefofWriteLocalVariable or ReadLocalVariable tobe oftype implicitInterRunnableVariable or explicitInterRunnableVariable.

Error Description

In a SwcInternalBehavior contaning a RunnableEntity, the VariableDataPrototype configured in the reference "LocalVariable" of the VariableAccessRef of the ReadLocalVariable, has to be defined in the role implicitInterRunnableVariable or explicitInterRunnableVariable.

How to Solve

Configure the VariableAccessRefofWriteLocalVariable or ReadLocalVariable tobe oftype implicitInterRunnableVariable or explicitInterRunnableVariable.

4.137 ASW4X_IB_0011 [Constr_2001]

Error Description

Itis possible butnotmandatory todefine aninitial value for a specificimplicitInterRunnableVariable or explicitInterRunnableVariable.

How to Solve

Configure the initial value for the ImplicitInterRunnableVaiable or ExplicitInterRunnableVariable.

4.138 ASW4X_IB_0012 [Constr_1182]

Error Description

The only allowed values for the attribute swImplPolicy of meta-class InternalTriggeringPoint are either STANDARD (in which case the processing of the internal triggering does not use a queue) or QUEUED (in which case the processingofinternal triggeringpositively uses a queue.

How to Solve

Reconfigure the SwImplPolicy value toeither STANDARD or QUEUED.

4.139 ASW4X_IB_0013 [Constr_1141]

Error Description

The attribute scope of meta-class VariableAccess shall only be applied with respect to the aggregation of VariableAccess in the following roles: dataReadAccess, dataWriteAccess, dataSendPoint , dataReceivePointByValue, dataReceivePointByArgument

How to Solve

a. Delete the value for the attribute Scope.
b. Reconfigure the the value of Scope to be any of the following - dataReadAccess, dataWriteAccess, dataSendPoint, dataReceivePointByValue, dataReceivePointByArgument.

4.140 ASW4X_IB_0014 [Constr_2002, 2005]

a. Error Description

In a RunnableEntity,A VariableAccess inthe role dataReadAccess shall refer toanRPortPrototype.

b. How to Solve

Reconfigure the VariableInAtomicSw Type Instance torefer RPortPrototye.

c. **Error Description**

Ina RunnableEntity, A VariableAccess inthe role dataReadAccess shall refer toanRPortPrototype whichis typed by a SenderReceiver interface or anNvDataInterface only.

d. **How to Solve**

Reconfigure the VariableInAtomicSw Type Instance torefer RPortPrototye ofSenderReceiver or NvDataInterface.

e. **Error Description**

Ina RunnableEntity,A VariableAccess inthe role dataReadAccess shall refer toanPRPortPrototype.

f. **How to Solve**

Reconfigure the VariableInAtomicSwTypeInstance torefer PRPortPrototye.

g. **Error Description**

Ina RunnableEntity, A VariableAccess inthe role dataReadAccess shall refer toanPRPortPrototype whichis typed by a SenderReceiver interface or anNvDataInterface only.

h. **How to Solve**

Reconfigure the VariableInAtomicSwTypeInstance torefer PRPortPrototye ofSenderReceiver or NvDataInterface

4.141 ASW4X_IB_0015 [Constr_2003, 2004]

a. **Error Description**

Ina RunnableEntity,A VariableAccess inthe role DataWriteAccess shall refer toanRPortPrototype.

How to Solve

Reconfigure the VariableInAtomicSw Type Instance torefer RPortPrototye.

b. **Error Description**

Ina RunnableEntity, A VariableAccess inthe role DataWriteAccess shall refer toanPPortPrototype whichis typed by a SenderReceiver interface or anNvDataInterface only.

How to Solve

Reconfigure the VariableInAtomicSw Type Instance torefer RPortPrototye ofSenderReceiver or NvDataInterface

c. Error Description

Ina RunnableEntity,A VariableAccess inthe role DataWriteAccess shall refer toanPRPortPrototype.

How to Solve

Reconfigure the VariableInAtomicSw Type Instance torefer PRPortPrototye.

d. Error Description

Ina RunnableEntity, A VariableAccess inthe role DataWriteAccessshall refer toanPRPortPrototype whichis typed by a SenderReceiver interface or anNvDataInterface only.

How to Solve

Reconfigure the VariableInAtomic Sw Type Instance to refer PRPortPrototye of SenderReceiver or NvDataInterface.

4.142 ASW4X_IB_0016 [Constr_2020]

Error Description

The value of SwImplyPolicy of a VariableDataPrototype referenced by the Variable Access in DataReadAccess shouldnotbe QUEUED.

How to Solve

Delete the SwImplPolicy.

4.143 ASW4X_IB_0017 [Constr_2021]

Error Description

A WaitPoint referencing a DataReceivedEvent is permitted if and only if the swImplPolicy of the VariableDataPrototype referencedby this DataReceivedEventis settoqueued.

How to Solve

a. Reconfigure the DataReceivedEvents torefer VariableDataPrototype whichhas SwImplPolicy QUEUED
b. Delete the SwImplPolicy inVariableDataPrototype
c. Reconfigure the value ofSwImplPolicy inVariableDataPrototype toQUEUED

4.144 ASW4X_IB_0018 [Constr_2032]

Error Description

In a PPortPrototype, if a SenderComSpec does specify a transmissionAcknowledge there shall be also a DataSendCompletedEvent specified, whose VariableAccess references the same VariableDataPrototype as the SenderComSpec.

How to Solve

Configure a DataSendCompletedEventwithTargetDataprototype value same as DataElementofSenderComSpec.

4.145 ASW4X_IB_0019 [Constr_2033]

Error Description

The timeoutvalue ofa WaitPointassociatedwitha DataSendCompletedEvent-shall have the same value as the correspondingtimeoutvalue ofTransmissionAcknowledgementRequest.

How to Solve

a. Delete the timeoutValues
b. Reconfigure the timeout value of tansmissionAcknowledgementRequest to match the timeout value of asso ciatedWaitpoint.

4.146 ASW4X_IB_0020 [Constr_2022]

Error Description

A ClientServerOperationofa particular RPortPrototype a shall be mutually exclusive andreferencedby either SynchronousServerCallPoints or AsynchronousServerCallPoints, notinboth.

How to Solve

Reconfigure suchthatthe ClientServerOperationwill be referedeither inthe SynchronousServerCallPointor AsynchronousServerCallPoint.

4.147 ASW4X_IB_0021 [Constr_2023]

Error Description

The timeout values of all ServerCallPoints referencing the same instance of ClientServerOperation in a RPortPrototype shall be identical.

How to Solve

Reconfigure the timeoutvalues ofthe ServerCallPoints tobe the same.

4.148 ASW4X_IB_0022 [Constr_2006]

Error Description

The AsynchronousServer-CallPointhas tobe referencedby exactly one AsynchronousServerCall-ResultPoint. This means that only the RunnableEntity with this AsynchronousServerCallResultPointcan fetchthe resultof the asynchronous server invocationofthis particular AsynchronousServerCallPoint.

How to Solve

Reconfigure the AsynchronousServerCallResulttorefer differentAsynchronousServerCallPoints.

4.149 ASW4X_IB_0023 [Constr_2030]

Error Description

The Wait-Pointwhich references a AsynchronousServerCallReturnsEvent andthe AsynchronousServerCallResultPoint which is referenced by this AsynchronousServerCallReturnsEvent shall be aggregated by the same RunnableEntity.

How to Solve

Reconfigure AsynchronousServerCallReturns,AsynchronousServerCallResultandWaitPointtobe inthe same Runnable Entity.

4.150 ASW4X_IB_0024 [Constr_2024, 2028]

a) Error Description

The definition of a PortAPIOption withenableTakeAddress set to true is only permitted for software-components where the attribute SwcInternalBehavior.supportsMultipleInstantiationis settofalse.

How to Solve

Reconfigure the attribute EnableTakeAddress tosetittofalse.

b) Error Description

The staticMemory is only supported if the attribute supportsMultipleInstantiation of the owning SwcInternalBehavior is settofalse.

How to Solve

Reconfigure the supportsMultipleInstantiationtosetittofalse

4.151 ASW4X_IB_0025 [Constr_1150]

Error Description

The valueType associatedwithPortDefinedArgumentValue shall be ofcategory VALUEor TYPE_REFERENCE. The latter case is only supportedifthe value ofcategory ofthe targetdata type is settoVALUE.

How to Solve

Reconfigure the ValueType ofthe PortDefinedArgumentValue toVALUEor TYPE_REFERENCE.

4.152 ASW4X_IB_0026 [Constr_2007]

Error Description

In the SwcInternalBehavior, ifthe attribute ofall the PerInstanceMemory are identical, thenthe attribute "TypeDefinition"shouldalsobe identical.

How to Solve

Reconfigure the TypeDefinitionofall PerInstanceMemory tobe identical.

4.153 ASW4X_IB_0028 [Constr_2027]

Error Description

"A PortPrototype that is referenced by a SwcServiceDependency via assigned Port shall be typed by a PortInterface that has isService set to true. This rule does not apply to PortPrototypes used in the context of NV data management, i.e. for connections betweenanApplicationSwComponentType andNvBlockSwComponentType.

How to Solve

Reconfigure the AssignedPorttopointtoa PortInterface thathas attribute "isService"settotrue.

4.154 ASW4X_IB_0029

a) Error Description

The assignedPortdefinedwithrespecttoa DiagnosticEventNeeds contaningDiagEventDebounceMonitorInternal may only refer toanRPortPrototype.

How to Solve

Reconfigure the As-

signedPorttorefer

anRPortPrototype

b) Error Description

InSwcServiceDependency, the existence ofthe combinationofAssignPorts andDiagnosticEventDebounceAlgorithmconfiguredinDiagnosticEventNeeds is invalidexceptfor DiagonosticEventDebouncemonitorInternal.

How to Solve

Reconfigure the DiagnosticEventNeeds tocontainDiagEventDebounceMonitorInternal

4.155 ASW4X_IB_0030 [Constr_2053]

a) Error Description

Ina SwcInternalBehavior containingSwcServiceDependency configuredwithObdRatioServiceNeeds thathas the attribute “ConnectionType”settoAPIUSEthen, the value ofthe attribute “Role”ofRoleBasedPortAssignment shouldbe settoIUMPRNumerator.

How to Solve

a. Reconfigure the ConnectionType toa value other thanAPI-USE.
b. Reconfigure the value ofthe attribute "Role"ofReleBasedPortAssignment.

b) Error Description

Ina SwcInternalBehavior containingSwcServiceDependency configuredwithObdRatioServiceNeeds thathas the attribute “ConnectionType”settoAPI-USEthen, the value ofthe attribute “Role”ofRoleBasedPortAssignmentcannotbe settoIUMPRNumerator.

How to Solve

a. Reconfigure the ConnectionType toa value other thanOBSERVER.
b. Delete the value ofattribute Role ofRoleBasedPortAssignment.

4.156 ASW4X_IB_0031 [Constr_1101]

Error Description

A RPortPrototype configuredwithModeSwitchInterface shall notbe referredinmore thanone SwConnector ofthe CompositionSwComponentType.

How to Solve

Reconfigure the value ofthe PortInterface tohave ModeSwitchInterface thatis notreferredinmore thanone SwConnector ofCompositionSwComponentType.

4.157 ASW4X_IB_0032 [Constr_1179, 1180]

a) Error Description

Inthe ModeDeclarationGroupcontaningthe ModeDeclarations, the attribute, "value"shouldbe configuredfor all the ModeDeclarations or notconfiguredatall.

How to Solve

Reconfigure the attribute value ofall the ModeDeclarations.

b) Error Description

IfModeDeclarations define the value attribute, the ModeDeclarationGroupshall alsodefine the attribute ModeDeclarationGroup.onTransitionValue.

How to Solve

Configure a value for the attribute OnTransitionValue inModeDeclarationGroup.

4.158 ASW4X_IB_0033

Error Description

The values oftwovalue attributes withinone ModeDeclarationGroupare notallowedtohave the same numerical value. Itis notallowedthatthe numerical value ofthe ModeDeclarationGroup.onTransitionValue attribute andthe numerical value ofone ofthe correspondingvalue attributes are identical.

How to Solve

Reconfigure the value ofthe attribute "OnTransitionValue"sothatithas differentvalue fromthe attribute "Value" ofModeDeclarations.

4.159 ASW4X_IB_0034 [Constr_4003]

a) Error Description

Inthe SwcModeSwitchEvent, ifthe attribute "Activation"is settoON-Transitionthenitshouldrefer to2different modes whichbelongs tosame ModeDeclarationGroup.

How to Solve

a. Change the value ofattribute activationtoother thanON-TRANSITION.
b. Reconfigure the SwcModeSwitchEventtorefer twodifferentModeDeclarations.

b) Error Description

Inthe SwcModeSwitchEventifthe attribute "Activation"is settoother thanON-Transition, then SwcModeSwitchEventshall refer toexactly one ModeDeclaration.

How to Solve

a. Change the value ofattribute activationtoother thanON-TRANSITION.
b. Reconfigure the SwcModeSwitchEventtorefer single ModeDeclaration.

4.160 ASW4X_IB_0035 [Constr_4011]

a) Error Description

In a PPortprototype, if there is a ModeSwitchSenderComSpec configured with ModeSwitchedAckRequest, then there should be a ModeSwitchedAckEvent withevent source ModeSwitchPoint with reference TargetModeGroup, same as ModeDeclarationGroupPrototype configuredinModeSwitchSenderComSpec.

How to Solve

Reconfigure the modeGroupsothatthe value is same as the TargetMode-Groupofthe ModeSwitchPointreferedin ModeSwitchedAckEvent.

b) Error Description

InPPortPrototype containingModeSwitchSenderComSpec, the value ofthe attribute “timeout”ofModeSwitchAckRequestshouldbe equal tovalue ofthe attribute “timeout”inWaitPointofRunnableEntity.

How to Solve

Reconfigure the timeoutvalue ofthe ModeSwitchAckRequestofModeSwitch-SenderComSpectobe equal tovalue of“timeout”inWaitPointofRunnableEntity.

4.161 ASW4X_IB_0036 [Constr_1098]

Error Description

A SwcModeSwitchEventshall notsimultaneously reference tothe same ModeDeclarationinboththe roles mode anddisabledMode.

How to Solve

Reconfigure the targetModedeclarationinMode anddisabledMode tohave a differentvalue.

4.162 ASW422_IB_0042 [Constr_1386]

Error Description

A PortAPIOptionconfiguredwithPortDefinedArgumentValue shall refer only toa ProvidedPortPrototype ofClientServerInterface.

How to Solve

Configure the PortInterface toClientServerInterface

4.163 ASW422_IB_0043 [Constr_1234]

Error Description

IfRunnable Entity ofNvMSwComponentType is referedinthe operationInvokedEventandifRunnable Entity.Symbol is configuredwitharbitrary whichis notassociatedwithNvMthenitis anerror

How to Solve

Configure the RunnableEntity.Symbol toAPI name associatedwiththe NvM

4.164 ASW421_IB_0045 [Constr_1363]

Error Description

IfDiagnosticValueNeeds is aggregatedby a SwcServiceDependency inthe role service-Needs thenthe attributes DiagnosticValueNeeds.diagnosticValueAccess DiagnosticValueNeeds.dataLengthshall notexist.

How to Solve

a. Delete the value for DataLength
b. Delete the value for DiagnosticValueAccess
c. Delete values for boththe attributes

4.165 ASW421_IB_0047 [Constr_1323]

Error Description

If in the contextofanApplicationSwComponentType the attribute SwcServiceDependency.serviceNeeds is implementedby anNvBlockNeeds thenthe followingattributes NvBlockNeeds.storeCyclic, NvBlockNeeds.cyclicWritingPeriod,NvBlockNeeds.storeEmergency, NvBlockNeeds.storeImmediate shall only existif inthe contextofthe same SwcServiceDependency a SwcServiceDependency.assignedPortexists thathas the attribute role settothe value NvDataPort.

How to Solve

Setthe value ofthe assignedPorttotoa PortPrototype oftype NvDataInterface

4.166 ASW421_IB_0048_MemorySection [Constr_1311]

Error Description

Ifthe Listofvalues storedinthe attributes MemorySection.optiondoes notinclude any ofthe values {safetyQM,safetyAsilA, safetyAsilB,safetyAsilC,safetyAsilD}, thenitis anerror

How to Solve

a. Configure optionvalue as safetyQM
b. Configure optionvalue as safetyAsilA
c. Configure optionvalue as safetyAsilB
d. Configure optionvalue as safetyAsilC
e. Configure optionvalue as safetyAsilD

f. Configure optionwithsafetyQM, safetyAsilA, safetyAsilB, safetyAsilC, safetyAsilD

4.167 ASW421_IB_0048_SwAddrMethod [Constr_1311]

Error Description

Ifthe Listofvalues storedinthe attributes SwAddrMethod.optiondoes notinclude any ofthe values {safetyQM,safetyAsilA, safetyAsilB,safetyAsilC,safetyAsilD}, thenitis anerror

How to Solve

a. Configure optionvalue as safetyQM
b. Configure optionvalue as safetyAsilA
c. Configure optionvalue as safetyAsilB
d. Configure optionvalue as safetyAsilC
e. Configure optionvalue as safetyAsilD
f. Configure optionwithsafetyQM, safetyAsilA, safetyAsilB, safetyAsilC, safetyAsilD

4.168 ASW421_IB_0049 [Constr_1364]

Error Description

IfSwcServiceDependency contains the DiagnosticIoControlNeeds inthe role ServiceNeeds andthe attributes diagnosticValueAccess anddataLengthare configured, thenitis anerror.

How to Solve

a. Delete the value for freezeCurrentStateSupported
b. Delete the value for shortTermAdjustmentSupported
c. Delete values for boththe attributes

4.169 ASW421_IB_0051 [Constr_1300]

Error Description

InVariableAndParameterInterfaceMappingor ClientServerInterface-Mappinghavingthe DataPrototypeMapping the firstDataPrototype andsecondDataPrototype referingtoAutosarDataType oftype VALUEor TYPE_ REFERENCEwhichresolves toVALUE, shouldbe oftype primitive data type.

How to Solve

Configure Primitive type for both.

Error Description

InVariableAndParameterInterfaceMappingor ClientServerInterface-Mappingcontains SubElementMappingwithin DataPrototypeMapping, shouldnotrefer toImplementationDataTypeElementonServer/requestor side thatis mappedtoanApplicationCompositeDataPrototype

How to Solve

Refer elementother thanImplementationDataTypeElementandApplicationCompositeDataPrototype.

Error Description

InVariableAndParameterInterfaceMappingor ClientServerInterface-Mappingcontains SubElementMappingwithin DataPrototypeMapping, shouldnothave more thanone FirstElementor SecondElement.

How to Solve

Delete extra firstElementandsecondElement.

4.170 ASW422_IB_0042 [Constr_1386]

Error Description

A PortAPIOptionconfiguredwithPortDefinedArgumentValue shall refer only toa ProvidedPortPrototype ofClientServerInterface.

How to Solve

Configure the PortInterface toClientServerInterface

4.171 ASW422_IB_0043 [Constr_1234]

Error Description

IfRunnable Entity ofNvMSwComponentType is referedinthe operationInvokedEventandifRunnable Entity.Symbol is configuredwitharbitrary whichis notassociatedwithNvMthenitis anerror

How to Solve

Configure the RunnableEntity.Symbol toAPI name associatedwiththe NvM

4.172 ASW422_IB_0044 [Constr_1402]

Error Description

If the attribute optionofSwAddrMethodor MemorySectionis settocoreLocal thenthe attribute sectionInitializationPolicy ofthe same SwAddrMethodor MemorySectionshall be either settoINIT or CLEARED.

How to Solve

a. Setthe attribute sectionInitializationPolicy value toINIT.
b. Setthe attribute sectionInitializationPolicy value toCLEARED.

4.173 ASW422_IB_0052[Constr-1301]

Error Description

IfSwcServiceDependency is configuredwithRoleBasedDataAssignmentwiththe role defaulValue then RoleBasedDataTypeAssignmentcannotbe configured

How to Solve

Delete the configurationfor RoleBasedDataTypeAssignment.

4.174 ASW4X_IB_1773 [constr_1773]

Error Description

All the dataSendPoints that refer to the same accessedVariable (ie. referring to the same datatypePrototype) should have same value for the attribute returnValueProvision

How to Solve

Reconfigure dataSendPoints such that, all the dataSendPoints referring to the same accessedVariable, should have same value for the attribute returnValueProvision

4.175 ASW4X_IB_1774 [constr_1774]

Error Description

All the dataReceivePointByArgument that refer to the same accessedVariable (ie. referring to the same datatypePrototype) should have same value for the attribute returnValueProvision

How to Solve

Reconfigure dataReceivePointByArgument such that, all the dataReceivePointByArgument referring to the same accessedVariable, should have same value for the attribute returnValueProvision

4.176 ASW4X_IB_1775 [constr_1775]

Error Description

All the serverCallPoints that refer to the same operation (ie. referring to the same datatypePrototype) should have same value for the attribute returnValueProvision

How to Solve

Reconfigure serverCallPoints such that, all the serverCallPoints referring to the same operation, should have same value for the attribute returnValueProvision

4.177 ASW4X_IB_1776 [constr_1776]

Error Description

All the serverCallPoints that refer to the same operation (ie. referring to the same datatypePrototype) should have same value for the attribute returnValueProvision

How to Solve

Reconfigure serverCallPoints such that, all the serverCallPoints referring to the same operation, should have same value for the attribute returnValueProvision

4.178 EndToEndDesc_DataLength_4X

Error Description

a. InanEndToEndDescription, ifthe "Category"attribute is configured-with"PROFILE_01"thenthe "DataLength" attribute shouldbe within[0-240] range andalsodivisible by 8.
b. InanEndToEndDescription, if the "Category"attribute is configured witheither "PROFILE_02"or "PROFILE_03" thenthe "DataLength"attribute shouldbe within[0-65535] range andalsodivisible by 8.

How to Solve

Configure the DataLengthattribute basedonthe following:

a. IfCategory = "PROFILE_01", thenDataLengthvalue shouldbe within[0-240] range andalsodivisible by 8.
b. IfCategory = "PROFILE_02"or "PROFILE_03, thenDataLengthvalue shouldbe within[0-65535] range and alsodivisible by 8.

4.179 EndToEndDesc_MaxDeltaCounterInit_4X

Error Description

1. In an EndToEndDescription, if the "Category" attribute is configured with "PRO-FILE_02" then the "MaxDeltaCounterInit"attribute shouldbe within[0-15] range.
2. In an EndToEndDescription, if the "Category" attribute is configured witheither "PRO-FILE_01" o- r "PROFILE_ 03"thenthe "MaxDeltaCounterInit"attribute shouldbe within[0-14] range.

How to Solve

Configure the "MaxDeltaCounterInit"attribute basedonthe following

1. IfCategory = "PROFILE_02", then"MaxDeltaCounterInit"value shouldbe within[0-15] range.
2. IfCategory = "PROFILE_01"or "PROFILE_03, then"MaxDeltaCounterInit"value shouldbe within[0-14] range.

4.180 ASW4X_OTHR_0001 [Constr_1110]

Error Description

In an EndToEndDescription, the "Category" attribute is should be configured with one among the {"NONE", "PROFILE_01", "PROFILE_02", "PROFILE_03"}.

How to Solve

1. Dele the EndToEndDescription.
2. Re configure the Category attribute ofEndToEndDescriptiontoone amongthe {"NONE", "PROF- ILE_01", "PROFILE_02", "PROFILE_03"}.

4.181 ASW4X_OTHR_0002 [Constr_1001]

Error Description

InanEndToEndDescription, there shouldnotbe any DataIdconfiguredmore thanonce.

How to Solve

Delete the DataId.

4.182 ASW4X_OTHR_0003_10 [Constr_1113]

Error Description

In an EndToEndDescription, if the "Category" attribute is configured with either "PROFILE_01" or "PROFILE_02" thendataId, dataLengthandmaxDeltaCounterInitmustbe configured.

How to Solve

Configure the mandatory attributes {"DataId", "DataLength", "MaxDeltaCounterInit"} Based on which ever element is missed configuring

4.183 ASW4X_OTHR_0004_14 [Constr_1111]

Error Description

In an EndToEndDescription, if the "Category" attribute is configured with either "PROFILE_01" or "PROFILE_03" thenthe constraints {"DataIds size shouldbe 1", "DataID value shouldbe within[0-65535] range"}shouldbe satisfied.

How to Solve

1.If there are more than one DataIDs set then retain the value for the first one alone and remove the values for rest of dataIDs
2.Allow the user to se the range of dataID between 0-65535

4.184 ASW4X_OTHR_0005 [Constr_1112]

Error Description

InanEndToEndDescription, ifthe "Category"attribute is configuredwitheither "PROFILE_01"thenthe DataIdMode attribute value shouldbe within[0-2]range.

How to Solve

1. Re configure the Category attribute ofEndToEndDescriptiontoother than"PROFILE_01"
2. Configure the DataModeIdattribute suchthatitis within[0-2]range.

4.185 ASW4X_OTHR_0006_7 [Constr_1114, 1115]

Error Description

InanEndToEndDescription, ifthe "Category" attribute is configuredwith"PROFILE_01"thenthe value of "CrcOffset"and"CounterOffset"attribute shouldbe within[0-65535] range andalsodivisible by 4.

How to Solve

1. Re configure the Category attribute ofEndToEndDescriptiontoother than"PROFILE_01"
2. Configure the CrcOffset, CounterOffsetattributes suchthatitis within[0-65535] range andalsodivisible by 4.

4.186 ASW4X_OTHR_0011 [Constr_1120]

Error Description

In an EndToEndDescription, if the "Category" attribute is configured with "PROFILE_02" the the constraints {"DataIds size mustbe 16", "DataID value shouldbe within[0-255] range"}shouldbe satisfied.

How to Solve

Configure the DataId such that it satisfies both the constraints
1.DataIds size must be 16 (DataIDs should added or removed based on less than or greater than 16 elements)
2.Configure the DataIdsuchthatitsatisfies boththe constraints {"DataIds size mustbe 16", "Data ID value shouldbe within[0-255] range"}.

4.187 ASW4X_OTHR_0017

Error Description

Ina ServiceSwComponentType, the PortPrototypes shouldbe configuredwithPortInterfaces whose "isService" attribute is settoTRUE.

How to Solve

1. Delete the PortPrototype.
2. Reconfigure the PortInterface.
3. Setthe "isService"attribute ofPortInterface toTRUE.

4.188 ASW4X_OTHR_0018

Error Description

A SensorActuatorSwComponentType whichis mappedthroughComponentInSystemInstanceRefofSwcToEcuMapping has an EcuIntance configured, the same EcuInstance will also be configured in ECUMapping where the SensorActuatorSwComponentType is run. This ECUMapping has reference to HwElement which should be same as the configured"SensorActuator"attribute ofthe SensorActuatorSwComponentType.

How to Solve

1. Delete the ECUMapping.
2. Delete the SwcToEcuMapping.
3. Reconfigure the HwElementofECUMappingandSensorActuatorSwComponentType tothe same.
4. MapdifferentSwComponentType tothe SwcToEcuMappingthroughComponentInSystemInstanceRef.

4.189 ASW4X_OTHR_0019 [Constr_1095]

Error Description

InanNvBlockDescriptor, whichhas NvBlockNeeds configured, ifthe value ofNDataSets is greater thanzerothen the Reliability attribute shouldnotbe setto"ERROR-CORRECTION".

How to Solve

1. Delete the Reliability attribute.
2. Delete the NvBlockNeeds element.
3. Reconfigure the value ofNDataSets tozero.
4. Reconfigure the value ofReliability attribute toother than"ERROR-CORRECTION".

4.190 ASW4X_OTHR_0021 [Constr_1000]

Error Description

In an EndToEndProtection, which has EndToEndProtectionVariablePrototype configured will have Sender/Recievers configured. The PortPrototype referred in the Sender/Recievers should be configured only with "SenderRecieverInterface".

How to Solve

Diasplay all the dataelements which are of type SenderReceiver Interface so that user can select out of the available list.

4.191 ASW4X_OTHR_0029 [Constr_1222, 1223, 1224]

Error Description

1. In a DataPrototype(VariableDataPrototype or ParameterDataPrototype), if the 'Category' attribute of the referredDataType is STRING, thenthe DataPrototype should be configured with ApplicationValueSpecification as its InitValue.
2. For a DataPrototype (VariableDataPrototype or ParameterDataPrototype), which is referring to ApplicationRecordType as its Type, shouldbe configuredwithRecordValueSpecificationas its InitValue.
3. For a DataPrototype (VariableDataPrototype or ParameterDataPrototype), which is referring to ApplicationArrayType as its Type, should be configured with either ArrayValueSpecification or ApplicationRuleBasedValueSpecificationas its InitValue.

How to Solve

1. FirstScenario
   a. Delete DataPrototype.
   b. Re configure the InitValue withApplciationValueSpecification.
2. SecondScenario
   a. Delete DataPrototype.
   b. Re configure the InitValue withRecordValueSpecification.
3. ThirdScenario
   a. Delete DataPrototype.
   b. Re configure the InitValue witheither ArrayValueSpecificationor ApplicationRuleBasedValue Specification.

4.192 ASW4X_OTHR_0030 [Constr_1273]

Error Description

1. The DataPrototype (VariableDataPrototype, ParameterDataPrototype) referringtoa Applic-
ationArrayDataType, andthe initvalue configuredfor the DataPrototype is ArrayValueSpecification. Thenthe ele-
ments (Fields) inthe ArrayValueSpecificationshouldbe identical tothe ApplicationArrayElementinthe ApplicationArrayDataType.

1.

How toSolve

1. Notpossible since changes neededatmany elements.

4.193 ASW4X_OTHR_0030 [Constr_1274]

Error Description

1. The DataPrototype (VariableDataPrototype, ParameterDataPrototype) referringtoa ImplementationDataType withCategory “ARRAY”, andthe initvalue configuredfor the DataPrototype is ArrayValueSpecification. Thenthe elements (Fields) inthe ArrayValueSpecificationshouldbe identical*tothe subElements inthe ImplementationDataType.

How to Solve

1. Notpossible since changes neededatmany elements.

4.194 ASW4X_OTHR_0031 [Constr_1400]

Error Description

A specificData-Transformationshall only be referencedby either

1. A DataPrototypeMappinginthe role firstToSecondDataTransformationor
2. AnISignal inthe role dataTransformationor
3. AnISignalGroupinthe role comBasedSignalGroupTransformation

How to Solve

1.within the DataPrototypeMapping, allow the user to remove the reference for the FirstDataPrototype or the secondDataprototype, which is referred by Isignal within DataTransformationRefConditional

2.Allow the user to correct the Isignal by selecting a reference for DataTransformation

4.195 ASW422_OTHR_0005[Constr-1112]

Error Description

InanEndToEndDescription, ifthe "Category"attribute is configuredwitheither "PROFILE_01"thenthe DataIdMode attribute value shouldbe within[0-3]range.

How to Solve

1. Allow the user to enter the DataModeId attribute to have any value between the range 0 to 3.

4.196 ASW422_OTHR_0032[Constr-1383]

Error Description

The ImplementationDataType of category TYPE_ REFERENCE can have the ImplementationDataType.swDataDefProps.compuMethod and ImplementationDataType.swDataDefProps.dataConstr values configured only if ImplementationDataType attribute refers to an ImplementationDataType of category VALUEandall the TYPE_REFERENCEs are resolved.

How to Solve

Allow the user to configure any initValue other than ApplicationValueSpecification.

4.197 ASW422_OTHR_0032_1 [Constr_1401]

Error Description

Restrictions onthe relationbetweenDataPrototypeMappingandDataTransformation

How to Solve

1. Delete or change the VDPreference fromData Mapping
2. Delete or change the VDPreference fromIsignal

4.198 ASW422_OTHR_0035[Constr-1388]

Error Description

InVariationPointProxy ifthe category is VALUEthenbothPostbuildValueAccess andattribute valueAccess cannot be configuredtogether.

How to Solve

1. Delete any one ofconfiguredvalues for PostBuildValueAccess or ValuesAccess
2. Configure the ImplementationDatatype
3. Configure PostBuildValueAccess
4. Enter a positive integer value inValueAccess

4.199 ASW422_OTHR_0037_SwAddressMethod[Constr-1381]

Error Description

Ifthe attribute SwAddrMethod.optionshouldconfiguredwithbothcoreLocal and-coreGlobal

How to Solve

1.Configure optionvalue as CoreLocal

2.Configure optionvalue as CoreGlobal

3.Configure optionvalue withbothCoreGlobal andCoreLocal

4.200 ASW422_OTHR_0037_MemorySection[Constr-1381]

Error Description

Ifthe attribute SwAddrMethod.optionshouldconfiguredwithbothcoreLocal and-coreGlobal

How to Solve

1.Configure optionvalue as CoreLocal

2.Configure optionvalue as CoreGlobal

3.Configure optionvalue withbothCoreGlobal andCoreLocal

4.201 ASW422_OTHR_0037[Constr-1395]

Error Description

NvBlockDataMappingshall be complete, IfanNvBlock-DataMappingrefers tosub-elements or leafelements of the NvDataInterface.nv-Data inthe contextofa particular PortPrototype thenall remainingsub-elements or leaf elements shall effectively be mapped

How to Solve

1. InAutosar-VariableRef.autosarVariable, configure the autosarVariable.targetDataPrototype tonvDataInterface.nvData.
2. Ifautosar varibale notexistthencreate one andredirecttonvDatainterface.

4.202 3.37 ASW44_DataTypesCompatibility[Constr-1050]

Error Description

For eachImplementationDatatype havingsubelement, the attribute isOptional ofsubelementshouldhave (any one shouldbe satisfiedie ORcondition)

i. Same values atbothPportandRportside - or ii. Value false ifitpresentonly onone side - oriii. isOptional shouldnotbe presentonbothsides.

2)IfDataPrototypemappingis present, thenone ImplementationDatatype shouldbe referedinFirstDataPrototype andanother ImplementationDatatype shouldbe referedinSecondDataPrototype

3)DataPrototypemappingexists suchthatSecondDataProtoype refers toanImplementationDatatype, and FirstDataPrototype refers toanImplementationDataType havinga subelement. For Implementationdatatype withsubelement(firstdataPrototype), there shoulda ImplementationDatatypeSubelementRefinthe role FirstElementthatrefers tothe subelement.

How to Solve

1.Remove the value ofthe attribute isOptional onbothPandRPorts

2.Configure identical value for the attribute isOptional

3)Quick fix notpossible since itneeds changes atmany elements 4)Quick fix notpossible since itneeds changes atmany elements

4.203 DataTypesCompatibility[Constr-2013]

Error Description

Definingtwoattributes bitfieldTextTableMaskNvBlockDescriptor andattribute bitfieldTextTableMaskPortPrototype is additional change fromthe earlier implementation.

4.204 DataTypesCompatibility[Constr-1051]

Error Description

BothSwDataDefProps aggregate a ValueSpecificationinthe role invalidValue or bothdoes notdefine ValueSpecificationinthe role inValidVaue.

4.205 DataTypesCompatibility[Constr-1048]

Error Description

For eachApplicationRecordDatatype havingsubelement, the attribute isOptional ofsub-elementshouldbe (ORconditionie any one shouldbe satisfied)

i. settoidentical values inbothsides PportandRportor ii. settofalse ifitpresentonly onone side or iii. IsOptional shouldnotbe presentonbothsides.

How to Solve

remove the isOptional value where itis set

4.206 DataTypesCompatibility[Constr-1078]

Error Description

BothClientServerOperations has identical value for the attribute diagArgIntegrity or bothdoes notconfigure this attribute.

How to Solve

Seta defaultvalue true toother ClientServerOperation.

4.207 DataTypesCompatibility[Constr-1662]

Error Description

This validation is applicable only when both ApplicationRecordDataType and Implementationdatatype are Optional Structure element.

An ApplicationRecordDataType having RecordElements with isOptional value set to true can only be mapped or connected to an ImplementationDatatype of category structure that represent optional element structure (ie isStructWithElementsetto-true)ifonly correspondingpair ofelements has same values for isOptional.

For ImplementationDatatype –the attribute isStructWithOptionalElementshouldbe settotrue. Ifthe attribute is notsetor settothe value False thenthe respective Implementationdatatypeelementis mandatory tobe configuredandthis validationis notapplicable.

How toSolve

Setthe isOptional attribute value tobe true ofImplementationDataType.

4.208 ASW4X_COMP_1860 [Constr_1860]

Error Description

In a CompositionSwComponentType, the Provider and Requestor Inner Ports and Target ports of the DelegationSwConnector should be configured.

How to Solve

Provide option for user to select PPOrtInCompositionInstanceRef or RPOrt-InCompositionInstanceRef and targetpports or targetrports within Delegationconnector.

4.209 ASW4X_COMP_1861 [Constr_1861]

Error Description

In a CompositionSwComponentType, the Provider and Requestor Outer Ports of the DelegationSwConnector should be configured.

How to Solve

Provide option for user to select Outer Ports within DelegationConnector.

4.210 ASW4X_COMP_1862 [Constr_1862]

Error Description

In a CompositionSwComponentType, the Required Outer Ports of the PassThroughSwConnector should be configured.

How to Solve

Provide option for user to select Outer Ports within PassThroughSwConnector.

5. System Validations

This sectiondeals withthe error descriptionofsystemvalidations. SystemValidations are specific-tosystemtemplate inautosar.

5.1 ASW4X_SYS_0001 [Constr_3008]

a) Error Description

In CanCommunicationConnector, the ComController reference should not be empty or null and it should refer a CanCommunicationController available inthe same ECUInstance,where Connector is configured.

How to Solve

Configure CommController reference for the CanCommunicationConnector which is present within same ECUInstance.

b) Error Description

In CanCommunicationConnector, the ComController reference should be configured only to CanCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Reconfigure CommController reference to refer a CanCommunicationController which is present within same ECUInstance.

c) Error Description

InCanCommunicationConnector, the ComController reference shouldbe configuredtoa CanCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Reconfigure CommController reference to refer a CanCommunicationController which is present within same ECUInstance.

5.2 ASW4X_SYS_0002 [Constr_3008]

a) Error Description

InFlexrayCommunicationConnector, the ComController reference shouldnotbe empty or null anditshouldrefer a FlexrayCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Configure CommController reference for the FlexRayCommunicationConnector which is present within same ECUInstance

b) Error Description

In FlexRayCommunicationConnector, the ComController reference should be configured only to FlexrayCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Reconfigure CommController reference torefer a FlexRayCommunicationController whichis presentwithinsame ECUInstance.

c)Error Description

InCanCommunicationConnector, the ComController reference shouldbe configuredtoa CanCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Reconfigure CommController reference torefer a FlexRayCommunicationController whichis presentwithinsame ECUInstance.

5.3 ASW4X_SYS_0003 [Constr_3008]

a) Error Description

InEthernetCommunicationConnector, the ComController reference shouldnotbe empty or null anditshouldrefer a EthernetCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Configure CommController reference for the EthernetCommunicationConnector present within same ECUInstance.

b) Error Description

In EthernetCommunicationConnector, the ComController reference should be configured only to EthernetCommunicationController available inthe same ECUInstance, where Connector is configured

How to Solve

Reconfigure CommController reference torefer a EthernetCommunicationController whichis presentwithinsame ECUInstance.

c) Error Description

In EthernetCommunicationConnector, the ComController reference should be configured to a EthernetCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Reconfigure CommController reference torefer a EthernetCommunicationController whichis presentwithinsame ECUInstance.

5.4 ASW4X_SYS_0004 [Constr_3008]

a) Error Description

InTTCanCommunicationConnector, the ComController reference shouldnotbe empty or null anditshouldrefer a TTCanCommunicationController available inthe same ECUInstance, where Connector is configured

How to Solve

Reconfigure CommController reference torefer a TTCanCommunicationConnector whichis presentwithinsame ECUInstance.

b) Error Description

In TTCanCommunicationConnector, the ComController reference should be configured only to TTCanCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Reconfigure CommController reference to refer a TTCanCommunicationController present within same ECUInstance.

c)

In TTCanCommunicationConnector, the ComController reference should be configured to TTCanCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Reconfigure CommController reference torefer a TTCanCommunicationController presentwithinsame ECUInstance.

5.5 ASW4X_SYS_0005 [Constr_3008]

a) Error Description

In LinCommunicationConnector, the ComController reference should not be empty or null and it should refer a LinCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Reconfigure CommController torefer a LinCommunicationConnector

presentwithinsame ECUInstance.

b) Error Description

In LinCommunicationConnector, the reference ComController should refer a LinCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Reconfigure CommController torefer a LinCommunicationConnector

presentwithinsame ECUInstance.

c) Error Description

InLinCommunicationConnector, the reference ComController shouldrefer a LinCommunicationController available inthe same ECUInstance, where Connector is configured.

How to Solve

Reconfigure CommController torefer a LinCommunicationConnector presentwithinsame ECUInstance.

5.6 ASW4X_SYS_0006 [Constr_3003]

Error Description

A CAN cluster shall containonly one PhysicalChannel.

How to Solve

InCanClusterConditional, delete additional CanPhysicalChannel andhave exactly one CanPhysicalChannel.

5.7 ASW4X_SYS_0007 [Constr_3018]

a) Error Description

A FlexrayCluster shall use one FlexrayPhysicalChannel with"ChannelName"attribute configuredas either "ChannelA"or "ChannelB".

How to Solve

Inthe FlexrayConditional configure the FlexrayPhysicalchannel withChannelName attribute setto"ChannelA"or "ChannelB".

b)Error Description

A FlexrayCluster shall have one FlexrayPhysicalChannel withattribute ChannelName configuredas either "ChannelA"or "ChannelB". Else twoFlexrayPhysicalChannels withone configuredtoChannelName as "ChannelA"and another configuredtoChannelName as "ChannelB".

How to Solve

Configure the FlexrayConditional tocontainonly twoFlexrayPhysicalchannels withone Channel havingattribute ChannelName settochannelA andthe other channel withchannelName attribute settochannelB.

c) Error Description

A FlexrayCluster shall have twoFlexrayPhysicalChannels withone channelName channelA andone channelName channelB.

How to Solve

1. Inthe FlexrayConditional,delete the additional FlexrayPhysicalchannels andhave exactly 2FlexrayPhysicalchannels.
2. Reconfigure the FlexrayConditional tocontainonly 2FlexrayPhysicalchannel withone channel havingattribute channelName settochannelA andthe other channel havingattribute channelName settochannelB.

5.8 ASW4X_SYS_0008 [Constr_3015]

Error Description

The LIN clusters shall containonly one PhysicalChannel.

How to Solve

InLinClusterConditional, delete additional LinPhysicalChannel andhave exactly one LinPhysicalChannel.

5.9 ASW4X_SYS_0009 [Constr_3016]

Error Description

InEthernetCluster, shall containexactly one EthernetPhysicalChannel.

How to Solve

InEthernetClusterConditional, Delete the additional EthernetPhysicalChannel tohave exactly one EthernetPhycialChannel.

5.10 ASW4X_SYS_0010 [Constr_3006]

a) Error Description

InECUMappingcontainingCommunicationMappingController, the Communication-Controller reference should refer toa connector available inthe same ECUInstance as referedinECUMapping.

How to Solve

Reconfigure the CommunicationController reference torefer toa connector thatis available inthe same ECUInstance.

b)Error Description

InECUMappingcontainingHwPortMapping, the CommunicationConnector reference shouldrefer toa connector available inthe same ECUInstance as referedinECUMap-ping.

How to Solve

Reconfigure the CommunicationConnector reference torefer toa connector thatis available inthe same ECUInstance.

c) Error Description

InECUMappingcontainingthe HwPortMapping, the CommunicationConnector refer-ence shouldrefer toa connector avaiable inthe same ECUInstance as referedi-nECUMapping.

How to Solve

Configure the CommunicationConnector reference torefer toa connector thatis available inthe same ECUInstance.

5.11 ASW4X_SYS_0011 [Constr_3002]

Error Description

In a SwcToImplMapping,the ComponentImplementation should refer to a SwcImple-mentation,whose InternalBehavior is available inside the AtomicSwComponentType whichis same as the SwComponentreferedinthe targetComponent.

How to Solve

Reconfigure the ComponentImplementationreference toa SwcImplementa-tion,whose Internal Behavior is available under the same AtomicSwComponentType referedintargetComponent.

5.12 ASW4X_SYS_0012 [Constr_3012]

a) Error Description

InFrame, the StartPositionvalue for all Pdus shouldnotbe Null or empty.

How to Solve

Configure the StartPositionvalue for

all PduToFrameMappings.

b) Error Description

InFrame, the StartPositionvalue for all Pdus whichare mappedtosame

Frame shouldnotbe same. **How to Solve**

Reconfigure the StartPositionvalue for all PduToFrame-

Mappings tohave unique value.

c) Error Description

InFrame, the StartPositionvalue for all Pdus whichare mappedtosame

Frame shouldnotoverlap. **How to Solve**

Reconfigure the StartPositionvalue for PduToFrameMappings sothatitdoes notover-lapwithother PDUs.

5.13 ASW4X_SYS_0013 [Constr_3013]

Error Description

1. In a Frame, the sum of the length of Pdus referred in the PduToFrameMappings should not be more than the lengthofthe Frame.
2. The Sum of Length of the Pdus referred in the PduToFrameMappings available under the Particular Frame shouldnotexceedthe FrameLengthspecifiedfor thatFrame.

How to Solve

1. Delete the extra Pdus.
2. Re-Configure the lengthofPdus.

5.14 ASW4X_SYS_0014 [Constr_3012]

a) Error Description

InFrame, the updateIndicationBitPositionvalue for all Pdus shouldbe unique.

How to Solve

1. Delete the UpdateIndicationBitPositionvalue.
2. Reconfigure the UpdateIndicationBitPositionfor eachPduToFrameMappings to-have unique value .

b) Error Description

InFrame, the updateIndicationBitPositionvalue for all Pdus shouldnotoverlap.

How to Solve

Reconfigure the UpdateIndicationBitPositionfor PduToFrameMappings sothatitdoes notoverlapwithother PDUs.

5.15 ASW4X_SYS_0015 [Constr_3009]

a) Error Description

Ina IsignalIPdu, the StartPositionvalue ofall the Pdus mappedtoa Frame should-

notbe Null or empty. **How to Solve**

Configure the StartPositionfor

IsignalToIPduMappings.

b) Error Description

Ina IsignalIPdu, the StartPositionvalue ofall the Pdus mappedtoa

Frame shouldbe unique. **How to Solve**

Configure unique StartPositionfor

IsignalToIPduMappings. **c) Error**

Description

Ina IsignalIPdu, the StartPositionvalue ofall the Pdus mappedtoa Frame

shouldnotoverlap. **How to Solve**

Reconfigure the StartPositionvalue for IsignalToIPduMappings sothatitdoes notover-lapwithother PDUs.

5.16 ASW4X_SYS_0016 [Constr_3010]

Error Description

InISignalIPdu, the sumoflengthall the Isignals andthe correspondingvalues ofattrib-ute "UpdateIndicationBitPosition"Shall notexceedthe lengthofISignalIPdu.

How to Solve

1.Delete the UpdateIndicationBitPositionvalue

2. Reconfigure the UpdateIndicationBitPositionsothatthe sumoflengthall the Isignals andthe correspondingval ues ofattribute ""UpdateIndicationBitPosition""does notexceedthe lengthofISignalIPdu.

5.17 ASW4X_SYS_0017 [Constr_3011]

a) Error Description

InISignalIPdu,the updateIndicationBitPositionvalue for a IsignalToIPduMapping-shouldbe unique.

How to Solve

1. Delete the UpdateIndicationBitPositionvalue.
2. Reconfigure the UpdateIndicationBitPositionvalue for eachIsignalToIPduMappings.

b) Error Description

InISignalIPdu, the updateIndicationBitPositionvalue for all IsignalToIPduMapping-shouldnotoverlap.

How to Solve

1. Reconfigure the UpdateIndicationBitPositionvalue for all IsignalToIPduMappings sothatitdoes notoverlapwith other PDUs.

5.18 ASW4X_SYS_0018 [Constr_3020]

a) Error Description

The value ofthe attribute CommunicationDirectionofcontainedISignal-IPduGroupmustbe identical tothe value of the attribute communicationDirection-ofthe enclosingISignalIPduGroup.

How to Solve

Configure the "CommunicationDirection" value for ISignalIPduGroup in the con-tainedISignalIPduGroup to be identical toenclosingISignalIPduGroup. **b) Error De-scription**

The value ofthe attribute CommunicationDirectionofcontainedISignal-IPduGroupmustbe identical tothe value of the attribute communicationDirection-ofthe enclosingISignalIPduGroup.

How to Solve

Configure the "CommunicationDirection"value for ISignalIPduGroupinthe con-tainedISignalIPduGrouptobe identical toenclosingISignalIPduGroup.

5.19 ASW4X_SYS_0019 [Constr_3026]

Error Description

Ifthe EmptySignalMappingrefer a SystemSignal, whichinturnis referredby multiple ISignals, thenall those ISignals lengthshouldbe 0.

How to Solve

Reconfigure the Isignals mappedtothe SystemSignal tosetthe value ofthe attribute lengthto0.

5.20 ASW4X_SYS_0020 [Constr_3000]

Error Description

The SystemSignal configuredtothe SenderRecArrayElementMappingofa SenderReceiverSignalGroupMapping shouldbe available inthe SignalGroupmappedtoit.

How to Solve

Reconfigure the value ofSystemSignal tobe available inthe SystemSignalGroupmappedtothe containingSenderReceiverToSignalGroupmapping.

5.21 ASW4X_SYS_0021 [Constr_3001]

Error Description

The SystemSignal configuredtothe ClientServerArrayElementMappingofa ClientServerToSignalGroupMappings shouldbe available inany ofthe SystemSignalGroupmappedtoeither RequestGroupor ResponseGroup.

How to Solve

Reconfigure the value of SystemSignal to be available in SystemSignalGroup mapped to attributes either RequestGroupor ResponseGroupofthe containingClientServerToSignalGroupmapping.

5.22 ASW4X_SYS_0022 [Constr_3005]

Error Description

Ina SystemMappingthatcontains the EcuResourceEstimation, the attribute "SwCompToEcuMapping"refers to SwcToEcuMapping. This SwcToEcuMappingshouldrefer the same EcuInstance thatis referredinthe attribute "EcuInstance"ofEcuResourceEstimation.

How to Solve

Reconfigure the EcuInstance ofthe EcuResourceEstimationtobe same as the EcuInstance ofthe refered SwcToEcuMapping.

5.23 ASW4X_SYS_0023 [Constr_3007]

Error Description

The selectorFieldCodes for the dynamicpartalternatives withinone MultiplexedIPdushall differ fromeachother.

How to Solve

Reconfigure the value ofselectorFieldCode ofall the DynamicPartAlternative tobe different.

5.24 ASW4X_SYS_0024 [Constr_3024]

Error Description

The values triggeredWithoutRepetitionandtrggeredOnChangeWithoutRepetitionshall notbe usedifthe ISignalToIPduMappingrefers toanISignalGroupor anISignal whichis partofanISignalGroup(groupsignal).

How to Solve

Reconfigure the attribute TransferProperty tohave a value other thantriggeredWithoutRepetitionand triggeredOnChange-WithoutRepetition.

5.25 ASW4X_SYS_0025 [Constr_3017]

Error Description

The sum of length of ISignalIPdus referred in DynamicPartAlternatives and StaticParts plus the SelectorFieldLengthofMultiplexedPdushoudbe less thanor equal tolengthofthatMultiplexedPdu.

How to Solve

Reconfigure the length ISignalIPdus referred in DynamicPartAlternatives and StaticParts plus the SelectorFieldLengthofMultiplexedPdushoudbe less thanor equal tolengthofthatMultiplexedPdu.

5.26 ASW4X_SYS_0026 [Constr_1367]

Error Description

The attribute periodicResponseUudtshall only refer toa DcmIPdu.

How to Solve

Reconfigure the attribute, PeriodicResponseUudt, sothatitrefers toa DcmIpdu.

5.27 ASW4X_SYS_0027 [Constr_3078]

Error Description

The value ofUdpNmCluster.nmNidPositionshall only be settoeither

bit0(byte 0)or bit8(byte 1). **How to Solve**

Reconfigure the nmNidPosition, sothatitis either 0or 1.

5.28 ASW4X_SYS_0028 [Constr_3079]

Error Description

The value ofUdpNmCluster.nmCbvPositionshall only be settoeither

bit0(byte 0)or bit8(byte 1). **How to Solve**

Reconfigure the nmCbvPosition, sothatitis either 0or 1.

5.29 ASW4X_SYS_0029 [Constr_3080]

Error Description

UdpNmCluster.nmCbvPositionandUdpNmCluster.nmNidPositionshall never

have the same value.

How to Solve

Reconfigure the twoattributes, sothatthey are different.

5.30 ASW4X_SYS_0030 [Constr_1368]

Error Description

The attributes PhysicalRequest, ResponseOnEvent, Response andFunctionalRequestshouldrefer toa TpConnectionIndentwhose parentis either CanTpConnectionor FlexrayTpConnectionor FlexrayArTpConnectionor DoIpTpConnection.

How to Solve

Reconfigure the attributes, sothatthe attributes PhysicalRequest, ResponseOnEvent, Response andFunctionalRequestrefer toa TpConnectionIndentwhose parentis either CanTpConnectionor FlexrayTpConnectionor FlexrayArTpConnectionor DoIpTpConnection.

5.31 ASW4X_SYS_0031 [Constr_3097]

Error Description

The segments definedby the SegmentPositionelements ofone andthe same MultiplexedIPdu- aggregatedvia StaticPartandDynamicPart- shall notoverlap.

How to Solve

Enter the value inSegmentPosition, sothatthe values don'toverlap.

5.32 ASW4X_SYS_0032 [Constr_3098]

Error Description

The segments definedby the SegmentPositionelements ofone andthe same MultiplexedIPdu- aggregatedvia StaticPartandDynamicPart- shall notexceedthe lengthofthe MultiplexedIPdu.

How to Solve

Enter the SegmentPositionvalue suchthatthe total lengthdoesn'texceedthe lengthofthe MultiplexedIPdu.

5.33 ASW4X_SYS_0033 [Constr_3099]

Error Description

The segments definedby the SegmentPositionelements aggregatedinthe DynamicPartofa MultiplexedIPdushall notexceedthe lengthofany DynamicPartAlternative.iPdu.

How to Solve

Enter the SegmentPositionvalue suchthat, the aggregatedlengthinthe DynamicPartofa MultiplexedIPdushall notexceedthe lengthofany DynamicPartAlternative.iPdu.

5.34 ASW4X_SYS_0034 [Constr_3100]

Error Description

The segments definedby the SegmentPositionelements aggregatedinthe StaticPartofa MultiplexedIPdushall not exceedthe lengthofthe StaticPart.iPdu

How to Solve

Enter the SegmentPositionvalue suchthat, the aggregatedlengthinthe StaticPartofa MultiplexedIPdushall not exceedthe lengthofthe StaticPart.iPdu

5.35 ASW4X_SYS_0035 [Constr_3101]

Error Description

Every ISignalIPduthatis referencedby the DynamicPartAlternative shall containanISignal thatrepresents the selector field. The selector fieldsignal shall be locatedatthe positionthatis describedby the selectorFieldLengthandselectorFieldStartPosition.

How to Solve

a)the selector fieldsignal shall be locatedatthe positiondescribedby selectorFieldStartPosition.

b)The lengthofthe signal shouldbe same as selectorFieldLength.

5.36 ASW4X_SYS_0036 [Constr_3051]

Error Description

InISignalMapping, ifthe SourceSignal references anISignal thenthe TargetSignal shall alsoreference anISignal.

How to Solve

Make sure bothsource andtargetsignal refer anISignal.

5.37 ASW4X_SYS_0037 [Constr_3052]

Error Description

IfanISignalMappingtoanISignal thatis a member ofa ISignalGroupexists thenan-ISignalMappingtothe enclosingISignalGroupshall existas well.

How to Solve

Make sure the ISignalGroupalsohas a mapping.

5.38 ASW4X_SYS_0038 [Constr_3053]

Error Description

InanISignalMapping, ifthe TargetSignal refers toanISignalGroup, thenevery signal inthatISignalGorupshould have anexplicitor implicitmapping.

How to Solve

Make sure every signal inthe ISignalGroup, whichis referredby the TargetSignal, has anexplicitor implicitmapping.

5.39 ASW4X_SYS_0039 [Constr_3095]

Error Description

The existence ofcanControllerFdAttributes andcanControllerFdRequirements is mutually exclusive. i.e., In CanControllerConfiguration, either CanControllerFdConfigurationshouldbe configuredor CanControllerFdConfigurationRequirements shouldbe configured. Bothcannotbe configured.

How to Solve

Configure either canControllerFdAttributes or canControllerFdRequirements .

5.40 ASW4X_SYS_0040 [Constr_3518]

Error Description

The value givenfor CanControllerFdConfiguration.paddingValue andCanControllerFdConfigurationRequirements.paddingValue shall be inthe range from0to255.

How to Solve

For the attributes CanControllerFdConfiguration.paddingValue andCanControllerFdConfigurationRequirements.paddingValue, enter the value between0to255.

5.41 ASW4X_SYS_0041 [Constr_3069]

Error Description

The value ofCanNmCluster.nmNidPositionshall only be settoeither

bit0(byte 0)or bit8(byte 1). How toSolve

Reconfigure the attribute nmNidPositionsothatitis either 0or 1.

5.42 ASW4X_SYS_0042 [Constr_3070]

Error Description

The value ofCanNmCluster.nmCbvPositionshall only be settoeither

bit0(byte 0)or bit8(byte 1). **How to Solve**

Reconfigure the attribute nmCbvPositionsothatitis either 0or 1.

5.43 ASW4X_SYS_0043 [Constr_3071]

Error Description

CanNmCluster.nmCbvPositionandCanNmCluster.nmNidPositionshall never have the same value.

How to Solve

Reconfigure nmCbvPositionandnmNidPositionsothatthey bothare difficult.

5.44 ASW4X_SYS_0044 [Constr_3036]

Error Description

CAN Frames andLIN Frames shall only containone Pdu.

How to Solve

Make sure CAN andLIN frames shall containonly one Pdu.

5.45 ASW4X_SYS_0045 [Constr_3206]

Error Description

A FrameTriggeringthatreferences a LinEventTriggeredFrame shall nothave a ref-

erence toa FramePort. **How to Solve**

Delete all the frame ports, whenthe FrameTriggeringrefers a LinEventTrig-geredFrame .

5.46 ASW4X_SYS_0046 [Constr_3037]

Error Description

For CAN the maximumframeLengthis 8bytes and64bytes incase ofCAN FD.

How to Solve

a)For CAN frame the maximumframeLengthshouldbe 8bytes.

b)For CAN FD frame the maximumframeLengthshouldbe 64bytes.

5.47 ASW4X_SYS_0047 [Constr_3037]

Error Description

For LIN the maximumframeLengthis 8bytes.

How to Solve

Make sure the maximumframeLengthis 8bytes.

5.48 ASW4X_SYS_0048 [Constr_3038]

Error Description

For FlexRay the maximumframeLengthis 254bytes.

How to Solve

Make sure the maximumframeLengthfor FlexRay frame is 254bytes.

5.49 ASW4X_SYS_0049 [Constr_3034]

Error Description

The values ofattributes ofLinSlaveConfigandLinSlave shall be identical for each-LinSlaveConfigthatpoints toa LinSlave.

How to Solve

1. The attribute ConfiguredNadinLinSlaveConfigandthe attribute ConfiguredNadinreferredLinSlave (through LinSlaveEcu) shouldbe equal.
2. The attribute FunctionIdinLinSlaveConfigandthe attribute FunctionIdinreferredLinSlave (through LinSlaveEcu) shouldbe equal.
3. The attribute SupplierIdinLinSlaveConfigandthe attribute SupplierIdinreferredLinSlave (through LinSlaveEcu) shouldbe equal.
4. The attribute VariantIdinLinSlaveConfigandthe attribute VariantIdinreferredLinSlave (throughLinSlaveEcu) shouldbe equal.

5.50 ASW4X_SYS_0050 [Constr_3508]

Error Description

The nmReadySleepTime value shall be a multiple ofcycle *nmRepetitionCycle.

How to Solve

Reconfigure the nmReadySleepTime sothatits value is the productofcycle and-nmRepetitionCycle.

5.51 ASW4X_SYS_0051 [Constr_3202 & Constr_3202]

Error Description

Withina PhysicalChannel a LinUnconditionalFrame shall be referencedby only one LinFrameTriggeringtoallow a derivationofthe identifier ofa substitutedFrame if:

1. The LinUnconditionalFrame is referencedby a LinEventTriggeredFrame inthe role lin-UnconditionalFrame.
2. The LinUnconditionalFrame is referencedby a LinSporadicFrame inthe role substitut-edFrame.

How to Solve

Ina PhysicalChannel, there shouldbe only a single reference toa LinUncondition-alFrame througha LinFrameTriggering.

5.52 ASW4X_SYS_0052 [Constr_3204]

Error Description

A LinUnconditionalFrame associatedwitha LinSporadicFrame shall notbe allo-catedinthe same LinScheduleTable as the LinSporadicFrame.

How to Solve

Reconfigure the LinScheduleTable sothata LinUnconditionalFrame whichis associat-edwitha LinSporadicFrame is notreferredinthe same LinScheduleTable .

5.53 ASW4X_SYS_0053 [Constr_3516]

Error Description

The Pdu.lengthofCAN PDUs shall be restrictedto0..8for classicCAN L-PDUs

The Pdu.lengthofCAN PDUs shall be restrictedto0..8, 12, 16, 20, 24, 32, 48,

64for CAN FD L-PDUs.

How to Solve

For CAN L-PDUs, reconfigure the lengthsothatitis inthe range 0..8.

For CAN FD L-PDUs, reconfigure the lengthsothatitis inthe range 0..8, 12, 16, 20, 24, 32, 48, 64.

5.54 ASW4X_SYS_0054 [Constr_3205]

Error Description

A FrameTriggeringthatreferences a LinSporadicFrame shall nothave a ref-

erence toa FramePort.

How to Solve

Delete all the FramePorts inthe FrameTriggering, ifthe FrameTriggeringrefers toa LinSporadicFrame .

5.55 ASW4X_SYS_0093 [Constr_3179]

Error Description

InEndToEndTransformationDescriptionfollowingrestrictionshall be respected: SumofMinOkStateInitandMaxErrorStateInitmustbe less thanor equals toWindowSize.

How to Solve

1.AutoUpdate the WndowsSize based on the sum of MinOkStateInit and MaxErrorStateInit attributes

2.Allow users to update the value for Windowsize

3.Provide the options to edit either MinOkStateInit

4.Provide the options to edit either MaxErrorStateInit

Hint: Tooltip showing formula on MinOKStateInt and MaxErrorStateInit

5.56 ASW4X_SYS_0093 [Constr_3180]

Error Description

InEndToEndTransformationDescriptionfollowingrestrictionshall be respected: MinOkStateValidandMaxErrorStateValidmustbe less thanor equals toWindowSize.

How to Solve

Configure appropriate values.

5.57 ASW411_SYS_0119 [Constr-3025]

Error Description

Incase several TpConnections use the same Frame ID for their communicationneeds only one NPduelementper Frame Idshall exist. This constraintapplies for all supportedAUTOSARtransportprotocols (CanTp, LinTp, FrTp, FrArTpandJ1939Tp)

How to Solve

Provide the dialogtoselectthe Npduwhichhas the same frameID

5.58 ASW4X_SYS_0093 [Constr_3181]

Error Description

InEndToEndTransformationDescriptionfollowingrestrictionshall be respected:minOk-StateInvalidandmaxErrorStateInvalidmustbe less thanor equals toWindowSize.

How to Solve

Configure appropriate values.

5.59 ASW421_SYS_0074[Constr-3096]

Error Description

InJ1939DcmIPduthe value ofthe attribute DiagnosticMessageType shall be inthe range 1to57

How to Solve

Configure the DiagnosticMessageType withinthe range 1to57

5.60 ASW421_SYS_0076[Constr_3104]

Error Description

InJ1939NmNode containingJ1939NodeName, the attribute function-

shouldbe inthe range 0to255. **How to Solve**

Setthe attribute functionrange within0to255.

5.61 ASW421_SYS_0076[Constr_3105]

Error Description

InJ1939NmNode containingJ1939NodeName, the attribute functionInstance

shouldbe inthe range 0to31. **How to Solve**

Setthe attribute functionInstance range within0to31.

5.62 ASW421_SYS_0076[Constr_3106]

Error Description

InJ1939NmNode containingJ1939NodeName, the attribute identitiyNumber

shouldbe inthe range 0to2097151.

How to Solve

Set the attribute identitiyNumber range within 0 to 2097151.

5.63 ASW421_SYS_0076[Constr_3107]

Error Description

InJ1939NmNode containingJ1939NodeName, the attribute industryGroup-shouldbe inthe range 0to7.

How to Solve

Setthe attribute industryGrouprange within0to7.

5.64 ASW421_SYS_0076[Constr_3108]

Error Description

InJ1939NmNode containingJ1939NodeName, the attribute manufacturerCode shouldbe inthe range 0to2047.

How to Solve

Setthe attribute manufacturerCode range within0to2047.

5.65 ASW421_SYS_0076[Constr_3109]

Error Description

InJ1939NmNode containingJ1939NodeName, the attribute vehicleSys-temshouldbe inthe range 0to127. **How to Solve**

Setthe attribute vehicleSystemrange within0to127.

5.66 ASW421_SYS_0076[Constr_3110]

Error Description

InJ1939NmNode containingJ1939NodeName, the attribute vehicleSystemInstance shouldbe inthe range 0to 15.

How to Solve

Setthe attribute vehicleSystemInstance range within0to15.

5.67 ASW421_SYS_0085[Constr-3116]

Error Description

The ClientID range ofany ECUInstance shouldnotoverlapwithother ECUInstance withinthe System

How to Solve

Change the ClientIdRange ofone ofthe ECUInstance

5.68 ASW421_SYS_0086[Constr-3118]

Error Description

InClientIdDefinition, containingClientServerOperationthe attribute contextPorts-houldrefer toRPortPrototype

How to Solve

Change the ClientServerOperation.Contexttorefer anRPortPrototype

5.69 ASW421_SYS_0086[Constr-3118]

Error Description

The attribute byteOrder ofSOMEIPTransformationDescriptionshall be notbe opaque.

How to Solve

1. Change the byteOrder value toMOST-SIGNIFICANT-BYTE-FIRST.
2. Change the byteOrder value toMOST-SIGNIFICANT-BYTE-LAST.

5.70 ASW421_SYS_0090 [Constr-3124]

Error Description

The attribute needsOriginalData ofa TransformationTechnology shall only be usedfor the non-firsttransformers inthe transformer chain.

How to Solve

1. Change the order ofTransformationTechnologies inTransformerChains
2. Delete the configurationofneedsOriginalData attribute for firstTransformationTech-nology

5.71 ASW421_SYS_0091 [Constr-3125]

Error Description

InDataTransformationifthe attribute trasformerChainhas firsttransformer reference, ie. TransformationTechnology whichcontains bufferProperties, thenthe attribute inPlace shall be settofalse

How to Solve

1. Change the order ofTransformationTechnologies inTransformerChains
2. Setthe inPlace attribute ofBufferProperties for firstTransformationTechnology tofalse

5.72 ASW421_SYS_0106 [Constr-3127]

Error Description

The TransformerHardErrorEventshouldcontaineither ofthe attributes Operationor Trigger. The attributes OperationandTrigger shall be mutually exclusive.

How to Solve

1. Assigna DataTransformationinIsignal.
2. Create a new DataTransformationandinIsignal assignthe same DataTransformation.

5.73 ASW421_SYS_0107 [Constr-3128]

Error Description

For eachTransformation-Descriptionvariantthatis a SOMEIPTransformationDescrip-tion

1. Attribute protocol ofTransformationTechnology shall be settoSOMEIP
2. Attribute versionofTransformationTechnology shall be setto1
3. Attribute transformerClass ofTransformationTechnology shall be settoserializer
4. Attribute headerLengthofBufferProperties shall be setto64(bits)

How to Solve

1. Setprotocol attribute toSOMEIP.
2. Setthe versionattribute to1.
3. Setthe transformerClass attribute toSERIALIZER.
4. Setthe BufferProperties.HeaderLengthto64bits.

5.74 ASW421_SYS_0108[Constr-3130]

Error Description

The attribute InterfaceVersionofSOMEIPTransformationDescriptionshouldbe inthe range 0to255.

How to Solve

Setvalue ofattribute InterfaceVersionofSOMEIPTransformationDescriptioninthe range 0to255.

5.75 ASW421_SYS_0116 [Constr-3132]

Error Description

Ifa ISignalGrouphas a reference tothe DataTransformationelementinthe role com-BasedSignalGroupTransformationthenthis DataTransformationshall be the han-dledby the COMBasedTransformer

How to Solve

1.Refer the ComBasedTransformationinthe DataTransformationRefConditional

2.Remove the reference inthe DataTransformationRefConditional ifthe transformationis notComBasedTransformation

5.76 ASW421_SYS_3137 [Constr-3137]

Error Description

The IPduPort.rxSecurityVerificationattribute shall only be usedinIPduPorts withthe communicationDirection= in

How to Solve

Configure CommunicationDirectiontoIN

5.77 ASW421_SYS_3138 [Constr-3138]

Error Description

The IPdu-Port.rxSecurityVerificationinformationis only validfor SecuredIPdus

How to Solve

Configure SecureIPDU

5.78 ASW421_SYS_3140 [Constr-3140]

Error Description

The values ofSystem.containerIPduHeaderByte- Order are restrictedtoByteOrderEnum.mostSignificantByteFirst andByte- OrderEnum.mostSignificantByteLast. I.e. the value ByteOrderEnum.opaque is notallowed.

How to Solve

1. Configure ContainerIPduHeaderByteOrder toMOST_SIGNIFICANT_BYTE_FIRST
2. Configure ContainerIPduHeaderByteOrder toMOST_SIGNIFICANT_BYTE_LAST

5.79 ASW422_SYS_0055[Constr-3198]

Error Description

IfSystemMappingcontains more thanone PncMappingconfiguredwithShortLabel, thenShortLabel shouldbe Unique tothe system.

How to Solve

Modify PncMapping.ShortLabel tobe unique.

5.80 ASW422_SYS_0056[Constr-3199]

Error Description

InISignal ifthe attribute DataTypePolicy is settoTransforming-I-Signal thenISignal shouldreference a DataTransformation.

How to Solve

1. Create a DataTransformationfor DataTransformationRefConditional.
2. Refer existing.

5.81 ASW422_SYS_0057[Constr-3201]

Error Description

The EventHandler configuredwithvalues for the attribute ConsumedEventGroups, shouldhave identical values for EventGroupIdentifier.

How to Solve

Setthe EventGroupIdentifier ofConsumedEventGroups tobe identicalSetthe Event-GroupIdentifier ofConsumedEventGroups tobe identical.

5.82 ASW422_SYS_0058[Constr-3207]

Error Description

InEthernetPhysicalChannel containingthe SoAdConfigwithSocketConnection, the Sock-
etConnectionIpduIdentifier shouldrefer toa PDU triggeringwhichis usedinClientServ-erCommunication

How to Solve

Setthe attribute PDUTriggeringofSocketConnectionIpduIdentifier toPDUTriggerin-gusedinClientServerCommunication

5.83 ASW422_SYS_0060 [Constr-3209]

Error Description

InCanFrameTriggeringifthe attribute identifier is identical tothe attribute PGN ofJ1939configuredin J1939TpConfigthenthe attribute requestable alsoshouldbe identical withJ1939

How to Solve

Setthe value J1939requestable same as the attribute requestable ofJ1939TpPgele-ment

5.84 ASW422_SYS_0068 [Constr-3219]

Error Description

LinSlaves shall notbe partofthe EcuInstance ofthe correspondingLinMaster.

How to Solve

Remove LinSlave fromthe EcuInstance

5.85 ASW422_SYS_0071[Constr-3519]

Error Description

Ifthe value ofthe category ofGlobalTimeDomainshouldbe SYNCHRONIZED or OFFSET

How to Solve

1. Setthe value ofcategory toSYNCHRONIZED
2. Setthe value ofcategory toOFFSET

5.86 ASW422_SYS_0072[Constr-3520]

Error Description

1. IfinGlobalTimeDomain, the attribute offsetTimeDomainrefers toanother Global-TimeDomainthentargetreference shall have domainID inthe range 0-15.
2. IfinGlobalTimeDomainthe attribute offsetTimeDomainrefers toGlobalTimeDomain, thensource reference shall have domainID inthe range 16-31

How to Solve

1. Setthe DomainIdrange within0-15
2. Setthe DomainIdrange within16-31

5.87 ASW422_SYS_0092[Constr-1374]

Error Description

TwoGlobalTimeGateways referingtothe same ECUInstance, cannotrefer tothe same GlobalTimeMaster.

How to Solve

1. Change the Master for one ofthe GlobalTimeGateway
2. Remove the value ofMaster for one ofthe GlobalTimeGateway

5.88 ASW422_SYS_0097 [Constr-3021]

Error Description

Only SwComponentPrototypes thatare typedby SensorActuatorSwComponentType shall be mappedtoa HwElementwithcategory SensorActuator via the controlledHwElementrelation

How to Solve

Change the TargetComponenttorefer toa SwComponentPrototype oftype SensorActuatorSwComponentType

5.89 ASW422_SYS_0098 [Constr-3027]

Error Description

Incase the category ofthe Systemis SYSTEM_EXTRACT or ECU_EXTRACT the ecuExtractVersionattribute shall be defined

How to Solve

Configure EcuExtractVersion

5.90 ASW422_SYS_0099[Constr-3121]

Error Description

InDataTransformationthe attribute TransformerChains shouldshouldbe limitedto255transformerTechnologies.

How to Solve

Modify TransformerChains tocontainonly 255TransformerTechnologies.

5.91 ASW422_SYS_0100[Constr-3122]

Error Description

InDataTransformation, the trasformerChainshoudnotrefer tomore thanone TransformationTechnologies whose attribute trasformer class is notcustom, butvalues are identical

How to Solve

1.Remove the reference - Provide a pop up list of all the <TransfomationTechnology_Shortname> within a table

2.Update the Category to Custom for transformation techonology

5.92 ASW422_SYS_0101[Constr-3123]

Error Description

InDataTransformationthe attribute trasformerChainshouldhave firsttransformer reference whose attribute trasformerClass settoserializer

How to Solve

If there are no searializer contents then

a).Add new serializer transformer reference in the TransformerChain

or

If already serializer transformer is available then

b).Move any existing trasformer reference with Transformerclass attribute serializer to the first element.

5.93 ASW422_SYS_0101 [Constr-1323]

Error Description

InDataTransformationifthe attribute trasformerChaincontains firsttransformer reference whose attribute trasformerClass is notserializer,thenitis anerror

How to Solve

Setthe value ofusesEndToEndProtectiontofalse

5.94 ASW422_SYS_0101 [Constr-1387]

Error Description

Transformer is usedfor the transmissionofa VariableSize Array Data Types, butthe Variable-Size Array Data Type is not'new-world' variable-size array data type

How to Solve

Modify the ApplicationArrayDatatype tobe New-Worldvariable-size array data type

5.95 ASW422_SYS_0102 [Constr-3506]

Error Description

WhenISignalGrouprefers toa E2ETransformer/SOMEIPTransformer,inSystemMappingthatcontains SenderReceiverToSignalGroupMapping,the targetDataPrototype shouldnotrefers toa CompositeDataType.

How to Solve

Remove the E2ETransformer withinISignalGroup

5.96 ASW422_SYS_0105 [Constr-3153]

Error Description

For ISignalGroup, InDataTransformationthe attribute TranformerChainwhichcontains toCOMBasedtransformers andE2Etransformers withprofile name ofE2ETransformers as
(PROFILE1,PROFILE2,PROFILE4,PROFILE5,PROFILE6) thenbufferProperties.Header-Lengthofthe E2Etransformer shouldbe (16,12,16,96,24,40) bits respectively

How to Solve

Configure correctvalue for HeaderLength, Hint: Tooltipshowingthe range ofbits for HeaderLength

5.97 ASW422_SYS_Constr_3208 [Constr-3208]

Error Description

IftwoIsignals refers tothe same systemsignal, thennoIsignal shouldrefer toDatatrans-formationwhose attribute executeDespiteDataUnavailability is settoTRUE.

How to Solve

Setthe attribute executeDespiteDataUnavailability tofalse

5.98 ASW422_SYS_Constr_3210 [Constr-3210]

Error Description

IfJ1939TpConfigwhichhas J1939TpConnectioncontains more thanone J1939TpPgwithidentical values for the attribute pgn, butthe attribute Requestable are notidentical thenitis anerror.

How to Solve

Setattribute requestable ofall J1939TpPgtohave identical values where pgnare iden-tical.

5.99 ASW422_SYS_Constr_3211 [Constr-3211]

Error Description

IfPduTriggeringis configuredwithTriggerIPduSendCondition, butthe attribute IPdudoes notrefer ISignalIPdus, thenitis anerror.

How to Solve

Provide anISignalIPdureference for attribute PduTriggering.IPdu

5.100 ASW422_SYS_Constr_3213 [Constr-3213]

Error Description

InIsignal ifthe TransformationISignalProps defines the attribute TransformationTechnology transfomer class is equal toserializer andTransformationISignalProps is transportedvia ClientServerCommunication, butthe attribue CSErrorReactionis notconfigured, thenthe attribue CSErrorReactionshouldalsobe configured.

How to Solve

Configure value for CsErrorReaction

5.101 ASW422_SYS_Constr_3214 [Constr-3214]

Error Description

InIsignal ifthe TransformationISignalProps defines the attribute TransformationTechnology whose transformer class is notequal toSerializer thenthe attribute CSErrorReactionshouldnotbe configured

How to Solve

1.Setthe trasformer class toserializer inTrasformationTechnology

2.Delete the value ofattribute CsErrorReaction

5.102 ASW422_SYS_Constr_3216 [Constr-3216]

Error Description

IfanIsignal refers toa SystemSignal, whichis notMappedvia SenderReceiverToSignalMappingbutin SOMEIPTransformationISignalPropsConditional ofIsignal if, the attribute SessionHandlingSRis configured, thenit is anerror.

How to Solve

Isignal whichReferences systemsignal shouldbe mappedvia a SenderReceiverToSignalMapping

5.103 ASW422_SYS_Constr_3218 [Constr-3218]

Error Description

InIsignal havingSOMEIPTransformationISignalPropsConditional ifthe attribute sizeOfArrayLengthFields does not have either ofthe values 0,1,2or 4, thenitis anerror.

How to Solve

Configure length

5.104 ASW422_SYS_Constr_3220 [Constr-3220]

Error Description

InIsignal havingSOMEIPTransformationISignalPropsConditional ifthe attribute sizeOfStructLengthFields does not have either ofthe values 0,1,2or 4, thenitis anerror.

How to Solve

Configure length

5.105 ASW422_SYS_Constr_3221 [Constr-3221]

Error Description

InIsignal havingSOMEIPTransformationISignalPropsConditional ifthe attribute sizeOfUnionLengthFields does not have either ofthe values 0,1,2or 4, thenitis anerror.

How to Solve

Configure length

5.106 ASW430_SYS_3060 [Constr-3221]

Error Description

CompuMethod is configured under the iSignal and not under the System Signal.

How toSolve

Fix the compuMethod reference from iSignal to System Signal

5.107 ASW4X_SYS_5055 [Constr_5055]

Error Description

PRPortPrototypes are notpermittedtobe configuredinDataMappingwithSystemSignals insystemelementInother words the usage ofPRPortPrototypes for inter-Ecucommunicationis notsupportedby AUTOSAR.

How toSolve

Remove the DataMapping

5.108 ASW4X_SYS_5030 [Constr_5030]

Error Description

InLinCommunicationConnector the attribute index configuredwithinLinOrderedConfigurableFrame shouldbe unique.

How to Solve

Allow the user to enter a unique value to attribute "index".

5.109 ASW4X_SYS_5183 [Constr_5183]

Error Description

PortElementToCommunicationResourceMappingshouldreference parameterDataPrototype or modeDeclarationGroupPrototype or trigger or ClientServerOperationor VariableDataPrototype.

How to Solve

Notpossible since creationofelements required.

5.110 ASW4X_SYS_1441 [Constr_1441]

Error Description

IfanImplementationDataType ofcategory STRUCT, havinga subelementofcategory union, is mappedtoSystemSignal indatamapping, thenISignal thatreferences SystemSignal shouldcontainSOMEIPTransformationISignalProps.

5.111 ASW4X_SYS_3314 [Constr_3314]

Error Description

ConfiguringbufferComputationwithinBuffer properties is mandatory.

5.112 ASW4X_SYS_1207 [Constr_1207]

Error Description

WithinDataMappingfor a PPortPrototype, the attribute DataMapping.communicationDirectionifitexists itshouldbe setto“out”.

For a RPortprototype, the value shouldbe setto“in”.

How toSolve

Toggle the value for attribute CommunicationDirection

5.113 ASW4X_SYS_3299 [Constr_3299]

Error Description

SocketAddress.pathMtuDiscoveryEnabledshouldbe settoTRUEonly ifEthernet-CommunicationConnector.pathMtuEnabledis TRUE.

How toSolve

Setthe value for attribute pathMtuDiscoveryEnabledtofalse.

5.114 ASW4X_SYS_3311 [Constr_3311]

Error Description

SocketAddress.flowLabel shouldonly be usedifthe aggregatedApplicationEnd-pointrefers toa NetworkEndpoint withanIpv6Configuration.

How toSolve

Remove value offlowLabel.

5.115 ASW4X_SYS_3516 [Constr_3516]

Error Description

The value ofthe attribute frameLengthofa frame shouldbe from0to8for classicCAN PDUs and0to8,12, 16, 20, 24, 32, 48, 64for CAN FD L-PDUs.

How to Solve

Allow the user tosetthe value (systemshouldallow only the values withinthe range 0to8for CanFrame and0to8, 12, 16, 20, 24, 32, 48, 64for CanFD frame).

5.116 ASW4X_SYS_3012 [Constr_3012]

Error Description

PDUs mappedtoa FlexrayFrame shall NOT overlap. The Startpositionofa PDU shouldnotbe same andalsoitshouldnotbe withinthe lengthofanother PDU.

How toSolve

i)Startposition+ lengthofPDU shouldbe the startpositionofsecondPDUToFrame Mapping.

5.117 ASW4X_SYS_3013 [Constr_3013]

Error Description

The combinedlengthofall Pdus thatare mappedintoa FlexrayFrame shall notexceedthe definedFlexrayFrame length.

How toSolve

i)Adjustthe lengthofthe frame by addingthe lengthofall PDUs under the frame
ii)User shouldnotbe allowedtoenter value less thancombinedlengthofall frames

5.118 ASW4X_SYS_3267 [Constr_3267]

Error Description

InSD StaticSocketConnectionthe PDUTriggeringshouldrefer toa GeneralPurpose PDU withCategory as SD Note:

StaticSocketConnectioncanbe calledSD StaticSocketConnectionwhenithas following configuration

- StaticSocketConnectionshouldaggregate UDPSocketAddress
- StaticSocketConnectionshouldrefer toa GeneralPurposePDU ofCategory SD (via SoConIPduIdentifier)

How to Solve

Update the category toSD.

5.119 ASW4X_SYS_3268 [Constr_328]

Error Description

The SD StaticSocketConnectionshouldbe aggregatedby anApplicationEndpointthatdefines a UdpPort

Criteria for validationtobe applicable: SD StaticSocketConnectionshouldrefer toa GeneralPurpose PDU ofcategory SD Note:

StaticSocketConnectioncanbe calledSD StaticSocketConnectionwhenithas following configuration

- StaticSocketConnectionshouldaggregate UDPSocketAddress

- StaticSocketConnectionshouldrefer toa GeneralPurposePDU ofCategory SD (via SoConIPduIdentifier)

5.120 ASW4X_SYS_3269 [Constr_3269]

Error Description

The SD StaticSocketConnectionshouldbe aggregatedby anApplicationEndpointthatdefines a UdpPortthatconfigures Port.

How to Solve

Setthe UDPPortnumber to0.

5.121 ASW4X_SYS_3270 [Constr_3270]

Error Description

EachSD StaticSocketConnectiondefinedshouldrefer withthe remoteAddress reference toanApplicationEndpointthatpoints toa NetworkEndpointthatdefines anIPAddress ANY (IPv4or IPv6).

How toSolve

Setthe IPaddress toANY inseconderror message.

5.122 ASW4X_SYS_3272 [Constr_3272]

Error Description

The SoConIPduIdentifier.headerIdofSD StaticSocket-Connections shouldalways be setto0xFFFF8100(decimal equivalent= 4294934784) for SD messages

StaticSocketConnectioncanbe calledSD StaticSocketConnectionwhenithas following configuration

- StaticSocketConnectionshouldaggregate UDPSocketAddress
- StaticSocketConnectionshouldrefer toa GeneralPurposePDU ofCategory SD (via SoConIPduIdentifier)

How to Solve

Setthe headerID to0xFFFF8100.

5.123 ASW4X_SYS_3273 [Constr_3273]

Error Description

The SD StaticSocketConnectionfor multicastshouldcontainanApplicationEndpointthatpoints toa NetworkEndpointthatdefines anIPMulticastAddress.

StaticSocketConnectioncanbe calledSD StaticSocketConnectionwhenithas following configuration

- StaticSocketConnectionshouldaggregate UDPSocketAddress
- StaticSocketConnectionshouldrefer toa GeneralPurposePDU ofCategory SD (via SoConIPduIdentifier)

Note: SD StaticSocketConnectionis Multicastwhenthe PDU triggeringrefredinthe SoconIPDUIdentier has IpDUPortthathas communicationDirectionas IN.

5.124 ASW4X_SYS_3274 [Constr_3274]

Error Description

The SD StaticSocketConnectionfor UnicastshouldcontainanApplicationEndpointthatpoints toa NetworkEndpoint thatdefines anIPUnicastAddress.

StaticSocketConnectioncanbe calledSD StaticSocketConnectionwhenithas following configuration

- StaticSocketConnectionshouldaggregate UDPSocketAddress
- StaticSocketConnectionshouldrefer toa GeneralPurposePDU ofCategory SD (via SoConIPduIdentifier)

Note: SD StaticSocketConnectionis Unicastwhenthe twodifferentPDU triggeringreferredinthe SoconIPDUIdentier has Atleastone IN andone OUT IpDUPort.The SD StaticSocketConnectionfor UnicastshouldcontainanApplicationEndpointthatpoints toa NetworkEndpointthatdefines anIPUnicastAddress. StaticSocketConnectioncanbe calledSD StaticSocketConnectionwhenithas following configuration

- StaticSocketConnectionshouldaggregate UDPSocketAddress
- StaticSocketConnectionshouldrefer toa GeneralPurposePDU ofCategory SD (via SoConIPduIdentifier)

Note: SD StaticSocketConnectionis Unicastwhenthe twodifferentPDU triggeringreferredinthe SoconIPDUIdentier has Atleastone IN andone OUT IpDUPort.

5.125 ASW4X_SYS_3276 [Constr_3276]

Error Description

Ifthe ApplicationEndPointrefers a NetworkEndPointthathas IPV4Configuration, thenthe SocketAddress thatowns the ApplicationEndPointshouldnotconfigure attribute allowedIPv6ExtHeaders.

How toSolve

Remove the reference for attribute allowedIPv6ExtHeaders.

5.126 ASW4X_SYS_3277 [Constr_3277]

Error Description

Ifthe ApplicationEndPointrefers a NetworkEndPointthathas IPV4Configuration, thenthe SocketAddress thatowns the ApplicationEndPointshouldnotconfigure attribute allowedIPv6ExtHeaders.

5.127 ASW4X_SYS_3297 [Constr_3297]

Error Description

A SocketAddress thatcontains anApplicationEndPointwithUDPaddress, cannotconfigure allowedTCPOptions.

How toSolve

Remove the value ofallowedTcpOptions.

5.128 ASW4X_SYS_3322 [Constr_3322]

Error Description

Withinthe contextofone SocketAddress, the value ofattribute pduCollectionSemantics ofall the SoConIPduIdentifier.

How to Solve

Update pduCollectionSemantics inall the SoConIPduIdentifiers (Allow the user tochoose one value).

5.129 ASW4X_SYS_3325 [Constr_3325]

Error Description

For the elements SecureCommunicationFreshnessProps, SecureCommunicationAuthenticationProps and CryptoServicePrimitive the validvalues basedondifferentProfiles is as giveninthe table below.

Criteria:

SecuredIPDU shouldhave SecureCommunicationFreshnessProps, SecureCommunicationAuthenticationProps,
CryptoServicePrimitive, SecOcCryptoServiceMapping. Ifso, thenCategory ofSecureCommunicationFreshnessProps, SecureCommunicationAuthenticationProps andCryptoServicePrimitive (ifconfigured) tobe checkedNote:

Profile_01, Profile_02, Profile_03tobe addedinmessages.properties file only for SecureCommunicationFreshnessProps, SecureCommunicationAuthenticationProps andnotfor CryptoServicePrimitive. Error marker tobe shownonSecureCommunicationFreshnessProps, SecureCommunicationAuthenticationProps

How to Solve

1)Setthe correctattribute value as definedby Autosar specfor specificprofile <Basedoncurrentprofile set>

2)Provide optionchange the Category toother twoprofiles andupdate the attribute values accordingly

5.130 ASW4X_SYS_3379 [Constr_3379]

Error Description

Ifthere are twoor more SocketAddress entities withinthe scope ofone SoAdConfiginthe scope ofone EcuInstance thathave the same static(fixedatconfigurationtime) IPAddress, Protocol andPortinthe aggregatedApplicationEndpointandNetworkEndpoint, thenonly one ofthese SocketAddress elements shall be referencedby ProvidedServiceInstances/ConsumedServiceInstances in the role localUnicastAddress

5.131 ASW4X_SYS_3464 [Constr_3464]

Error Description

1. PduTriggering is referenced by BusMirrorChannelMappingCan in the role targetPduTriggering, but the corresponding Pdu is not of the type GeneralPurposeIPdu
2. PduTriggering is referenced by BusMirrorChannelMappingCan in the role targetPduTriggering, but the category of the corresponding GeneralPurposeIPdu is not BUS_MIRRORING

How to Solve

Configure the busMirrorNetworkId within the BusMirrorChannels refering to the same Physical channel to be identical

5.132 ASW4X_SYS_3465 [Constr_3465]

Error Description

The attribute BusMirrorChannel.busMirrorNetworkId is not identical in all BusMirrorChannels that are referencing the same PhysicalChannel in the scope of the System.

How to Solve

Configure the busMirrorNetworkId within the BusMirrorChannels refering to the same Physical channel to be identical

5.133 ASW4X_SYS_3466 [Constr_3466]

Error Description

The attribute BusMirrorChannel.busMirrorNetworkId associated with PhysicalChannels that have the same specialization shall have unique BusMirrorChannel.busMirrorNetworkId within the scope of the System.

How to Solve

Configure the busMirrorNetworkId within the BusMirrorChannels refering to the Physical channel of same specialization to be unique

5.134 ASW4X_SYS_3467 [Constr_3467]

Error Description

The BusMirrorChannel that is aggregated by BusMirrorChannelMappingCan references a physicalChannel which is other than CanPhysicalChannel in the role targetChannel

How to Solve

Configure the reference TargetChannel to refer only CanPhysicalChannel

5.135 ASW4X_SYS_3468 [Constr_3468]

Error Description

BusMirrorChannelMappingCan configures more than one PduTriggering in the role targetPduTriggering

How to Solve

Delete any one of the additional targetPduTriggering within BusMirrorChannelMappingCan

5.136 ASW4X_SYS_3469 [Constr_3469]

Error Description

The CanFrameTriggering of a Frame that contains a Pdu of which, the PduTriggering is referenced by BusMirrorChannelMappingCan in the role, targetPduTriggering has the attribute txMask set to zero.

How to Solve

Configure the value of the attribute TxMask within PduTriggering to zero

5.137 ASW4X_SYS_3470 [Constr_3470]

Error Description

1. The CanFrameTriggering does not have any FramePorts with CommunicationDirection as OUT

2. The value of the attribute canFrameTxBehavior of CanFrameTriggering of the frame that contains tagetPduTriggering is set to CAN-FD and The CanFrameTriggering has a FramePort with CommunicationDirection as OUT but CanCommunicationController does not contain CanControllerFdConfiguration or CanControllerFdConfigurationRequirements

3. The value of the attribute paddingValue of CanControllerFdConfiguration or CanControllerFdConfigurationRequirements is not set to Zero

How to Solve

1. The CanFrameTriggering should contain FramePorts with CommunicationDirection as OUT
2. The value of the attribute paddingValue of CanControllerFdConfiguration or CanControllerFdConfigurationRequirements should be set to Zero

5.138 ASW4X_SYS_3506 [Constr_3506]

Error Description

Entry criteria: SystemSignalGroupshouldnothave Datatransformation(ExceptComBasedTransformer) .

Signal configuredwithinSenderReceiverToSignalGroupMappingshouldbe partofthe systemSignalGroup

Configurationwhere this constraintis notapplicable: For Sender-

RecArrayTypeMappinginSenderReceiverToSignalMapping, ifthere is a Base type is Uint8, thenthis constraintis notapplicable

5.139 ASW4X_SYS_3695 [Constr_3695]

Error Description

For a CANCommunicationController, Either canControllerXlAttributes or canControllerXlRequirements can be configured at a time and configuration for both the elements are not allowed as per autosar.

How to Solve

- Delete any one of the canControllerXlAttributes or canControllerXlRequirements within the CanControllerConfiguration.

5.140 ASW4X_SYS_3696 [Constr_3696]

Error Description

CanControllerConfiguration configuring a CAN XL controller shall aggregate

CanControllerXlConfiguration with the following attributes

- errorSignalingEnabled
- propSeg
- syncJumpWidth
- timeSeg1
- timeSeg2
- trcvPwmModeEnabled

How to Solve

For CanControllerXlConfiguration, configure all of the following mandatory attributes: errorSignalingEnabled, propSeg,syncJumpWidth,timeSeg1,time-Seg2,trcvPwmModeEnabled

5.141 ASW4X_SYS_3697 [Constr_3697]

Error Description

For a CANCommunicationController, either canControllerXlAttributes or canControl-lerXlRequirements has to be compulsorily configured.

How to Solve

Configure any one of the canControllerXlAttributes or canControllerXlRequirements with mandatory attributes within the CanControllerConfiguration

5.142 ASW4X_SYS_3698 [Constr_3698]

Error Description

The attribute errorSignalingEnabled should be set to FALSE if trcvPwmModeEnabled is set to TRUE

How to Solve

Configure the attribute errorSignalingEnabled to FALSE when the attribute trcvPwmModeEnabled is set to TRUE

5.143 ASW4X_SYS_3699_3700_3701 [Constr_3699_3700_3701]

Error Description

The attributes pwmL,pwmO,pwmS should be configured if trcvPwmModeEnabled is set to TRUE

How to Solve

Configure value for the attribute pwmL,pwmO,pwmS when the attribute trcvPwmModeEnabled is set to TRUE

5.144 ASW4X_SYS_3702 [Constr_3702]

Error Description

If the category of EthernetCommunicationController is equal to CAN_XL, only then the attributes macLayerType and macUnicastAddress can be configured.

How to Solve

- Set the category value of EthernetCommunicationController to CAN_XL
- Delete the values for the attribuites macLayerType and macUni-castAddress

5.145 ASW4X_SYS_3703 [Constr_3703]

Error Description

If the category of EthernetCommunicationController is equal to CAN_XL, then the reference canXlConfig should refer to CanCommunicationController that aggregates CanControllerConfiguration that inturn configures CanControllerXlConfiguration.

How to Solve

Configure the reference canXlConfig to CanCommunicationController the contains CanControllerConfiguration with CanControllerXlConfiguration, If the category of EthernetCommunicationController is equal to CAN_XL.

5.146 ASW4X_SYS_3704 [Constr_3704]

Error Description

If the class CanXlFrameTriggeringProps is aggregated by a CanFrameTriggering, then the CanCommunicationController which is referenced through commController by a CanCommunicationConnector, which in turn is referenced through commConnector by a CanPhysicalChannel that aggregates the CanFrameTriggering should aggregate at least one of CanControllerConfiguration with CanControllerXlConfiguration aggregated or CanControllerConfigurationRequirements with CanControllerXlConfigurationRequirements aggregated.

How to Solve

In CanCommunicationController, configure either CanControllerXlConfiguration or CanControllerXlConfigurationRequirements when CanPhysicalChannel aggregates CanFrameTriggering containing CanXlFrameTriggeringProps.

5.147 ASW4X_SYS_3705_3706_3707_3713 [Constr_3705_3706_3707_3713]

Error Description

For any CanXlFrameTriggeringProps the value of these attributes should be within the defined range:

priorityId should be in the range between 0 and 2047

sduType should be in the range between 0 and 255

vcid should be in the range between 0 and 255

acceptanceField should be in the range between 0 and 4294967295

How to Solve

- Configure priorityId in the range between 0 and 2047
- Configure sduType in the range between 0 and 255
- Configure vcid in the range between 0 and 255
- Configure acceptanceField in the range between 0 and 4294967295

5.148 ASW4X_SYS_3708 [Constr_3708]

Error Description

UDP Network Management is configured for an EthernetPhysicalChannel that is connected to an EthernetCommunicationController of category CAN_XL.

How to Solve

Remove the VLAN configuration reference to the If the EthernetPhysicalChannel of CAN_XL category.

5.149 ASW4X_SYS_5029 [Constr_5029]

Error Description

TtCanCluster is notallowedtobe referencedwithinJ1939NmCluster

How toSolve

Remove TtcanCluster value fromCommunicationCluster and/Or Provide anoptionofuser tosetvalue other than TtcanCluster.

5.150 ASW4X_SYS_5031 [Constr_5031]

Error Description

Withinthe AssignFrameIdRange, the value ofthe FramePid.index shouldbe unique

5.151 ASW4X_SYS_5032 [Constr_5032]

Error Description

Referringmore thanone NmConfigelementwiththe FibexElementRefConditional ofthe Systemelementis not allowed.

5.152 ASW4X_SYS_5049 [Constr_5049]

Error Description

CouplingPortDetails canhave followingconfigurations

- Only One CouplingPortTrafficClassAssignmentwithnoPriorities set
- 8CouplingPortTrafficClassAssignmentwithPriorities set

How to Solve

Remove the value for Priorities whenthere is only one CouplingPortTrafficClassAssignment.

5.153 ASW4X_SYS_5051 [Constr_5051]

For a Canframe Triggering, havinga Pduwhichhas PduTriggeringreferencedwithinBusMirrorMappingCaninthe role targetPduTriggeringshouldnotdefine anidentifier

How to Solve

Remove the value ofidentifier.

5.154 ASW4X_SYS_5053 [Constr_5053]

Error Description

IfanISignalPorthas a networkRepresentationProps.invalidValue (obtainedvia the ISignal as shownbelow), thenthe attribute communicationDirection-shouldbe settoIN

How toSolve

Setthe CommunicationDirectionvalue toIN.

5.155 ASW4X_SYS_5058 [Constr_5058]

Error Description

InCryptoServiceQueue the attribute QueueSize value shouldbe greater thanor equal to1ifconfigured

How toSolve

Allow the user tosetother values.

5.156 ASW4X_SYS_5060 [Constr_5060]

Error Description

SecuredIPducannotbe mappedtoa

LinFrame via the PduTriggering How

toSolve

Remove the PduToFrameMappingwhichcontains the SecuredIPdu.

5.157 ASW4X_SYS_5061 [Constr_5061]

Error Description

The EthernetCommunicationConnectors referencedwithinthe SocketAddress (as connectors or MulticastConnectors) andthe EthernetPhysicalChannel thataggregates the SoAdConfig(thatinturnaggregates the SocketAddress) shouldbe same

How toSolve

1. Remove the reference ofEthernetCommunicationConnector fromSocket-Adddress

2. Allow the user toaddvalidEtherEthernetCommunicationConnector toSocket-Address attributes(multicastconnector andconnector)

5.158 ASW4X_SYS_5062 [Constr_5062]

Error Description

When there are more thanone ProvidedServiceInstance whichhas same serviceIdentifier andthe same majorVersionanddifferentinstanceIdentifiers, thenitshouldnotbe mappedtosame Portnumber andIPaddress combination-aggregatedby ApplicationEndPointwithrespecttoOne Ecu, throughUnicastor Multicastaddress

5.159 ASW4X_SYS_5063 [Constr_5063]

Error Description

InProvidedServiceInstance, configuringthe attribute serviceIdentifier is mandatory

How to Solve

Allow the user to set value.

5.160 ASW4X_SYS_5064 [Constr_5064]

Error Description

InProvidedServiceInstance, configuringthe attribute MajorVersionis mandatory

How to Solve

Allow the user tosetvalue.

5.161 ASW4X_SYS_5065 [Constr_5065]

Error Description

InProvidedServiceInstance, configuringthe attribute MinorVersionis mandatory

How toSolve

Allow the user tosetvalue.

5.162 ASW4X_SYS_5066 [Constr_5066]

Error Description

InProvidedServiceInstance, configuringthe attribute InstanceIdentifier is mandatory

How to Solve

Allow the user to set value.

5.163 ASW4X_SYS_5067 [Constr_5067]

Error Description

Ina VLan, all the providedServerInstance shouldhave unique value combinationfor serviceIdentifier,instanceIdentifier, majorVersion

5.164 ASW4X_SYS_5068 [Constr_5068]

Error Description

ProvidedServiceInstance.LocalUnicastAddress shouldpointtoIPunicastaddress

How to Solve

Remove the Local unicastAddress (ApplicationEndpointRefConditional).

5.165 ASW4X_SYS_5069 [Constr_5069]

Error Description

ProvidedServiceInstance.remoteUnicastAddress shouldpointtoIPunicastaddress

How to Solve

Remove the Local unicastAddress (ApplicationEndpointRefConditional).

5.166 ASW4X_SYS_5070 [Constr_5070]

Error Description

The headerID betweenthe SoConIPDUIdentifiers shouldbe unique for a communicationdirectioninor out

Note: whenPDU identifier is inconflictwithother PDU identifier, the validationfor other PDU identifier shouldnotbe triggeredagain.

5.167 ASW4X_SYS_5071 [Constr_5071]

Error Description

The ApplicationEndpointthatis referencedby anEventHandler inthe role eventMulticastAddress shall reference a NetworkEndpointthatdefines anIP-MulticastAddress

How to Solve

Remove thatinvalidEventMulticastAddress (ApplicationEndpointRefConditional) fromEventhandler.

5.168 ASW4X_SYS_5072 [Constr_5072]

Error Description

IfanEventHandler thatis aggregatedby a ProvidedServiceInstance does nothave a definedeventMulticastAddress thenthe multicastThresholdshall be settothe value 0

How toSolve

SetmulticastThresholdvalue tozero.

5.169 ASW4X_SYS_5073 [Constr_5073]

Error Description

Withinthe EventHandler, Ifa PduActivationRoutingGrouphas the EventGroupControlType settoactivationUnicastor triggerUnicastor activationAndTriggerUnicastthenthe aggregatedProvidedServiceInstance shouldconfigure localUnicastAddress thatpoints toanIPUnicastaddress

How toSolve

Populate all possible ApplicationEndPointwhichrefers toIPunicastaddress sothatuser canchoose the value. For the chosenapplicationEndPoint, create ApplicationEndpointRefConditional under ProviderServiceInstance andshouldpointtoApplicationEndPoint.

5.170 ASW4X_SYS_5074 [Constr_5074]

Error Description

AnEventHandler thatconfigures a PduActivationRoutingGroupwiththe PduActivationRoutingGroup.eventGroupControlType settoactivationMulticastshouldhave aneventMulticastAddress reference, thatpoints toa “remote IPMulticastAddress. The ProvidedServiceInstance thataggregates the

EventHandler shall have a localUnicastAddress reference toa “local”UDPApplicationEndpointNote: Connector for IPunicastandMutlicastConnectors for IP-multicast

5.171 ASW4X_SYS_5075 [Constr_5075]

Error Description

A PduActivationRoutingGroupwitheventGroupControlType settoactivationMulticastcantoreference SoConIPduIdentifiers only throughiPduIdentifierUdprole

How toSolve

Remove the reference withinthe IPduIdentifierstcp.

5.172 ASW4X_SYS_5076 [Constr_5076]

Error Description

Ifthe PduActivationRoutingGroupcontains the iPduIdentifierTcpreference thenthe aggregatingProvidedServiceInstance shouldcontaina localUnicastAddress reference toanApplicationEndpointthatdefines a TCPaddress

How to Solve

Remove the reference inLocalunicastwhichdoes nothave TCPaddress andprovide optionfor user toconfigure ApplictionEndPointthathas TCPaddress.

5.173 ASW4X_SYS_5077 [Constr_5077]

Error Description

Ifthe PduActivationRoutingGroupcontains the iPduIdentifierTcpreference thenthe aggregatingProvidedServiceInstance shouldcontaina localUnicastAddress reference toanApplicationEndpointthatdefines a UDPaddress

How to Solve

Remove the reference inLocalunicastwhichdoes nothave UDPaddress andprovide optionfor user toconfigure ApplictionEndPointthathas UDPaddress.

5.174 ASW4X_SYS_5078 [Constr_5078]

Error Description

EachPduTriggeringReferredwithinthe PduActivationRoutingGroupofthe ServiceInstance (ProviderServiceInstance andConsumedServiceInstance) shouldbelongtothe same EthernetPhysicalChannel

Error scenario: Scenariois applicable tobothProviderServiceInstance andConsumedServiceInstance

PduTriggeringReferreddoes notbelongtothe same EthernetPhysicalChannel as thatofPduActivationRoutingGroup

5.175 ASW4X_SYS_5079 [Constr_5079]

Error Description

All the SocketAddress elements (ApplicationEndpointand/or StaticSocketConnection) thatare referencedwithin AbstractServiceInstance (ProvidedServiceInstance and/or ConsumedServiceInstance) shouldbelongtosame VLAN.

5.176 ASW4X_SYS_5080 [Constr_5080]

Error Description

The Vlanofthe applicationEndPointreferencedwithinProvidedServiceInstance andthe applicationEndPointreferencedwithinthe EventHandler shouldbelongtosame VLan.

5.177 ASW4X_SYS_5081 [Constr_5081]

Error Description

InProvidedServiceInstance, configuringthe attribute serviceIdentifier is mandatory

Note: This validationis applicable fromAR19-11andabove only. This validationshouldnotbe triggeredfor AR4.4version.

How tomake itAR19-11versionspecifictobe clarifiedduringimplementation.

How to Solve

Allow the user to setvalue.

5.178 ASW4X_SYS_5082 [Constr_5082]

Error Description

InConsumedServiceInstance, configuringthe attribute MajorVersionis mandatory

Note: This validationis applicable fromAR19-11andabove only. This validationshouldnotbe triggeredfor AR4.4version.

How to Solve

Allow the user tosetvalue.

5.179 ASW4X_SYS_5083 [Constr_5083]

Error Description

InConsumedServiceInstance, configuringthe attribute MajorVersionis mandatory

Note: This validationis applicable fromAR19-11andabove only. This validationshouldnotbe triggeredfor AR4.4version.

How to Solve

Allow the user tosetvalue.

5.180 ASW4X_SYS_5084 [Constr_5084]

Error Description

InConsumedServiceInstance, configuringthe attribute InstanceIdentifier is mandatory

Note: This validationis applicable fromAR19-11andabove only. This validationshouldnotbe triggeredfor AR4.4version.

How toSolve

Allow the user tosetvalue.

5.181 ASW4X_SYS_5085 [Constr_5085]

Error Description

ConsumedServiceInstance.LocalUnicastAddress

shouldpointtoIPunicastaddress.

How toSolve

Remove the Local unicastAddress (ApplicationEndpointRefConditional).

5.182 ASW4X_SYS_5086 [Constr_5086]

Error Description

ConsumedServiceInstance.remoteUnicastAddress

shouldpointtoIPunicastaddress.

How to Solve

Remove the Local unicastAddress (ApplicationEndpointRefConditional).

5.183 ASW4X_SYS_5087 [Constr_5087]

Error Description

Ifa PduActivationRoutingGrouphas the EventGroupControlType settoactivationUnicastor triggerUnicastor activationAndTriggerUnicastthenthe referencedby ConsumedEventGroupshouldconfigure localUnicastAddress that points toanIPUnicastaddress.

How to Solve

Populate all possible ApplicationEndPointwhichrefers toIPunicastaddress sothatuser canchoose the value. For the chosenapplicationEndPoint, create ApplicationEndpointRefConditional under ProviderServiceInstance andshouldpointtoApplicationEndPoint.

5.184 ASW4X_SYS_5088 [Constr_5088]

Error Description

Ifthe PduActivationRoutingGroupcontains the iPduIdentifierTcpreference thenthe aggregatingConsumedServiceInstance shouldcontaina localUnicastAddress reference toanApplicationEndpointthatdefines a TCPaddress.

How to Solve

Remove the reference inLocalunicastwhichdoes nothave TCPaddress andprovide optionfor user toconfigure ApplictionEndPointthathas TCPaddress.

5.185 ASW4X_SYS_5089 [Constr_5089]

Error Description

Ifthe PduActivationRoutingGroupcontains the iPduIdentifierTcpreference thenthe aggregatingConsumedServiceInstance shouldcontaina localUnicastAddress reference toanApplicationEndpointthatdefines a UDPaddress.

How to Solve

Remove the reference inLocalunicastwhichdoes nothave UDPaddress andprovide optionfor user toconfigure ApplictionEndPointthathas UDPaddress.

5.186 ASW4X_SYS_5090 [Constr_5090]

Error Description

The Vlanofthe applicationEndPointreferencedwithinConsumedServiceInstance andthe ApplicationEndPointreferencedwithinthe ConsumedEventGroups shouldbelongtosame VLan.

5.187 ASW4X_SYS_5091 [Constr_5091]

Error Description

The attribute tcpRole is only relevantifthe StaticSocketConnectionis aggregatedby a SocketAddress thatdefines a TCPPortinthe aggregatedApplicationEndpoint.

How to Solve

Remove the value for tcpRole.

5.188 ASW4X_SYS_5092 [Constr_5092]

Error Description

The transportprotocol thatis definedby the SocketAddress thataggregates the StaticSocketConnectionshall be the same inthe SocketAddress thatis referencedby the same StaticSocket-Connectioninthe role remoteAddress.

How to Solve

Remove the TPconfigurationfromSocketAddress <ShortName>

Remove the TPconfigurationfromremoteAddress <ShortName>.

5.189 ASW4X_SYS_5105 [Constr_5105]

Error Description

Only the lastPduina FlexrayFrame is allowedtobe a PduwithhasDynamicLength= true.

How to Solve

Setthe hasDynamicLengthtofalse

5.190 ASW422_SYS_3019 [Constr_3019]

Error Description

InFlatEcuExtract, all the VariableDataPrototypes specifiedby SenderReceiverInterface ofRPortPrototype are not connectedtoPPortPrototype.

How to Solve

Create connectionfor all the VariableDataPrototypes specifiedby SenderReceiverInterface ofRPortPrototype to PPortPrototype.

5.191 ASW4X_SYS_5177 [Constr_5177]

Error Description

Same software cluster thatis referencedwithinthe CpSoftwareClusterToEcuInstanceMapping.swCluster should alsobe referredwithinthe System.SwCluster.

How to Solve

Show the user the listofSwClusters toconfigure.

5.192 ASW4X_SYS_5178 [Constr_5178]

Error Description

For eachCpSoftwareClusterResource, value for globalResourceIdis mandatory.

How to Solve

Allow the user to enter a value.

5.193 ASW4X_SYS_5179 [Constr_5179]

Error Description

For eachCpSoftwareClusterResource, value for attribute ismandatory needtobe defined.

How to Solve

Allow the user toenter a value.

5.194 ASW4X_SYS_5180 [Constr_5180]

Error Description

The attribute global resource idofCpSoftwareClusterCommunicationResource or CpSoftwareClusterServiceResource withthe CpSoftwareClusterResource shouldnotbe settozero.

How to Solve

Allow the user toenter a value other thanzero.

5.195 ASW4X_SYS_5181 [Constr_5181]

Error Description

For eachCpSoftwareClusterServiceResource, value for attribute category needtobe defined.

How to Solve

Allow the user toenter a value.

5.196 ASW4X_SYS_5182 [Constr_5182]

Error Description

CpSoftwareClusterCommunicationResource mappedtoa PortElementToCommunicationResourceMappingis not allowedtorefer PRport.

How to Solve

Allow the user toconfigure the reference apartfromPRPort.

5.197 ASW4X_SYS_5183 [Constr_5183]

Error Description

PortElementToCommunicationResourceMappingshouldreference parameterDataPrototype or modeDeclarationGroupPrototype or trigger or ClientServerOperationor VariableDataPrototype.

5.198 ASW4X_SYS_5184 [Constr_5184]

Error Description

CpSoftwareClusterServiceResource is notallowedtobe mappedby more thanone CpSoftwareClusterToResourceMappingthatrefers CpSoftwareClusters inprovider role, mappedtosame EcuInstance.

5.199 ASW4X_SYS_5208 [Constr_5208]

Error Description

Same software cluster thatis referencedwithinthe CpSoftwareClusterToEcuInstanceMapping.swCluster should alsobe referredwithinthe System.SwCluster.

5.200 ASW4X_SYS_5209 [Constr_5209]

Error Description

SystemelementwithCategory SW_CLUSTER_SYSTEM_DESCRIPTION,shouldde-fine the reference System.swCluster.swComponentAssignmentswComponen-tatleastonce, atthe time whenthe software cluster extractionis finished.

How to Solve

Allow the user to select the available configurationfor swComponentAssign-mentswComponent.

5.201 ASW4X_SYS_5210 [Constr_5210]

Error Description

SystemelementwithCategory SW_CLUSTER_SYSTEM_DESCRIPTION,shouldcon-figure PortElementToCommunicationResourceMappingatleastonce atthe time whenthe software cluster extractionis finished.

5.202 ASW4X_SYS_5211 [Constr_5211]

Error Description

SystemelementwithCategory SW_CLUSTER_SYSTEM_DESCRIPTION,shouldcon-figure the attribute communicationresource withinPortElementToCommuni-cationResourceMappingatleastonce atthe time whenthe software cluster ex-tractionis finished.

How to Solve

Allow the user toselectthe available configurationfor communicationresource.

5.203 ASW4X_SYS_5212 [Constr_5212]

Error Description

SystemelementwithCategory SW_CLUSTER_SYSTEM_DESCRIPTION,shouldcon-figure CpSoftwareClusterResourceToApplicationPartitionMapping, atthe time whenthe software cluster extractionis finished.

5.204 ASW4X_SYS_5213 [Constr_5213]

Error Description

SystemelementwithCategory SW_CLUSTER_SYSTEM_DESCRIPTION,should-configure the attribute applicationPartitionwithinCpSoftwareClusterRe-sourceToApplicationPartitionMapping, atthe time whenthe software clus-ter extractionis finished.

How to Solve

Allow the user toselectthe available configurationfor applicationPartition.

5.205 ASW4X_SYS_5214 [Constr_5214]

Error Description

SystemelementwithCategory SW_CLUSTER_SYSTEM_DESCRIPTION,shouldconfigure the attribute resource withinCpSoftwareClusterResourceToApplicationPartitionMapping, atthe time whenthe software cluster extractionis finished.

How to Solve

Allow the user toselectthe available configurationfor resource.

5.206 ASW4X_SYS_5215 [Constr_5215]

Error Description

SystemelementwithCategory SW_CLUSTER_SYSTEM_DESCRIPTION,shouldconfigure the attribute serviceResource withinCpSoftwareClusterToResourceMappingatthe time whenthe software cluster extractionis finished.

How to Solve

Allow the user toselectthe available configurationfor service resource.

5.207 ASW4X_SYS_5216 [Constr_5216]

Error Description

SystemelementwithCategory SW_CLUSTER_SYSTEM_DESCRIPTION,shouldconfigure the wither the requester or provider withinCpSoftwareClusterToResourceMapping, atthe time whenthe software cluster extractionis finished.

5.208 ASW4X_SYS_5384 [Constr_5384]

Error Description

For each of the BusMirrorChannelMapping the attribute mirroringProtocol should be configured

How to Solve

For each of the BusMirrorChannelMapping element configure the value of mirroringProtocol to either VERSION-1 or NONE

5.209 ASW4X_SYS_5495_5496_5497 [Constr_5495_5496_5497]

Error Description

For each BusMirrorCanIdRangeMapping, the attributes destinationBaseId, sourceCanIdCode and the sourceCanIdMask should be configured

How to Solve

In BusMirrorCanIdRangeMapping configure the value of attributes destinationBaseId, sourceCanIdCode and sourceCanIdMask

5.210 ASW4X_SYS_5498 [Constr_5498]

Error Description

For each of the BusMirrorCanIdToCanIdMapping, the attribute remappedCanId should be configured

How to Solve

In BusMirrorCanIdToCanIdMapping configure the value of the attribute remappedCanId

5.211 ASW4X_SYS_5499 [Constr_5499]

Error Description

For each of the BusMirrorLinPidToCanIdMapping, the attribute remappedCanId should be configured

How to Solve

In BusMirrorLinPidToCanIdMapping configure the value of the attribute remappedCanId

5.212 ASW4X_ETH_0001 [Constr_3047]

Error Description

A MacMulticastAddress shall be unique ina particular EthernetCluster.

How to Solve

Reconfigure the MacMulticastAddress , sothatthey are unique inthe EthernetCluster.

5.213 ASW4X_ETH_0002 [Constr_3048]

Error Description

VlanIdentifier ina VlanConfiginanEthernetPhysicalChannel shouldbe between0to4095.

How to Solve

Reconfigure the VlanIdentifier, sothatitis between 0 to 4095.

5.214 ASW4X_ETH_0003 [Constr_3062]

Error Description

The EcuInstance thatis referencedfroma specificCouplingElementshall be connectedtothe same EthernetCluster as the specificCouplingElement.

How to Solve

Reconfigure the EthernetCluster sothatitis connectedtothe same ECU as specifiedinthe CouplingElement.

5.215 ASW4X_ETH_0004 [Constr_3133]

Error Description

The PhysicalLayerType the CouplingPorts whichare connectedby CouplingPortConnection, shouldbe always equal.

How to Solve

Reconfigure the PhysicalLayerType , sothatthey are equal for the CouplingPorts connectedby CouplingPortConnection.

5.216 ASW4X_ETH_0005 [Constr_3134]

Error Description

The ConnectionNegotiationBehavior oftwoCouplingPorts whichare connectedvia a CouplingPortConnectionshall notbe bothsettoMASTER.

How to Solve

Reconfigure the ConnectionNegotiationBehavior ofthe CouplingPorts , sothatthey are notequal.

5.217 ASW4X_ETH_0006 [Constr_3515]

Error Description

Ifthe CouplingPortDetails.ethernetPriorityRegenerationexists, thenitshoulcontainexactly 8EthernetPriorityRegeneration.

How to Solve

Delete the extra EthernetPriorityRegeneration, sothatthere are exactly 8EthernetPriorityRegeneration.

5.218 ASW4X_ETH_0007 [Constr_3133]

Error Description

EthernetFrame shall nothave a PduToFrameMapping

How to Solve

Delete the PduToFrameMappinginanEthernetFrame

5.219 ASW4X_ETH_0008 [Constr_3063]

Error Description

Usage ofPortNumber andDynamicallyAssignedwithvalue “true”is mutually exclusive.

How to Solve

Either setconfigure PortNumber or setDynamicallyAssignedto"true".

5.220 ASW4X_ETH_0009 [Constr_3212]

Error Description

DoIpTpConnectionshall only reference PduTriggerings ofDcmIPdus inthe role tpSdu.

How to Solve

Reconfigure sothat, the TpSdushouldrefer toa IPduTriggering, inwhichthe Ipduis a DcmIpdu.

5.221 ASW4X_ETH_0010 [Constr_3135]

Error Description

The connectionNegotiationBehavior oftwoCouplingPorts whichare connectedvia a CouplingPortConnectionshall notbe bothsettoslave

How to Solve

Reconfigure the ConnectionNegotiationBehavior ofthe CouplingPorts , sothat-they are notequal.

5.222 ASW421_ETH_0007[Constr-3113]

Error Description

A PDU cannotbe mappedtoa EthernetFrame

How to Solve

Remove the PDU mappingfor EthernetFrame

5.223 ASW4X_SYS_5106 [Constr_5106]

Error Description

Both the ISignalGroup and the ISignals that are referenced by that ISignalGroup, cannot be referenced within an ISignalTriggering

How to Solve

Remove the ISignals references from ISignalTriggering

5.224 ASW4X_SYS_5118 [constr 5118]

Error Description

If defined, the value of UdpProps.udpTtl shall be in the range of 1..255

How to Solve

Remove the ISignals references from ISignalTriggering

5.225 ASW4X_SYS_5119 [constr 5119]

Error Description

If defined, the value of TcpProps.tcpTtl shall be in the range of 1..255

How to Solve

Reconfigure the value of TcpProps.tcpDelayedAckTimeout in the range of 1..255.

5.226 ASW4X_SYS_5120 [constr 5120]

Error Description

If defined, the value of TcpProps.tcpDelayedAckTimeout shall be in the range of 0..0.5.

How to Solve

Reconfigure the value of TcpProps.tcpDelayedAckTimeout in the range of 0..0.5.

5.227 ASW4X_SYS_5121 [constr 5121]

Error Description

If defined, the value of TcpProps.tcpSynMaxRtx shall be in the range of 0..255.

How to Solve

Reconfigure the value of TcpProps.tcpSynMaxRtx in the range of 0..255

5.228 ASW4X_SYS_5122 [constr 5122]

Error Description

If defined, the value of TcpProps.tcpMaxRtx shall be in the range of 0..255

How to Solve

Reconfigure the value of TcpProps.tcpMaxRtx in the range of 0..255

5.229 ASW4X_SYS_5123 [constr 5123]

Error Description

If defined, the value of TcpProps.tcpKeepAliveProbesMax shall be in the range of 0..65535

How to Solve

Reconfigure the value of TcpProps.tcpKeepAliveProbesMax in the range of 0..65535

5.230 ASW4X_SYS_5124 [constr 5124]

Error Description

If defined, the value of TcpProps. tcpReceiveWindowMax shall be in the range of 0..65535

How to Solve

Reconfigure the value of TcpProps. tcpReceiveWindowMax in the range of 0..65535

5.231 ASW4X_SYS_5125 [constr 5125]

Error Description

If defined, the value of TcpIpIcmpv4Props.tcpIpIcmpV4Ttl shall be in the range of 1..255

How to Solve

Reconfigure the value of TcpIpIcmpv4Props.tcpIpIcmpV4Ttl in the range of 1..255

5.232 ASW4X_SYS_5126 [constr 5126]

Error Description

If defined, the value of Ipv4ArpProps.tcpIpArpNumGratuitousArpOnStartup shall be in the range of 0..255

How to Solve

Reconfigure the value of Ipv4ArpProps.tcpIpArpNumGratuitousArpOnStartup in the range of 0..255

5.233 ASW4X_SYS_5127 [constr 5127]

Error Description

If defined, the value of Ipv4FragmentationProps. tcpIpIpNumFragments shall be in the range of 0..255

How to Solve

Reconfigure the value of Ipv4FragmentationProps. tcpIpIpNumFragments in the range of 0..255.

5.234 ASW4X_SYS_5128 [constr 5128]

Error Description

If defined, the value of Ipv4FragmentationProps.tcpIpIpNumReassDgrams shall be in the range of 0..65535.

How to Solve

Reconfigure the value of Ipv4FragmentationProps.tcpIpIpNumReassDgrams in the range of 0..65535.

5.235 ASW4X_SYS_5129 [constr 5129]

Error Description

If defined, the value of Ipv6FragmentationProps.tcpIpIpReassemblyBuffer-Count shall be in the range of 0 to 255.

How to Solve

Reconfigure the value of Ipv6FragmentationProps.tcpIpIpReassemblyBuffer-Count in the range of 0..255.

5.236 ASW4X_SYS_5130 [constr 5130]

Error Description

If defined, the value of Ipv6FragmentationProps.tcpIpIpReassemblyBufferSize shall be in the range of **1500..65535.**

How to Solve

Reconfigure the value of Ipv6FragmentationProps.tcpIpIpReassemblyBuffer-Size in the range of 1500..65535.

5.237 ASW4X_SYS_5131 [constr 5131]

Error Description

If defined, the value of Ipv6FragmentationProps.tcpIpIpReassemblyTimeout shall be in the range of 0.001..100.

How to Solve

Reconfigure the value of Ipv6FragmentationProps.tcpIpIpReassemblyTimeout in the range of 0.001..100.

5.238 ASW4X_SYS_5132 [constr 5132]

Error Description

If defined, the value of Ipv6FragmentationProps.tcpIpIpReassemblySegmentCount shall be in the range of 1..255.

How to Solve

Reconfigure the value of Ipv6FragmentationProps.tcpIpIpReassemblySegmentCount in the range of 1..255.

5.239 ASW4X_SYS_5133 [constr 5133]

Error Description

If defined, the value of Ipv6FragmentationProps.tcpIpIpTxFragmentBufferCount shall be in the range of 1..1000.

How to Solve

Reconfigure the value of Ipv6FragmentationProps.tcpIpIpTxFragmentBufferCount in the range of 1..1000.

5.240 ASW4X_SYS_5134 [constr 5134]

Error Description

If defined, the value of Ipv6FragmentationProps.tcpIpIpTxFragmentBufferSize shall be in the range of 1500..65535.

How to Solve

Reconfigure the value of Ipv6FragmentationProps.tcpIpIpTxFragmentBufferSize shall be in the range of 1500..65535.

5.241 ASW4X_SYS_5135 [constr 5135]

Error Description

If defined, the value of Dhcpv6Props.tcpIpDhcpV6CnfDelayMin and the value of Dhcpv6Props.tcpIpDhcpV6CnfDelayMax shall be in the range of 0..100 and the value of

Dhcpv6Props.tcpIpDhcpV6CnfDelayMax shall be greater than the value of Dhcpv6Props.tcpIpDhcpV6CnfDelayMin.

How to Solve

Re configure the value of Dhcpv6Props.tcpIpDhcpV6CnfDelayMin and the value of Dhcpv6Props.tcpIpDhcpV6CnfDelayMax to be between 0 to 100 and with tcpIpDhcpV6CnfDelayMax having greater value

5.242 ASW4X_SYS_5136 [constr 5136]

Error Description

If defined, the value of Dhcpv6Props.tcpIpDhcpV6InfDelayMin and the value of Dhcpv6Props.tcpIpDhcpV6InfDelayMax shall be in the range of 0..100 and the value of Dhcpv6Props.tcpIpDhcpV6InfDelayMax shall be greater than the value of Dhcpv6Props.tcpIpDhcpV6InfDelayMin.

How to Solve

Reconfigure the value of tcpIpDhcpV6InfDelayMin and tcpIpDhcpV6InfDelayMax to be in the range 0 to 100, with tcpIpDhcpV6InfDelayMax having greater value

5.243 ASW4X_SYS_5137 [constr 5137]

Error Description

If defined, the value of Dhcpv6Props.tcpIpDhcpV6SolDelayMin and the value of Dhcpv6Props.tcpIpDhcpV6SolDelayMax shall be in the range of 0..100 and the value of Dhcpv6Props.tcpIpDhcpV6SolDelayMax shall be greater than the value of Dhcpv6Props.tcpIpDhcpV6SolDelayMin.

How to Solve

Reconfigure the value of tcpIpDhcpV6SolDelayMin and tcpIpDhcpV6Sol-DelayMax to be in the range 0 to 100, with tcpIpDhcpV6SolDelayMax having greater value

5.244 ASW4X_SYS_5138 [constr 5138]

Error Description

If defined, the value of Ipv6NdpProps.tcpIpNdpSlaacDadNumberOfTransmis-sions shall be in the range of 0..254.

How to Solve

Reconfigure the value of Ipv6NdpProps.tcpIpNdpSlaacDadNumberOfTransmis-sions shall be in the range of 0..254.

5.245 ASW4X_SYS_5139 [constr 5139]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpSlaacDadRetransmissionDelay-shall be in the range of 0..10.

How to Solve

Reconfigure the value of Ipv6NdpProps. tcpIpNdpSlaacDadRetransmissionDe-layshall be in the range of 0..10.

5.246 ASW4X_SYS_5140 [constr 5140]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpDefaultReachableTime be in the range of 0..120.

How to Solve

Reconfigure the value Ipv6NdpProps. tcpIpNdpDefaultReachableTime in the range of 0..120.

5.247 ASW4X_SYS_5141 [constr 5141]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpDefaultRetransTimer be in the range of 0..60.

How to Solve

Reconfigure the value of Ipv6NdpProps. tcpIpNdpDefaultRetransTimer be-tween 0 to 60

5.248 ASW4X_SYS_5142 [constr 5142]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpNumUnicastSolicitations be in the range of 0..255.

How to Solve

Reconfigure the value of Ipv6NdpProps. tcpIpNdpNumMulticastSolicitations in the range of 0..255.

5.249 ASW4X_SYS_5143 [constr 5143]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpNumMulticastSolicitations be in the range of 0..255.

How to Solve

Reconfigure the value Ipv6NdpProps. tcpIpNdpNumMulticastSolicitations in the range of 0..255

5.250 ASW4X_SYS_5144 [constr 5144]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpDelayFirstProbeTime be in the range of 0..60.

How to Solve

Reconfigure the value Ipv6NdpProps. tcpIpNdpDelayFirstProbeTime in the range of 0..60

5.251 ASW4X_SYS_5145 [constr 5145]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpMinRandomFactor be in the range of 0..100.

How to Solve

Reconfigure the value of Ipv6NdpProps. tcpIpNdpMaxRandomFactor in the range of 0..100.

5.252 ASW4X_SYS_5146 [constr 5146]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpMaxRandomFactor be in the range of 0..100.

How to Solve

Reconfigure the value of Ipv6NdpProps. tcpIpNdpMaxRandomFactor in the range of 0..100.

5.253 ASW4X_SYS_5147 [constr 5147]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpDestinationCacheSizebe in the range of 1..254

How to Solve

Reconfigure the value of Ipv6NdpProps. tcpIpNdpDestinationCacheSizebe in the range of 1..254.

5.254 ASW4X_SYS_5148 [constr 5148]

Error Description

If defined, the value of Ipv6NdpProps.tcpIpNdpPrefixListSize in the range of 1..254.

How to Solve

Reconfigure the value of Ipv6NdpProps.tcpIpNdpPrefixListSize in the range of 1..254.

5.255 ASW4X_SYS_5149 [constr 5149]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpDefaultRouterListSize in the range of 2..254.

How to Solve

Reconfigure the value of Ipv6NdpProps. tcpIpNdpDefaultRouterListSize in the range of 2..254.

5.256 ASW4X_SYS_5151 [constr 5151]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpMaxRtrSolicitations in the range of 0..255.

How to Solve

Reconfigure the value of Ipv6NdpProps. tcpIpNdpMaxRtrSolicitations between 0 to 255

5.257 ASW4X_SYS_5152 [constr 5152]

Error Description

If defined, the value of Ipv6NdpProps. tcpIpNdpMaxRtrSolicitations in the range of 0.001..60

How to Solve

Reconfigure the value of Ipv6NdpProps. tcpIpNdpMaxRtrSolicitations in the range of 0.001..60

5.258 ASW4X_SYS_5153 [constr 5153]

Error Description

If defined, the value of Ipv6NdpProps.tcpIpNdpRtrSolicitationInterval in the range of 0.001..60

How to Solve

Reconfigure the value of Ipv6NdpProps.tcpIpNdpRtrSolicitationInterval in the range of 0.001..60

5.259 ASW4X_SYS_5154 [constr 5154]

Error Description

If defined, the value of TcpIpIcmpv6Props. tcpIpIcmpV6HopLimit shall be in the range of 1..255

How to Solve

Reconfigure the value of TcpIpIcmpv6Props. tcpIpIcmpV6HopLimit in the range of 1..255

5.260 ASW4X_SYS_105_2011[constr_3153]

Error Description

Specific Properties are expected for Specific Profile Names of DataTransformation

How to Solve

Configure correct value for HeaderLength

Hint: Tooltip showing the range of bits for HeaderLength

5.261 ASW4X_ETH_0009[constr_3212]

Error Description

The DoIpTpConnection.tpSdu referes to a PduTriggering.

The IPdu referenced by that PduTriggering should be always a DcmIpdu.

How to Solve

Reconfigure the value of TpSdu to refer a IPduTriggering, in which the Ipdu is a DcmIpdu.

Hint: Tooltip showing the range of bits for HeaderLength

5.262 ASW4X_SYS_5176 [constr_5176]

Error Description

There should be only one CpSoftwareCluster of category HOST_SOFTWARE_CLUSTER on one Ecu Instance.

How to Solve

Only one CpSoftwareCluster of category HOST_SOFTWARE_CLUSTER should be mapped to one EcuInstance.

5.263 ASW4X_SYS_5185 [constr_5185]

Error Description

For each of the BinaryManifestProvideResource the value of the attribute globalResourceId should exist.

How to Solve

Configure the attribute globalResourceId for BinaryManifestProvideResource.

5.264 ASW4X_SYS_5186 [constr_5186]

Error Description

For each of the BinaryManifestProvideResource the value of the attribute resourceGuardValue should exist.

How to Solve

Configure the attribute resourceGuardValue for BinaryManifestProvideResource.

5.265 ASW4X_SYS_5187 [constr_5187]

Error Description

For each of the BinaryManifestProvideResource the value of the attribute supportsMultipleNotifierSets should exist.

How to Solve

Configure a valid reference supportsMultipleNotifierSets for BinaryManifestProvideResource.

5.266 ASW4X_SYS_5188 [constr_5188]

Error Description

For each of the BinaryManifestProvideResource the value of the attribute numberOfNotifierSets should exist.

How to Solve

Configure a valid reference numberOfNotifierSets for BinaryManifestProvideResource.

5.267 ASW4X_SYS_5189 [constr_5189]

Error Description

For each of the BinaryManifestProvideResource the value of the attribute resourceDefinition should exist.

How to Solve

Configure a valid reference resourceDefinition for BinaryManifestProvideResource.

5.268 ASW4X_SYS_5190 [constr_5190]

Error Description

For each of the BinaryManifestProvideResource the role item should exist.

How to Solve

Configure the element BinaryManifestItem within BinaryManifestProvideResource

5.269 ASW4X_SYS_5191 [constr_5191]

Error Description

For each of the BinaryManifestProvideResource the role item should exist.

How to Solve

Configure the element BinaryManifestItem within BinaryManifestProvideResource

5.270 ASW4X_SYS_5192 [constr_5192]

Error Description

For each of the BinaryManifestRequireResource the value of the attribute globalResourceId should exist.

How to Solve

Configure a unique value for the attribute globalResourceId within BinaryManifestRequireResource

5.271 ASW4X_SYS_5193 [constr_5193]

Error Description

For each of the BinaryManifestRequireResource the value of the attribute resourceGuardValue should exist.

How to Solve

Configure a unique value for the attribute resourceGuardValue within BinaryManifestRequireResource

5.272 ASW4X_SYS_5194 [constr_5194]

Error Description

For each of the BinaryManifestRequireResource the value of the attribute resourceDefinition should exist.

How to Solve

Configure a valid reference for resourceDefinition within BinaryManifestRequireResource

5.273 ASW4X_SYS_5195 [constr_5195]

Error Description

For each of the BinaryManifestRequireResource the role item should exist.

How to Solve

Configure the element BinaryManifestItem within BinaryManifestRequireResource

5.274 ASW4X_SYS_5196 [constr_5196]

Error Description

If BinaryManifestProvideResource.item.category is one of the following three values

Scenario 1:

PROVIDER_HANDLE - Here if the isUnused attribute is false then the BinaryManifestRequireResource.item.defaultValue should be set

How to Solve

Configure the element BinaryManifestItem within BinaryManifestRequireResource

5.275 ASW4X_SYS_5198 [constr_5198]

Error Description

The BinaryManifestResource.resourceDefinition should only be referenced from a BinaryManifestResource aggregated within the same CpSoftwareClusterBinaryManifestDescriptor

How to Solve

Configure the element BinaryManifestItem within BinaryManifestRequireResource

5.276 ASW4X_SYS_5199 [constr_5199]

Error Description

If attribute BinaryManifestItem.auxiliaryField.category is set to value AUXILARY_CONNECTED_SW_CLUSTER_ID then attribute BinaryManifestItem.auxiliaryField.defaultValue should be configured.

How to Solve

Configure the attribute BinaryManifestItem.auxiliaryField.defaultValue.

5.277 ASW4X_SYS_5200 [constr_5200]

Error Description

For every BinaryManifestItemDefinition, the value of the attribute category should be configured

How to Solve

Configure a valid Category for BinaryManifestItemDefinition

5.278 ASW4X_SYS_5201 [constr_5201]

Error Description

For every BinaryManifestItemDefinition, the value of the attribute size should be configured

How to Solve

Configure a valid size for BinaryManifestItemDefinition

5.279 ASW4X_SYS_5202 [constr_5202]

Error Description

For every BinaryManifestItem, the value of the attribute BinaryManifestitem-NumericalValue.value should be configured

How to Solve

Configure a value for BinaryManifestitemNumericalValue.value

5.280 ASW4X_SYS_5203 [constr_5203]

Error Description

The BinaryManifestItemPointerValue within BinaryManifestItem, does not have the value of the attribute Symbol configured

How to Solve

Configure the attribute Symbol for BinaryManifestItemPointerValue

5.281 ASW4X_SYS_5204 [constr_5204]

Error Description

For each of the BinaryManifestMetaDataField the value of the attribute category is not configured

How to Solve

For each of the BinaryManifestMetaDataField configure the value of the attribute category.

5.282 ASW4X_SYS_5205 [constr_5205]

Error Description

For each of the BinaryManifestMetaDataField the value of the attribute size should be configured.

How to Solve

For each of the BinaryManifestMetaDataField configure the value of the attribute size.

5.283 ASW4X_SYS_5206 [constr_5206]

Error Description

For each of the BinaryManifestMetaDataField the value of the attribute symbol is not configured

How to Solve

For each of the BinaryManifestMetaDataField, configure the value of the attribute symbol.

5.284 ASW4X_SYS_5207 [constr_5207]

Error Description

For each of the BinaryManifestMetaDataField the value of the attribute address should be configured.

How to Solve

For each of the BinaryManifestMetaDataField, configure the value of the attribute address.

5.285 ASW4X_SYS_5217 [constr_5217]

Error Description

For each of the BinaryManifestMetaDataField of category IMMUTABLE_TABLES_CHECKSUM the value of the attribute value should be configured.

How to Solve

For each of the BinaryManifestMetaDataField of category IMMUTABLE_TABLES_CHECKSUM, configure the value of the attribute value.

5.286 ASW4X_SYS_5218 [constr_5218]

Error Description

.The BinaryManifestItemPointerValue within BinaryManifestItem, should have the value of the attribute address configured.

How to Solve

Configure the attribute address for BinaryManifestItemPointerValue

5.287 ASW4X_SYS_5271 [constr_5271]

Error Description

For every BinaryManifestItem, the value of the attribute isUnused should be configured

How to Solve

Configure the element BinaryManifestItem within BinaryManifestRequireResource

5.288 ASW4X_SYS_5272 [constr_5272]

Error Description

For every BinaryManifestItem, the value of the attribute isUnused can be set to true only if BinaryManifestItemDefinition has its attribute isOptional set to true

How to Solve

Configure the value of attribute isUnused to false

5.289 ASW4X_SYS_3244 [constr_3244]

Error Description

The reference SenderRecRecordElementMapping.implementationRecordElement can only be configured if the corresponding SenderReceiverToSignalGroupMapping refers to a dataprototype <shortname> which is of type implementation data type

How to Solve

Reconfigure the variable data prototype referring to Implementation Datatype

5.290 ASW4X_SYS_3245 [constr_3245]

Error Description

The reference IndexedArrayElement.implementationArrayElement can only be configured if the corresponding SenderReceiverToSignalGroupMapping refers to a dataprototype <shortname> which is of type implementation data type

How to Solve

Reconfigure the variable data prototype referring to Implementation Datatype

5.291 ASW4X_CPFLEX_0001 [Constr_02130]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_NVM_BASE_SOCKET should be referred only from SwCluCNvMBaseSocket in the role of SwCluCResourceRef but it is referred from other than SwCluCNvMBaseSocket.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_NVM_BASE_SOCKET shall be referred from appropriate SwCluCNvMBaseSocket BSW container.

5.292 ASW4X_CPFLEX_0001 [Constr_02131]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_NV_BLOCK- should be referred only from SwCluCNvMProxyNvBlock in the role of SwCluCResourceRef but it is referred from other than SwCluCNvMProxyNvBlock.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_NV_BLOCK should shall be referred from appropriate SwCluCNvMProxyNvBlock BSW container.

5.293 ASW4X_CPFLEX_0001 [Constr_02664]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_COM_BASE_SOCKET should be referred only from SwCluCComProxyBaseSocket in the role of SwCluCResourceRef but it is referred from other than SwCluCComProxyBaseSocket.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_COM_BASE_SOCKET should shall be referred from appropriate SwCluCComProxyBaseSocket BSW container.

5.294 ASW4X_CPFLEX_0001 [Constr_02553]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_LDCOM_BASE_SOCKET should be referred only from SwCluCLdComProxyBaseSocket in the role of SwCluCResourceRef but it is referred from other than SwCluCLdComProxyBaseSocket.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_LDCOM_BASE_SOCKET should shall be referred from appropriate SwCluCLdComProxyBaseSocket BSW container.

5.295 ASW4X_CPFLEX_0001 [Constr_02752]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DCM_BASE_SOCKET should be referred only from SwCluCDcmProxyBaseSocket in the role of SwCluCResourceRef but it is referred from other than SwCluCDcmProxyBaseSocket.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DCM_BASE_SOCKET should shall be referred from appropriate SwCluCDcmProxyBaseSocket BSW container.

5.296 ASW4X_CPFLEX_0001 [Constr_02427]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DIAGNOSTIC_DATA should be referred only from SwCluCDcmProxyDiagnosticData in the role of SwCluCResourceRef but it is referred from other than SwCluCDcmProxyDiagnosticData.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DIAGNOSTIC_DATA should shall be referred from appropriate SwCluCDcmProxyDiagnosticData BSW container.

5.297 ASW4X_CPFLEX_0001 [Constr_02428]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DCM_ROUTINE_CONTROL should be referred only from SwCluCDcmProxyRoutineControl in the role of SwCluCResourceRef but it is referred from other than SwCluCDcmProxyRoutineControl.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DCM_ROUTINE_CONTROL should shall be referred from appropriate SwCluCDcmProxyRoutineControl BSW container.

5.298 ASW4X_CPFLEX_0001 [Constr_02429]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DCM_SERVICE_REQUEST_NOTIFICATION should be referred only from SwCluCDcmProxyServiceRequestNotification in the role of SwCluCResourceRef but it is referred from other than SwCluCDcmProxyServiceRequestNotification.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DCM_SERVICE_REQUEST_NOTIFICATION should shall be referred from appropriate SwCluCDcmProxyServiceRequestNotification BSW container.

5.299 ASW4X_CPFLEX_0001 [Constr_02720]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DEM_BASE_SOCKET should be referred only from SwCluCDemProxyBaseSocket in the role of SwCluCResourceRef but it is referred from other than SwCluCDemProxyBaseSocket.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DEM_BASE_SOCKET should shall be referred from appropriate SwCluCDemProxyBaseSocket BSW container.

5.300 ASW4X_CPFLEX_0001 [Constr_02721]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DIAGNOSTIC_EVENT should be referred only from SwCluCDemProxyDiagnosticEvent in the role of SwCluCResourceRef but it is referred from other than SwCluCDemProxyDiagnosticEvent.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_DIAGNOSTIC_EVENT should shall be referred from appropriate SwCluCDemProxyDiagnosticEvent BSW container.

5.301 ASW4X_CPFLEX_0001 [Constr_03210]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_FIM_BASE_SOCKET should be referred only from SwCluCFiMProxyBaseSocket in the role of SwCluCResourceRef but it is referred from other than SwCluCFiMProxyBaseSocket.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_FIM_BASE_SOCKET should shall be referred from appropriate SwCluCFiMProxyBaseSocket BSW container.

5.302 ASW4X_CPFLEX_0001 [Constr_03211]

Error Description

The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_FID should be referred only from SwCluCFiMProxyFID in the role of SwCluCResourceRef but it is referred from other than SwCluCFiMProxyFID.

How to Solve

- The category of the CpSoftwareClusterServiceResource shall be modified to appropriate category based on the BswContainer it is referred from.
- The CpSoftwareClusterServiceResource of category SWCLUSTER_RES_FID should shall be referred from appropriate SwCluCFiMProxyFID BSW container.

5.303 ASW4X_MAC_5361[Constr_5361]

Error Description

Only a CouplingElement with couplingType set to switch is allowed to aggregate a CouplingPort that in turn aggregates the MacSecProps in the role macSecProps.

How to Solve

Reconfigure type of the CouplingElement to SWITCH

5.303 ASW4X_MAC_3726[Constr_3726]

Error Description

The configuration of MacSecProps within the CouplingPort can exist only once.

How to Solve

Retain only one MacSecProps configuration and remove the rest of the MacSecProps

5.304 ASW4X_SYS_5334[Constr_5334]

Error Description

The values defined for CryptoServiceKey.length shall be multiple of 8.

How to Solve

Configure the value CryptoServiceKey.length to be multiple of 8.

5.305 ASW4X_SYS_9206[Constr_9206]

Error Description

For each CryptoServiceKey, the attribute length shall exist.

How to Solve

Configure the value CryptoServiceKey.length to be multiple of 8.

6. BSWMDT Validations

This sectiondeals withthe error descriptionofBSWMDT validations. BSWMDT Validations are specifictoBSWMD template inautosar.

6.1 ASW4X_BSWMD_0001 [Constr_4020]

Error Description

Ina BswModuleDescriptionthe value ofthe attribute "Category"shouldbe any ofthe followingBSW_MODULE, BSW_CLUSTER, LIBRARY.Other values or anempty value are notallowed.

How to Solve

Reconfigure the value ofthe category toany ofthe followingBSW_MODULE, BSW_CLUSTER, LIBRARY.

6.2 ASW4X_BSWMD_0002 [Constr_4036]

Error Description

1. Ina BswModuleDescription, the ProvidedEntries containingthe BswModuleEntryRefConditional havingreference BSWModuleEntry, the value ofthe attribute “callType”shouldbe CALLBACK..
2. Ina BswModuleDescription, the OutgoingCallBacks containingthe BswModuleEntryRefConditional havingreference BSWModuleEntry, the value ofthe attribute “callType”shouldbe CALLBACK

How to Solve

Reconfigure the value ofthe attribute callType tobe CALLBACK.

6.3 ASW4X_BSWMD_0003 [Constr_4014]

a) Error Description

In a BswModuleEntry, if the value of the attribute "callType" is INTERRUPT, then the value of the attribute executionContextshouldbe other thanTASK or HOOK.

How to Solve

Reconfigure the value ofthe attribute callType

tobe Other thanTASK or HOOK.

b) Error Description

Ina BswModuleEntry, if the value of the attribute "callType" is SCHEDULED, then the value of the attribute executionContextshouldbe other thanINTER-RUPT-CAT1or INTERRUPT-CAT2.

How to Solve

Reconfigure the value ofthe attribute callType tobe Other thanINTERRUPT-CAT1or INTERRUPT-CAT2.

6.4 ASW4X_BSWMD_0004 [Constr_4052]

Error Description

Ina BswModuleEntry, containingthe ReturnType_SwServiceArgthe value ofthe attribute "Direction"cannotbe IN or INOUT.

How to Solve

Reconfigure the value ofDirectiontoOUT.

6.5 ASW4X_BSWMD_0005 [Constr_4039]

Error Description

Ina SwcBaseMapping, the value ofthe reference SwcBehavior shouldhave a componenttype ServiceSwComponentType, EcuAbstractionSwCompo-nentType or ComplexDeviceDriverSwComponentType.

How to Solve

Reconfigure the value ofthe SwcBehavior thatis configuredinthe compo-nenttype ofServiceSwComponent-Type, EcuAbstractionSwComponentType or ComplexDeviceDriverSwComponentType.

6.6 ASW4X_BSWMD_0006 [Constr_4040]

Error Description

1. Ina SwcBswSynchronizedModeGroupPrototype, ModeDeclarationGroupPrototypes referredby BswModeGroup andSwcModeGroup(via TargetModeGroupofPModeGroupInAtomicSwcInstanceRef) shouldrefer tosame ModeDeclarationGroups.
2. SwcBswSynchronizedModeGroupPrototype canonly refer toModeDeclarationGroupPrototypes, whichare referredtosame ModeDeclarationGroups.

How to Solve

1. Delete SwcBswSynchronizedModeGroupPrototype.
2. Re configure the ModeDeclarationGroupPrototype ofBswModeGroupandSwcModeGrouptobe identical.
3. Re configure the ModeDeclarationGroupPrototype ofBswModeGrouptobe withinthe ProvidedModeGroups.

6.7 ASW4X_BSWMD_0007 [Constr_4043]

Error Description

Ina BswInternalBehavior, the attribute "Period"ofBswTimingEventis notgreater thanzero.

How to Solve

Reconfigure the Periodvalue tobe greater thanzero.

6.8 ASW4X_BSWMD_0008 [Constr_4017]

a) Error Description

Ina BswSchedulableEntity, the attribute ImplementedEntry referingtoBSWModuleEntry shouldhave anattribute callType settoSCHEDULED.

How to Solve

Reconfigure the BSWModuleEntry referedinImplmentedentry tohave anattribute callType settoSCHEDULED. **b) Error Description**

Ina BswSchedulableEntity, the attribute ImplementedEntry referingtoBSWModuleEntry shouldhave anattribute executionContextsettoTASK.

How to Solve

Reconfigure the BSWModuleEntry referedinImplmentedentry tohave anattribute executionContextsettoTASK.

6.9 ASW4X_BSWMD_0009 [Constr_4016]

Error Description

In a BswCalledEntity, the attribute ImplementedEntry refering to BSWModuleEntry Should have an attribute callType settoSCHEDULED.

How to Solve

Reconfigure the BSWModuleEntry refered in Implmented entry to have an attribute executionContext set to REGULARor CALLBACK.

6.10 ASW4X_BSWMD_0010 [Constr_4018]

a) Error Description

In a BswInterruptEntry, the attribute ImplementedEntry refering to BSWModuleEntry should have an attribute "callType"is settoINTERRUPT.

How to Solve

econfigure the BSWModuleEntry referedinImplmentedentry tohave anattribute callType settoINTERRUPT.

b) Error Description

In a BswInterruptEntry, the attribute ImplementedEntry refers to BSWModuleEntry whose attribute "executionContext" can be set to INTERRUPT-CAT1, if and only if the attribute "interruptCategory" in BSWInterruptEntry is settoCAT-1.

How to Solve

Reconfigure the value ofInterruptCategory tobe settoCAT1.

c) Error Description

In a BswInterruptEntry, the attribute ImplementedEntry refers to BSWModuleEntry whose attribute "executionContext" can be set to INTERRUPT-CAT2, if and only if the attribute "interruptCategory" in BSWInterruptEntry is settoCAT-2.

How to Solve

Reconfigure the value ofInterruptCategory tobe settoCAT2.

6.11 ASW4X_BSWMD_0011 [Constr_4064]

Error Description

The value of the attribute SwImplPolicy of SwcBswSynchronizedTrigger. swcTrigger and the value of swImplPolicy ofSwcBswSynchronizedTrigger.bswTrigger shouldbe notidentical.

How to Solve

Reconfigure the SwImplPolicy ofthe BswTrigger tobe identical withswcTrigger.

6.12 ASW4X_BSWMD_0012 [Constr_4015]

a) Error Description

Ina BswCalledEntity the reference ImplementedEntry has the value BswModuleEntry, This BswModuleEntry can have the value ofthe attribute “executionContext”as INTERRRUPT-CAT2, ifonly the value ofthe BswModuleEntry.executionContextofBswModuleEntryRefConditional is same.

How to Solve

Reconfigure the value of"executionCon-

text"ofBswModuleEntry as INTERRUPT-CAT-2

b) Error Description

Ina BswCalledEntity the reference “ImplementedEntry”has the value BswModuleEntry, This BswModuleEntry can have the value ofthe attribute “callType”as CALLBACK or REGULAR.

How to Solve

Reconfigure the value of"callType"ofBswModuleEntry as CALLBACK or REGULAR.

6.13 ASW4X_BSWMD_0013 [Constr_4063]

Error Description

For every ModeDeclarationGroup referenced by a ModeDeclarationGroupPrototype used in a BswModuleDescriptiona ModeRequestTypeMapshall existthatpoints tothe ModeDeclarationGroupand alsotoan eligible ImplementationDataType. The ModeRequestTypeMapshall be aggregatedby a DataTypeMappingSetwhichis referencedfromthe BswInternalBehavior thatis aggregatedby the BswModuleDescription.

How to Solve

Create a ModRequestTypeMapinside the DataTypeMappingSettocontaina reference tothe ModeDeclarationGroupandanImplementationDatatype.

6.14 ASW4X_BSWMD_0014 [Constr_4041]

a) Error Description

In SwcBswMapping, for a SwcBswSynchronizedModeGroupPrototype the bswComponent providing the bswModeGroupshouldbe same as the bsw-Componentcontainingthe bswBehavior.

How to Solve

Reconfigure the bswModeGrouptohave the a ModeDeclarationPrototype whose enclosingcomponentis same as the componentcontainingthe bswInternalBehavior. **b) Error Description**

In SwcBswMapping, for a SwcBswSynchronizedModeGroupPrototype the SwcComponent providing the TargetModeGroupshouldbe same as the Swc-Componentcontainingthe swcInternalBehavior.

How to Solve

Reconfigure the TargetModeGrouptohave the a ModeDeclarationPrototype whose enclosingcomponentis same as the componentcontainingthe swcInternalBehavior.

6.15 ASW4X_BSWMD_0015 [Constr_4042]

a) Error Description

In SwcBswMapping for a SwcBswSynchronizedTrigger, the bsw component providing the bswTrigger should be same as the bswComponentcontainingbswBehavior.

How to Solve

Reconfigure the bswTrigger to have the a Trigger whose enclosing component is same as the component containingthe bswInternalBehavior. **b) Error Description**

In SwcBswMapping containing the SwcBswSynchronizedTrigger the swc component providing the swcTrigger shouldbe same as the swcComponentcontainingswcBehavior.

How to Solve

Reconfigure the swcTrigger to have the a Trigger whose enclosing component is same as the component containingthe swcInternalBehavior.

6.16 ASW4X_BSWMD_0016 [Constr_4060]

Error Description

The only allowedvalues for the attribute Trigger.swImplPolicy are either STANDARD (inwhichcase the Trigger processingdoes notuse a queue) or QUEUED (inwhichcase the processingofTriggers positively uses a queue).

How to Solve

1. Delete the value ofattribute SwImplPolicy.
2. Reconfigure the value ofthe attribute SwImplPolicy tobe either QUEUED or STANDARD.

6.17 ASW4X_BSWMD_0017 [Constr_4038]

a) Error Description

InBswModuleDescriptionthe BswModuleDescription.bswModuleDependency.targetModuleId(ifgiven) mustdiffer fromBswModuleDescription.moduleId. This does notholdifthe value is 254(usedfor IOHardware Abstractionmodules) or 255(usedfor Complex Driver modules).

How to Solve

1. Delete the value ofTargetModuleId.
2. Reconfigure the TargetModuleIdofBswModuleDependency andModuleIdofBswModuleDescriptiontohave unique values.

b) Error Description

InBswModuleDescriptionthe BswModuleDescription.bswModuleDependency.targetModuleId(ifgiven) mustdiffer fromBswModuleDescription.moduleId. This does notholdifthe value is 254(usedfor IOHardware Abstractionmodules) or 255(usedfor Complex Driver modules).

How to Solve

1. Delete the value ofTargetModuleId
2. Reconfigure the TargetModuleIdofBswModuleDependency tohave a value other than254or 255.

c) Error Description

InBswModuleDescriptionthe BswModuleDependency.targetModuleRef(ifgiven) mustdiffer fromthe package locationofthe BswModuleDescriptionthatowns the BswModuleDependency.

How to Solve

1. Delete the value ofBswModuleDescription
2. Reconfigure the value ofBswModuleDependency.targetModuleRefwitha BswModuleDescriptionthatdoes not ownthe same BswModuleDependency.

6.18 ASW4X_BSWMD_0018 [Constr_4037]

a) Error Description

InBswModuleDescriptioncontainingBswModuleDependency , the Required Entry BswModuleEntryRefConditional refers toBswModuleEntry whichhas anattribute "CallType"whichshouldbe settoREGULAR. **How to Solve**

Reconfigure the BswModuleEntry tohave the value ofthe

attribute CallType as REGULAR. **b) Error Description**

InBswModuleDescriptioncontainingBswModuleDependency, the Expected-Callback BswModuleEntryRefConditional refers toBswModuleEntry whichhas anattribute "CallType"whichshouldbe settoCALLBACK. **How to Solve**

Reconfigure the BswModuleEntry tohave the value ofthe attribute CallType as CALLBACK.

6.19 ASW4X_BSWMD_0019 [Constr_4065]

Error Description

Ina BswInternalBehavior containingthe BswInternalTriggeringPoint, the allowedvalue for the attribute "SwImplyPolicy"is either STANDARD or QUEUED

How to Solve

1. Delete the value for the attribute SwImplPolicy.
2. Reconfigure the value for the attribute ""SwImplyPolicy""toeither STANDARD or QUEUED.

6.20 ASW4X_BSWMD_0020 [Constr_4022]

a) Error Description

BswModuleEntity.implementedEntry mustrefer toanelementdeclaredas providedEntry or as bswModuleDependency expectedCallback ofthe enclosingBswModuleDescription.

How to Solve

Configure the attribute implementedEntry toprovidedEntry or as bswModuleDependency.expectedCallback ofthe enclosingBswModuleDescription.

b) Error Description

In BswCalledEntity containing BswDirectCallPoint,the reference callEntry should refer to a BswModuleEntry declared as outgoingCallback or providedEntry or declared as requiredEntry in ModuleDependency of enclosing BswModuleDescription.

How to Solve

Reconfigure the callEntry to refer a BswModuleEntry declared as outgoingCallback or providedEntry or declared as requiredEntry inModuleDependency ofenclosingBswModuleDescription.

c) Error Description

InBswCalledEntity containingBswAsynchronousServerCallPointthe reference callEntry shouldrefer toa BswModuleEntry thatis declaredas BswModuleClientServerEntry ofenclosingBswModuleDescription.

How to Solve

Reconfigure the callEntry torefer a BswModuleEntry thatis declaredas BswModuleClientServerEntry ofenclosing BswModuleDescription. **d) Error Description**

InBswCalledEntity containingBswSynchronousServerCallPoint the reference callEntry should refer to a BswModuleEntry thatis declaredas BswModuleClientServerEntry ofenclosingBswModuleDescription.

How to Solve

Reconfigure the callEntry torefer a BswModuleEntry thatis declaredas BswModuleClientServerEntry ofenclosing BswModuleDescription. **e) Error Description**

InBswCalledEntity containingBswAsynchronousServerCallResultPointthe reference AsynchronousServerCallPointshouldrefer toBswAsynchronousServerCallPointwhichcanbe configuredinthe same BswCalledEntity.

How to Solve

Configure the AsynchronousServerCallPointtorefer toBswAsynchronousServerCallPointwhichcanbe configured inthe same BswCalledEntity. **f) Error Description**

InBswCalledEntity containingBswAsynchronousServerCallResultPointthe reference AsynchronousServerCallPointshouldrefer toBswAsynchronousServerCallPointwhichcanbe configuredinthe same BswCalledEntity.

How to Solve

Configure the AsynchronousServerCallPoint to refer to BswAsynchronousServerCallPoint that is present in the same BswCalledEntity.

g) Error Description

InBswCalledEntity containingIssuedTriggers the reference Trigger shouldrefer ReleasedTrigger ofthe enclosing BswModuleDescription

How to Solve

Reconfigure the reference Trigger torefer toelementReleasedTrigger ofthe enclosingBswModuleDescription. **h) Error Description**

InBswCalledEntity containingManagedModeGroupthe reference ModeDeclarationGroupPrototype shouldrefer to elementdeclaredas providedModeGroupofthe enclosingBswModuleDescription.

How to Solve

Reconfigure the reference ModeDeclarationGroupPrototype torefer toelementdeclaredas providedModeGroup ofthe enclosingBswModuleDescription **i) Error Description**

InBswCalledEntity containingManagedModeGroupthe reference ModeDeclarationGroupPrototype shouldrefer to elementdeclaredas requiredModeGroupofthe enclosingBswModuleDescription.

How to Solve

Reconfigure the reference ModeDeclarationGroupPrototype to refer to element declared as requiredModeGroup ofthe enclosingBswModuleDescription. **j) Error Description**

In BswCalledEntity containing DataSendPoints the reference AccessedVariable should refer to an element VariableDataPrototype whichis declaredas ProvidedData ofthe enclosingBswModuleDescription.

How to Solve

Reconfigure the AccessedVariable torefer toVariableDataPrototype whichis declaredas ProvidedData ofthe enclosingBswModuleDescription. **k) Error Description**

InBswCalledEntity containingDataReveivePoints the reference AccessedVariable shouldrefer toanelementVariableDataPrototype whichis declaredas RequiredData ofthe enclosingBswModuleDescription

How to Solve

Reconfigure the AccessedVariable to refer to VariableDataPrototype which is declared as RequiredData of the enclosingBswModuleDescription. **l) Error Description**

InBswCalledEntity containingManagedModeGroupthe reference ModeDeclarationGroupPrototype shouldrefer to elementdeclaredas providedModeGroupofthe enclosingBswModuleDescription.

How to Solve

Reconfigure the reference ModeDeclarationGroupPrototype to refer to element declared as providedModeGroup ofthe enclosingBswModuleDescription.

6.21 ASW4X_BSWMD_0021 [Constr_4023]

Error Description

A BswExternal- TriggerOccurredEvent must refer to a Trigger that is declared via BswModuleDescriptor requiredTrigger for the same module.

How to Solve

1. Delete the value for the reference Trigger.
2. Reconfigure the value ofthe reference Trigger torefer a value whichbelongs tothe same BswModuleDescription

6.22 ASW4X_BSWMD_0022 [Constr_4029]

a) Error Description

Ina BswImplementationcontainingthe ResourceConsumption, inMeasured-StackUsage the value ofthe attribute "MinimumMemory Consumption"shouldbe less than"AverageMemoryConsumption"

How to Solve

1. Delete the value ofMinimumMemoryConsumption
2. Reconfigure the value ofMinimumMemoryConsumptiontobe less thanAverageMemoryConsumption.

b) Error Description

Ina BswImplementationcontainingthe ResourceConsumption, inMeasured-StackUsage the value ofthe attribute "AverageMemoryConsumption"shouldbe less than"MaximumMemory Consumption".

How to Solve

Reconfigure the value ofAverageMemoryConsumptiontobe less

thanMaximumMemoryConsumption.

c) Error Description

Ina BswImplementationcontainingthe ResourceConsumption,inMeasured-StackUsage the value ofthe attribute "AverageMemoryConsumption"shouldbe less than"MaximumMemory Consumption".

How to Solve

Reconfigure the value ofAverageMemoryConsumptiontobe less thanMaximumMemoryConsumption.

6.23 ASW4X_BSWMD_0023 [Constr_4030]

a) Error Description

Ina BswImplementationcontainingthe ResourceConsumption, inMeasuredHeapUsage the value ofthe attribute "MinimumMemory Consumption"shouldbe less than"AverageMemoryConsumption".

How to Solve

1. Delete the value ofMinimumMemoryConsumption
2. Reconfigure the value ofMinimumMemoryConsumptiontobe less thanAverageMemoryConsumption. **b) Error Description**

Ina BswImplementationcontainingthe ResourceConsumption, inMeasuredHeapUsage the value ofthe attribute "AverageMemoryConsumption"shouldbe less than"MaximumMemory Consumption".

How to Solve

Reconfigure the value ofAverageMemoryConsumptiontobe less

thanMaximumMemoryConsumption. **c) Error Description**

In a BswImplementation containing the ResourceConsumption,in MeasuredHeapUsage the value of the attribute "AverageMemoryConsumption"shouldbe less than"MaximumMemory Consumption".

How to Solve

Reconfigure the value ofAverageMemoryConsumptiontobe less thanMaximumMemoryConsumption.

6.24 ASW4X_BSWMD_0024 [Constr_4059]

a) Error Description

In BswModuleDescription containing the providedModeGroups refers to ModeDeclarationGroup, the same ModeDeclarationGroupshouldnotbe referedinrequiredModeGroups withthe same name andvise-versa.

How to Solve

Reconfigure the shortname ofModeDeclarationGroups referedby providedModeGroups and requiredModeGroups tohave a unique shortname. **b) Error Description**

In BswModuleDescription containing the requiredModeGroups refers to ModeDeclarationGroup, the same ModeDeclarationGroupshouldnotbe referedinprovidedModeGroups withthe same name.

How to Solve

Reconfigure the shortname ofModeDeclarationGroups referedby provided-ModeGroups and requiredModeGroups tohave a unique shortname.

6.25 ASW4X_BSWMD_0025 [Constr_4048]

a) Error Description

Ina BswImplementationthe reference PreconfiguredConfigurations canrefer toone or more only ifcategory of BswModuleDescriptionis BSW-CLUSTER.

How to Solve

Reconfigure the category ofBswMod-

uleDescriptionis BSW-CLUSTER. **b) Error**

Description

In a BswImplementation the reference PreconfiguredConfigurations cannot refer to BswImplementation which does notbelongtothe same Module.

How to Solve

Reconfigure the reference PreconfiguredConfigurations tohave a value that-belongs tosame Module.

6.26 ASW4X_BSWMD_0026 [Constr_4047]

Error Description

Ina BswImplementationthe reference VendorSpecificmoduleDefs cannotre-fer toone or more ModeDef, ifcategory ofunderlyingBswModuleDescrip-tionis notBSW-CLUSTER

How to Solve

1. Delete the values referedinVendorSpecificmodules.
2. Reconfigure the ModDefinthe vendorSpecificmodules.

6.27 ASW4X_BSWMD_0027 [Constr_4044]

a) Error Description

Ina BswImplementationcontainingMcSupportData configured-withMcSwEmulationMethodSupport, the baseReference shouldbe configuredwhenthe value ofthe category is DOUBLE_POINTERED.

How to Solve

Configure the value ofbase reference toa

validVariableDataPrototype.

b) Error Description

In a BswImplementation containing McSupportData configured with McSwEmulationMethodSupport, the ReferenceTable shouldbe configuredwhenthe value ofthe category is SINGLE_POINTERED.

How to Solve

Configure the value ofReferenceTable toa

validVariableDataPrototype. **c) Error De-**

scription

Ina BswImplementationcontainingMcSupportData configured-withMcSwEmulationMethodSupport, 1or more elementGroups shouldbe configuredwhenthe value ofthe category is INITIALIZED_RAM.

How to Solve

Configure one or more McParameterElementGroupinMcSwEmulation-MethodSupport.

6.28 ASW4X_BSWMD_0028 [Constr_4026]

Error Description

In a BswModeSwitchedAckEvent, the reference ModeGroup should refer to a ModeDeclarationGroupPrototype configuredas providedModeGroups inside the BSWModuleDescription.

How to Solve

Configure the modeGrouptoModeDeclarationGroupPrototype configuredas providedModeGroups inthe BSwModulDescription.

6.29 ASW4X_BSWMD_0029 [Constr_4025]

Error Description

In a BswModeSwitchEvent,the ModeDeclaration should be configured to a ModeDeclarationGroupPrototype presentinaccessedModeGroupinside BswModuleEntity.

How to Solve

Reconfigure the TargetMode tocontainModeDeclarationthatis presentinaccessedModeGroupinside BswModuleEntity.

6.30 ASW4X_BSWMD_0030 [Constr_4021]

a) Error Description

Ina SwServiceArgcontainingthe SwPointerTargetProps, the value ofthe reference FunctionPointerSignature shouldbe configuredtoBswModuleEntry inwhich, the attribute "SwServiceImplPolicy"shouldbe configuredto STANDARD.

How to Solve

Configure the value ofthe attribute SwServiceImplPolicy

inBswModuleEntry toSTANDARD. **b) Error Description**

Ina SwServiceArgcontainingthe SwPointerTargetProps, the value ofthe reference FunctionPointerSignature shouldbe configuredtoBswModuleEntry inwhich, the attribute "SwServiceImplPolicy"shouldbe configuredto STANDARD.

How to Solve

Reconfigure the value ofthe attribute SwServiceImplPolicy inBswModuleEntry toSTANDARD.

6.31 ASW4X_BSWMD_0031 [Constr_4045]

Error Description

In a BswImplementation the reference PreConfiguredConfigurations containing EcucModuleConfigurationValues, the reference implementConfigVariantshouldbe configuredtoPRECONFIGURED-CONFIGURATION.

How to Solve

Configure the reference implementConfigVariant inside EcucModuleConfigurationValues to PRECONFIGUREDCONFIGURATION.

6.32 ASW4X_BSWMD_0032 [Constr_4046]

Error Description

Ina BswImplementationthe reference RecommendedConfigurations containingEcucModuleConfigurationValues, the reference implementConfigVariantshouldbe configuredtoRECOMMENDED-CONFIGURATION.

How to Solve

Configure the reference implementConfigVariant inside EcucModuleConfigurationValues to RECOMMENDEDCONFIGURATION.

6.33 ASW4X_BSWMD_0033 [Constr_4034]

Error Description

InMcDataInstance containingthe ImplementationElementInParameterInstanceRefthe value ofthe targetshould be a subelementofthe ImplementationDatatype referedinthe contextParameterDataPrototype.

How to Solve

Reconfigure the value ofTargettocontaina ImplementationDataTypeElementofthe ImplementationDataType referedinthe ParameterDataPrototype.

6.34 ASW4X_BSWMD_0034 [Constr_4053]

Error Description

Ina BswModuleEntry containingthe SwServiceArg,ifthe attribute "Direction"is configuredtoOUT,withthe SwDataDefPropsConditional referingthe ImplementationDatatype,thencategory ofImplementationDatatype shouldbe configuredtoone ofthe following: <DATA_REFERENCE>, <FUNCTION_REFERENCE>, <TYPE_ REFERENCE>.

How to Solve

Reconfigure the category of the ImplementationDataType to one of the following <DATA_REFERENCE>, <FUNCTION_REFERENCE>, <TYPE_REFERENCE>.

6.35 ASW4X_BSWMD_0035 [Constr_4019]

a) Error Description

BswModuleDescription.moduleIdshall refer tothe identifier ofthe standardizedAUTOSARmodules. This identifier canalso be used to distinguish modules which are notstandardized (i.e. if they belong to the ECU Abstraction or are Complex Device Drivers), or toidentify ICC2clusters.

How to Solve

Configure a validvalue for the attribute ModuleID thatrefers tothe identifier

ofstandardizedAUTOSARmodules.

b) Error Description

In a BswModuleDescription with category as BSW_CLUSTER the value of the attribute ModuleID configured shouldrefer toa Non-StandardizedAutosar module.

How to Solve

Configure a validvalue for the attribute ModuleID thatrefers to the identifier ofNon-standardizedAUTOSAR modules.

6.36 ASW4X_BSWMD_0036 [Constr_4058]

Error Description

Ifthe BswModuleDescriptionandSwComponentType are mappedthroughSwcbswMappingthendifferentmode groups presentinside themcannothave similar names.

How to Solve

1. Delete the additional reference ofthe ModeDeclarationGrouphavingthe same name.
2. Reconfigure the mode groups inside the BswModuleDescriptionthatdoes nothave additional reference withthe same name .

6.37 ASW4X_BSWMD_0037 [Constr_4024]

a) Error Description

The BswModeSwitchEventvalue ofthe attribute "Activation"shouldnotbe NULL or Empty.

How to Solve

Configure the attribute activationtoany ofthe followingvalues: On-Entry, On-Exist, On-Transition.

b)Error Description

The BswModeSwitchEvent shouldcontain exactly one ModeInBswModuleDescriptionInstanceRefwhen the attribute "Activation"is configuredtoa value other thanOn-Transition.

How to Solve

Reconfigure the BswModeSwitchEventtoconfigure exactly one ModeInBswModuleDescriptionInstanceRef.

c) Error Description

IfBswModeSwitchEvent.activationhas the value onTransitionBswModeSwitchEventshouldrefer totwodifferent modes belongingtothe same instance ofModeDeclarationGroup.

How to Solve

Reconfigure two different ModeInBswModuleDescriptionInstanceRefs to refer the same instance of ModeDeclarationGroup.

6.38 ASW4X_BSWMD_0038 [Constr_4061]

a) Error Description

InMcDataInstance the reference TargetconfiguredinImplementationElementInParameterInstanceRefshouldbe identical withthe reference Contextconfiguredinthe SubelementMcDataInstance inside ImplementationElementInParameterInstanceRef.

How to Solve

Reconfigure the ImplementationDataTypeElementinboththe upper and-

lower DataInstances tobe identical.

b) Error Description

InMcSupportData, since the containedMcDataInstance has anInstanceInMemory,the containingMcDataInstance shouldalsohave anInstanceInMemory.

How to Solve

Configure anInstanceInMemory for the containingMcDataInstance.

6.39 ASW4X_BSWMD_0039 [Constr_4031]

Error Description

InAnalyzedExecutionTime the BestCaseExecutionTime shouldbe less thanor equal toWorstCaseExecutionTime

How to Solve

Reconfigure the value ofthe BestCaseExecutionTime tobe less thanor equal toWorstCaseExecutionTime

6.40 ASW4X_BSWMD_0040 [Constr_4054]

Error Description

In MemorySection the reference ExecutableEntities should be configured with BswCalledEntity whose reference SwAddrMethodis same as the SwAddrMethodvalue ofMemorySection.

How to Solve

Reconfigure the executable entities to have BswCalledEntity whose SwAddr-Method is same SwAddrMethod of MemorySection.

6.41 ASW4X_BSWMD_0041 [Constr_4032]

a) Error Description

InMeasuredExecutionTime the MinimumExecutionTime shouldbe less

thanor equal toNominalExecutionTime. **How to Solve**

Reconfigure the MinimumExecutionTime tobe less thanor

equal toNominalExecutionTime.

b) Error Description

InMeasuredExecutionTime the NominalExecutionTime shouldbe less thanor

equal toMaximumExecutionTime.

How to Solve

Reconfigure the NominalExecutionTime tobe less thanor equal toMaxi-mumExecutionTime.

6.42 ASW4X_BSWMD_0042 [Constr_4033]

a) Error Description

InSimulatedExecutionTime the MinimumExecutionTime shouldbe less

thanor equal toNominalExecutionTime. **How to Solve**

Reconfigure the MinimumExecutionTime tobe less thanor

equal toNominalExecutionTime. **b) Error Description**

InSimulatedExecutionTime the NominalExecutionTime shouldbe less thanor

equal toMaximumExecutionTime **How to Solve**

Reconfigure the NominalExecutionTime tobe less thanor equal toMaxi-mumExecutionTime.

6.43 ASW4X_BSWMD_0043 [Constr_4062]

Error Description

In BswInternalBehavior containing the BswServiceDependency for RoleBasedDataAssignment the TargetDataprototype configuredis notaninstance oflocalParameter.

How to Solve

Configure a validvalue for the attribute Symbol whichis the name ofthe elementinthe programminglanguage.

6.44 ASW4X_BSWMD_0044 [Constr_4051]

a) Error Description

Ina BswImplementationcontainingthe ResourceConsumption, inMeasuredStackUsage the value ofthe attribute "MinimumMemory Consumption"shouldbe less than"AverageMemoryConsumption".

How to Solve

Change the TargetDataprototype tobe

aninstance oflocal Parameter. **b) Error**

Description

InBswInternalBehavior containingthe BswServiceDependency for RoleBasedDataAssignmentthe TargetDataProtoType configuredis notaninstance oflocalVariable.

How to Solve

Change the TargetDataprototype tobe

aninstance oflocal Variable. **c) Error**

Description

UsedPimvalue shouldnotbe setsince RoleBasedDataAssignmentis usedincontextofServiceDependency. **How to Solve**

Delete the value ofthe reference usedPim.

7. ECUExtract Error Description

This sectiondeals withthe error descriptionofECUEXtractvalidations.

- EcuExtract_ERR_01

Error Description

Systemhas invalidcategory. Only category with“SYSTEM_DESCRIPTION,SYSTEM_EX-TRACT,RPT_ SYSTEM,SYSTEM_CONSTRAINTS,ABSTRACT_SYSTEM_DESCRIPTION,ECU_SYSTEM_DE-SCRIPTION andSW_ CLUSTER_SYSTEM_DESCRIPTION"are supportedby ECUExtract

How to Solve

1. Change the category ofSystemto“SYSTEM_DESCRIPTION”or “SYSTEM_EXTRACT.

- EcuExtract_ERR_02

Error Description

Systemis configuredwithmore thanone ECU Instance inSwcToEcuMapping. This is notyetsupportedby ECUExtract.

How to Solve

1. Invoke ECUExtractfromcorrespondingECU Instance insteadofSystem.

- EcuExtract_ERR_03

Error Description

Conflicts inCommunicationSpecificationfoundonSwConnector

How to Solve

1. Use "Ignore comspecconflicts"toproceedwithECUExtract, by ignoringcomspecerrors.

- EcuExtract_ERR_04

Error Description

PortPrototype is notconnectedtoany other component. NoSignal is mappedtoPortPro-totype for external ECU communication. Hence nodelegationconnectionanddelegat-edportis createdinflatview.

How to Solve

1. Configure DataMappingfor the PortPrototype.
2. Close the openPortPrototype.

- EcuExtract_ERR_05

Error Description

PortPrototypes are involvedinM:N Communication. i.e., PortPrototype is involvedin-both1:N communicationand N:1communication.

How toSolve

1. Try removingthe M:N configuration.
2. IfVariationpointis configuredwhichmightbreak this M:N communicationthenuse Preference option"Donot abortECUExtractfor M:N communicationifVariationpointis configured".

- EcuExtract_ERR_06

Error Description

Foundalternative configurationfoundinthe project, whichis notsupportedby EcuExtract. Elementwithsame shortname under one parentwithdifferentvariationpoint.

How to Solve

1. Check UI option"Ignore alternate configuration"toproceedwithECUExtractby ignoringalternate configuration.

- EcuExtract_ERR_07

Error Description

Twoor more elements has invalidshortname configuredunder a single parent.

How to Solve

1. Remove duplicate elements.

- EcuExtract_ERR_08

Error Description

NoSystemMappings foundunder the System.

How to Solve

1. Create a SystemMappingunder the System.

- EcuExtract_ERR_09

Error Description

Multiple SwcToEcuMappingreferringtosame ECUInstance, Partition-andProcessingunitfoundunder SwToEcuMapping.

How to Solve

1. Merge multiple SwcToEcuMappinghavingsame combinationofECUInstance, PartitionandProcessingunitinto one.

- EcuExtract_ERR_10

Error Description

SwcToEcuMappingcontains variationpointconfiguredunder systemMappingwhichis invalidconfiguration. Latest bindingtime ofSwcToEcuMappingis SYSTEM-DESIGN-TIME, soitis notallowedtohave itduringECUExtract. **How to Solve**

1. Remove variationpointfromSwcToEcuMapping.
2. Use variantresolver inECUExtract.

- EcuExtract_ERR_11

Error Description

VariationPointconfiguredunder SystemMappingwhichis notsupportedinECUExtract.

How to Solve

1. Remove variationpointconfigured.

- EcuExtract_ERR_12

Error Description

Duplicate SystemMappingwithsame shortname withsame shortname foundunder System.SystemMapping shouldhave a unique shortname.

How to Solve

1. Rename anyone ofthe SystemMappingtomaintainunique shortname.

- EcuExtract_ERR_13

Error Description

There is noRootSwCompositionPrototype foundunder the System.

How to Solve

1. Configure RootSwCompositionPrototype under Systemandrefer tothe TopLevel-Composition.

- EcuExtract_ERR_14

Error Description

Multiple RootSwCompositionPrototype foundunder the System. Whichis notallowed.

How to Solve

1. Maintainonly one RootSwCompositionPrototype under the System, check inall the splitSystemfor the extra RootSwCompositionPrototype.

- EcuExtract_ERR_15

Error Description

VariationPointis configuredunder RootSwCompositionPrototype. Latestbinding-time ofRootSwCompositionPrototype is SYSTEM-DESIGN-TIME, soitis notallowedtohave itduringECUExtract.

How to Solve

1. Remove the variationpointconfigured.
2. Use variantresolver inECUExtract.

- EcuExtract_ERR_16

Error Description

Toplevel compositionis notconfiguredinRootSwCompositionPrototype.

How to Solve

1. Refer a TopLevelCompositioninRootSwCompositionPrototype.

- EcuExtract_ERR_17

Error Description

EndToEndDescriptionhas beenconfiguredmultiple times for anEndToEndProtection. Only one EndToEndDescriptionshouldbe available under EndToEndProtection

How to Solve

1. Have only one EndToEndDescriptionconfiguredunder EndToEndProtection.

- EcuExtract_ERR_18

Error Description

As per AUTOSAR, EndToEndProtectionVariablePrototype available under EndToEndProtectionsupports PortPrototypes only withSenderReceiverInterface

How to Solve

1. Remove EndToEndProtectionconfiguredfor PortPrototype havingnon- SenderReceiverInterface.

- EcuExtract_ERR_19

Error Description

Data Mapping's communicationdirectionis either notconfiguredor configuredas "IN"whichis invalidfor portwith provider interface.

How to Solve

1. Change the directionofDataMappingtoOUT.

- EcuExtract_ERR_20

Error Description

Duplicate SystemMappingshortnames foundunder same System(includingSplitfiles). Butthis is notallowedas SystemMappingname shouldbe unique withinthe System

How to Solve

1. Rename SystemMappingname
2. ClubbothSystemMappingcontents intoone

- EcuExtract_ERR_21

Error Description

Error occuredwhile executingFlatInstanceDescriptor namingtemplates. The customizable scriptprovidedduring ECUExtractgenerationis erroneous

How to Solve

1. Scripts needs tobe correctedor defaultscripts couldbe usedby uncheckingoptionuse Customize scripts.

- EcuExtract_ERR_22

Error Description

DataMappinghas beenplacedona ServicePort. This is notallowedas per Autosar because service ports are not usedfor inter ECU communication

How to Solve

1. Remove DataMappingplacedonserive ports

- EcuExtract_ERR_23

Error Description

ContextpathsetinDataMappingis invalid. Some references mightbe missing

How to Solve

1. Use contextpathdialogfromProperties view or SystemDataMappingEditor tocorrectthe contextpath

- EcuExtract_ERR_24

Error Description

Contextpathsetunder SwcToEcuMappingis invalid. Some references mightbe missingor removed

How to Solve

1. Use SwcToEcuMappingEditor toconfigure validSwcToEcuMappingreferences

- EcuExtract_ERR_25

Error Description

InvalidConnections found. The connector mighthave proxy references or involvedinalternate or invalidvariant configuration

How to Solve

1. Use "**Ignore Alternate configuration**"optiontoproceedwithECUExtract

2. Ifconnections are involvedinalternate configuration, thenuse "Resolve Variants"frompreference toresolve the variants.

- EcuExtract_WARN_01

Error Description

SwComponentPrototype is notconfiguredwithany SwComponentType or itmay be proxy

How toSolve

1. Configure SwComponentType for SwComponentPrototype.

- EcuExtract_WARN_02

Error Description

Ports are involvedinM:N Communicationpattern. i.e., PortPrototype is involvedinboth1:N communicationand N:1communication

How to Solve

1. Try removingthe M:N configuration.

- EcuExtract_WARN_03

Error Description

Conflicts inCommunicationSpecificationfoundonSwConnector

How to Solve

1. Ensure noconflicts incommunicationspecification.

- EcuExtract_WARN_04

Error Description

FlatMapnotconfiguredor configuredtoproxy reference

How to Solve

1. Configure a validflatmapinextractedSystem.

- EcuExtract_WARN_05

Error Description

FlatView notconfiguredor configuredtoproxy reference

How to Solve

1. Configure a validflatview inextractedSystem

- EcuExtract_WARN_06

Error Description

SwcToEcuMappingavailable under SystemMappingdoesn'trefer toany

EcuInstance or itis proxy **How to Solve**

1. Use "SWC ToECU MappingEditor"toconfigure SwcToEcuMapping. This editor canbe invokedonSystem.

- EcuExtract_WARN_07

Error Description

NoValidSender or Receiver configuredfor EndToEndProtection.

How to Solve

1. Configure a validSender or Receiver under EndToEndProtection.

- EcuExtract_WARN_08

Error Description

InvalidconfigurationofEndToEndProtectionISignalIPduinanEndToEndProtection

How to Solve

1. Configure a validEndToEndProtectionISignalIPdu.

- EcuExtract_WARN_09

Error Description

Either all the InternalBehavior shouldbe configuredwithvariationpointor none ofthemshouldhave variationpint configured.

How to Solve

1. Remove all the variationpoints configuredfor InternalBehaviors.
2. Addvariationpoints toremainingInternalBehaviors.

- EcuExtract_WARN_10

Error Description

AutosarDataPrototype is notconfiguredtoany datatype or invalidreference

How to Solve

1. Configure a validDataType toAutosarDataPrototype.

- EcuExtract_WARN_11

Error Description

PortPrototype is notconfiguredwithany Interface or configuredtoproxy reference

How to Solve

1. Configure a validinterface toPortPrototype.

- EcuExtract_WARN_12

Error Description

SwComponentType is multiple instantiated. Butattribute under SwcInternalBe-haviour "SupportsMultipleInstantiation"is either settofalse or notset

How to Solve

1. Set"SupportsMultipleInstantiation"totrue inside SwcInternalBehaviour.

- EcuExtract_WARN_13

Error Description

ComponentInSystemInstanceRefunder swcToImplMappingDescriptionis invalid

How to Solve

1. Configure a validComponentInSystemInstanceRefunder SwcToImplMapping.

Use "SWC toECU Mapping Editor"toconfigure the SwcToImplMappings.

- EcuExtract_WARN_14

Error Description

SwcToImplMappingdoes nothave any SwcImplementationconfigured

How to Solve

1. Configure a validSwcImplementationinSwcToImplMapping. Use "SWC toECU MappingEditor"toconfigure the SwcToImplMappings.

- EcuExtract_WARN_15

Error Description

PortPrototype is placedonTopLevelComposition, butSystemcategory is not"SYSTEM_EXTRACT". Hence this Portis ignored. Andwill notbe consideredfor flattening

How to Solve

1. Change the Systemcategory to"SYSTEM_EXTRACT".
2. IfSystemcategory has tobe "SYSTEM_DESCRIPTION", thendonotdelegate Porttill toplevel.

- EcuExtract_WARN_16

Error Description

DelegationSwConnector is notcreatedinflatview compositionfor datamappingconfiguredfor the signal. DataMappingmightnotbelongtothe selectedECU or external ECU communication

How to Solve

1. Configure a validdataMappingusingData MappingEditor.

- EcuExtract_WARN_17

Error Description

DataMappingavailable under selectedsystemis nottakentoEcuExtractedSystem. Ignore this warningifitis not a validDataMappingfor selectedEcuInstance.

How to Solve

1. Use "SystemData MappingEditor"toconfigure the DataMappings.

- EcuExtract_WARN_18

Error Description

SwcToEcuMappingavailable under SystemMappingdoesn'trefer toany EcuInstance or itis invalidreference

How to Solve

Refer toa validECUInstance inSwcToEcuMapping. Use "SWC ToECU MappingEditor"toconfigure the SwcToEcuMapping.

- EcuExtract_WARN_19

Error Description

Invalid flatinstance descriptors has been removed from existing flatmap

How to Solve

NA

- EcuExtract_WARN_20

Error Description

Systemversionconfiguredunder Systemis mismatchingacross splitfiles

How to Solve

Configure same SystemVersionacross splitSystem.

- EcuExtract_WARN_21

Error Description

Systemversionis notspecified, defaultSystemExtractversionwill be addedtothe extractedSystem

How to Solve

1. Update SystemVersioninSystem.

- EcuExtract_WARN_22

Error Description

ArgumentDataPrototype configuredinClientServerInterfaceMappingis invalid

How to Solve

1. Inside ClientServerInterfaceMappingrefer toa validClientServerOperationwhichis usedby PortPrototypes involvedinconnection(referringtothe ClientServerInterfaceMapping).

- EcuExtract_WARN_23

Error Description

One ofthe AutosarDataPrototype reference available under ClientServerInterfaceMappingis notconfiguredor proxy

How to Solve

1. Configure a validDataPrototype under ClientServerInterface.

- EcuExtract_WARN_24

Error Description

ClientServerOperationMappingunder ClientServerInterfaceMappingcontains invalidconfiguration

How to Solve

1. Inside ClientServerInterfaceMappingrefer toa validClientServerOperationwhichis usedby PortPrototypes involvedinconnection(referringtothe ClientServerInterfaceMapping).

- EcuExtract_WARN_25

Error Description

InvalidPortInterfaceMappingreference

How to Solve

1. Refer tovalidClientServerOperationinClientServerInterfaceMapping.

- EcuExtract_WARN_26

Error Description

DataPrototypeMappingavailable under PortInterfaceMappingcontains invalidconfiguration

How to Solve

1. Inside VariableAndParameterInterfaceMappingrefer toa validDataPrototypes whichis usedby PortPrototypes involvedinconnection(referringtothe VariableAndParameterInterfaceMapping).

- EcuExtract_WARN_27

Error Description

PRPortPrototype is connectedtobothRPortPrototype andPPortPrototype, this configurationis notsupportedby ECUExtract

How to Solve

1. Use PRPortprototype inrole ofeither PPortPrototype or RPortPrototype atonce.

- EcuExtract_WARN_28

Error Description

NovalidDelegationSwConnector foundfor the PortPrototype inCompositionSwComponentType

How to Solve

1. Configure validPortInterfaceMappingifPorts are notcompatible. Use AutoDelegationconnectiontocreate Delegationconnections.

- EcuExtract_WARN_29

Error Description

SwcToEcuMappingcontextis invalid. The rootSwCompositionPrototype referredby SwcToEcuMappingis notvalid

How to Solve

1. Invoke SwcToEcuMappingEditor onSystemandmapcomponents toEcuInstance

- EcuExtract_WARN_30

Error Description

Unexpectederror occuredwhile clonningthe resources fromthe project. Please refer to error log for more information.

8. Importer Error Description

This section deals with the error description of Importers validations. Importers Validations are specific to Importers template in autosar.

8.1 Importer_ERR_01

Error Description

"ArPackage "+arpackage "has notbeencreateddue toException: "+e.getmessages ()

8.2 Importer_ERR_02

Error Description

"The ISignal "+ iSignal.getShortName()+ "presentinISignalGroup"+cre-ateISignalGroup.getShortName()+ " is notmappedtothe Pdu"+ selPdu.getShort-Name()

8.3 Importer_ERR_03

Error Description

"Error duringMappingRules Execution: "+e.getmessage ()

8.4 Importer_ERR_04

Error Description

"Index value ofCanController is outofbounds, only %s number ofCanControllers are defined"

8.5 Importer_ERR_05

Error Description

"Invalidvalue specifiedfor PackingByteOrder for signal"+SignalName, validvalues are 0or 1

8.6 Importer_ERR_06

Error Description

"For the MultiplexedPDU"+MultiplexedIPduName "DynamicPartis notcreateddue toException:"+e.getmessage
()

8.7 Importer_ERR_07

Error Description

"Error occurredwhile executingextensionscript: "+e

8.8 Importer_ERR_08

Error Description

Frame IdofMasterReqdiagnosticframe shouldalways have the value 60

8.9 Importer_ERR_09

Error Description

Frame Idof"SlaveRespdiagnosticframe shouldalways have the value 61

8.10 Importer_ERR_10

Error Description

EcuInstance withname (ecuinstance.getshortname())is notfound

8.11 Importer_ERR_11

Error Description

"Invalidfile name: "+filename "Please selecta fibex file witha validname"

9. RTTM Validations

This section deals with the error description of RTTM validations.

9.1 ASW4X_RTTM_0001

Error Description

InOSModule the DefinitionofSubcontainers shouldnotbe empty or proxy.

How to Solve

Configure validDefinitionfor subcontainer.

9.2 ASW4X_RTTM_0002

Error Description

InOSTask the OsTaskEventRefshouldrefer toanOSEvent.

How to Solve

Configure validvalue for the reference value OSTaskEventRef.

9.3 ASW4X_RTTM_0003

Error Description

InOSTask the OsTaskEventRefshouldrefer toanOSEvent.

How to Solve

Configure validOSEventfor OSTaskEventRefinOSTask.

9.4 ASW4X_RTTM_0004

Error Description

InOSTask all the references shouldhave the definitionconfigured.

How to Solve

Configure the Definitionwithvalidvalue

9.5 ASW4X_RTTM_0005

Error Description

In OSTask the IntegerParameter configured should be either OSTaskPriority or OSTaskActivation,Other parameters are notallowed.

How to Solve

Reconfigure the IntegerParameter tobe either OSTaskPriority or OSTaskActivation.

9.6 ASW4X_RTTM_0006

Error Description

InOSTask container the definitionofparameter shouldnotbe empty or proxy.

How to Solve

Configure validdefinitionfor the parameter inOSTask.

9.7 ASW4X_RTTM_0007

Error Description

InOSTask the value ofthe attribute OSTaskActivationshouldnotbe less thanzero.

How to Solve

Configure a value for OSTaskActivationwhichis greater thanzero.

9.8 ASW4X_RTTM_0008

Error Description

InOSTask the value ofthe attribute OSTaskActivationshouldnotbe empty.

How to Solve

Configure a value for OSTaskActivationwhichis greater thanzero.

9.9 ASW4X_RTTM_0009

Error Description

InOSTask container there shouldnotbe more thanone OSTaskActivationconfiguredwhichis invalid.

How to Solve

Delete the additional OSTaskActivationconfiguredandretainonly one OSTaskActivation.

9.10 ASW4X_RTTM_0010

Error Description

InOSTask the value ofthe attribute OSTaskPriority shouldnotbe less thanzero.

How to Solve

Configure a value for OSTaskPriority whichis greater thanzero.

9.11 ASW4X_RTTM_0011

Error Description

InOSTask the value ofthe attribute OSTaskPriority shouldnotbe empty.

How to Solve

Configure a value for OSTaskPriority whichis greater thanzero.

9.12 ASW4X_RTTM_0012

Error Description

InOSTask container there shouldnotbe more thanone OSTaskPriority configuredwhichis invalid.

How to Solve

Delete the additional OSTaskPriority configuredandretainonly one OSTaskActivation.

9.13 ASW4X_RTTM_0013

Error Description

InOSTask container the value OSTaskPriority shouldbe configured.

How to Solve

Configure a value for OSTaskPriority whichis greater thanzero.

9.14 ASW4X_RTTM_0014

Error Description

InOSTask container the value ofthe attribute OSTaskActivationshouldnotbe empty.

How to Solve

Configure a value for OSTaskActivationwhichis greater thanzero.

9.15 ASW4X_RTTM_0017

Error Description

InOSModule, the OSTask shouldhave shortname configured.

How to Solve

Configure shortname for OSTask.

9.16 ASW4X_RTTM_0018

Error Description

InOSModule, the OSEventshouldhave shortname configured.

How to Solve

Configure shortname for OSEvent.

9.17 ASW4X_RTTM_0020

Error Description

InEcucModuleConfigurationValues NvMshouldhave a definitionconfigured.

How to Solve

Configure the definitionfor EcucValueContainer.

9.18 ASW4X_RTTM_0023

Error Description

InRTEModule, the definitionofsubcontainer shouldnotbe empty or proxy.

How to Solve

Configure the Definitionwithvalidvalue.

9.19 ASW4X_RTTM_0024

Error Description

InRTEModule there shouldbe only one RTEGenerationcontainer.

How to Solve

Delete the additional RTEGenerationcontainer toretainonly one RTEGenera-tionConfigure shortname for OSEvent.

9.20 ASW4X_RTTM_0027

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteExclusiveAreaImplementation , the EcucReferenceValue shouldbe oftype ExclusiveAreaRef.

How to Solve

Reconfigure the Value ofthe RteExclusiveAreaReftobe oftype ExclusiveAreaRef.

9.21 ASW4X_RTTM_0028

Error Description

InRTEModule containingthe RteSwComponentInstance configuredwithRteExclusiveAreaImplementationshould nothave more thanone ExclusiveAreaRefconfigured.

How to Solve

Delete the additional RteExclusiveAreaReftoretainonly one EcucReferenceValue.

9.22 ASW4X_RTTM_0029

Error Description

InRTEModule containingthe RteSwComponentInstance configuredwithRteExclusiveAreaImplementationshould have one ExclusiveAreaRefconfigured.

How to Solve

Configure the RteExclusiveAreaRefinRteExclusiveAreaImplementation.

9.23 ASW4X_RTTM_0031

Error Description

InRTEModule containingthe RteSwComponentInstance configuredwithRteExclusiveAreaImplementation should have ExclusiveAreaRef value configured that is from the same SwcInternalBehavior configured for RteSwComponentInstance.

How to Solve

Configure the ExclusiveAreaRef value that is from the same SwcInternalBehavior configured for RteSwComponentInstance.

9.24 ASW4X_RTTM_0037

Error Description

The RteNVRamAllocationshouldnothave more thanone

RteSwNvRamMappingReference configured.

How to Solve

Delete the additional RteSwNvRamMappingReference andretainonly one RteSwNvRamMappingReference.

9.25 ASW4X_RTTM_0038

Error Description

The RteNVRamAllocationshouldhave a RteSwNvRamMappingReference configured.

How to Solve

Configure the RteSwNvRamMappingReference for RteNVRamAllocation.

9.26 ASW4X_RTTM_0039

Error Description

The RteNVRamAllocationshouldnothave more thanone RteNvmBlockRefconfigured.

How to Solve

Delete the additional RteNvmBlockRefandretainonly one RteNvmBlockRef.

9.27 ASW4X_RTTM_0041

Error Description

The RteNVRamAllocationdoes nothave the RteSwNvRamMappingReference

andRteNVBlockRefconfigured

How to Solve

Configure the RteSwNvRamMappingReference andRteNVBlockReffor RteNVRamAllocation.

9.28 ASW4X_RTTM_0040

Error Description

The RteNVRamAllocationshouldhave a NvmBlockRefconfigured.

How to Solve

Configure the RteNvmBlockReffor RteNVRamAllocation.

9.29 ASW4X_RTTM_0042

Error Description

InRTEModule containingthe RteSwComponentInstance is configuredwithduplicate RteNVRamAllocation.

How to Solve

Delete the duplicate containers inthe RteNVRamAllocation.

9.30 ASW4X_RTTM_0043

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteNVRamAllocation the reference RteSwNvRamMappingRef should refer to a SwcNvBlockNeed which is from the same SwcInternalBehavior configuredfor RteSwComponentInstance.

How to Solve

1.Reconfigure the reference RteSwNvRamMappingReftorefer toa SwcNvBlock-Needwhichis fromthe same SwcInternalBehavior configuredfor RteRteSwCom-ponentInstance.

9.31 ASW4X_RTTM_0044

Error Description

InRTEModule containingthe RteSwComponentInstance configuredwithRteN-VRamAllocationthe RteNvBlockRef shouldrefer toa RteNvBlockDescriptor whichis fromthe same NvMmodule configuredfor EcucValueCollection.

How to Solve

Reconfigure RteNvBlockReftorefer toa RteNvBlockDescriptor whichis fromthe same NvMmodule configuredfor EcucValueCollection.

9.32 ASW4X_RTTM_0045

Error Description

InRTEModule containingthe RteSwComponentInstance configuredwithRteN-VRamAllocationthe RteNvBlockRef shouldrefer toa RteNvBlockDescriptor whichis fromthe same NvMmodule configuredfor EcucValueCollection.

How to Solve

Reconfigure RteNvBlockReftorefer toa RteNvBlockDescriptor whichis fromthe same NvMmodule configuredfor EcucValueCollection.

9.33 ASW4X_RTTM_0046

Error Description

InRTEModule containingthe RteSwComponentInstance configuredwithRteN-VRamAllocationthe RteNvBlockRef shouldrefer toa RteNvBlockDescriptor whichis fromthe same NvMmodule configuredfor EcucValueCollection.

How to Solve

Reconfigure RteNvBlockReftorefer toa RteNvBlockDescriptor whichis fromthe same NvMmodule configuredfor EcucValueCollection.

9.34 ASW4X_RTTM_0049

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteNVRamAllocation the reference RteSwNvRamMappingRef should refer to a SwcNvBlockNeed which is from the same SwcInternalBehavior configuredfor RteSwComponentInstance.

How to Solve

Reconfigure the reference RteSwNvRamMappingReftorefer toa SwcNvBlock-Needwhichis fromthe same SwcInternalBehavior configuredfor RteRteSwCom-ponentInstance.

9.35 ASW4X_RTTM_0050

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteNVRamAllocation the reference RteSwNvRamMappingRef should refer to a SwcNvBlockNeed which is from the same SwcInternalBehavior configuredfor RteSwComponentInstance.

How to Solve

Reconfigure the reference RteSwNvRamMappingReftorefer toa SwcNvBlock-Need.

9.36 ASW4X_RTTM_0051

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteNVRamAllocation the reference RteSwNvRamMappingRef should refer to a SwcNvBlockNeed and RteNvBlockRef should refer to a RteNvBlockDescriptor.

How to Solve

Configure the reference RteSwNvRamMappingReftorefer toa SwcNvBlock-NeedandRteNvBlockReftorefer toa RteNvBlockDescriptor.

9.37 ASW4X_RTTM_0054

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping the reference RteEventshouldbe configured.

How to Solve

Configure the RteEventReftoa timingevent.

9.38 ASW4X_RTTM_0055

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping should not have more thanone RteEventRefconfig-ured.

How to Solve

Delete the additional RteEventRefandretainonly one RteEventRef.

9.39 ASW4X_RTTM_0056

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping containers shouldhave RteMappedToTaskRefconfig-ured.

How to Solve

Reconfigure the RteMappedToTaskReftoa OSTask.

9.40 ASW4X_RTTM_0057

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping should not have more thanone RteMappedTo-TaskRefconfigured.

How to Solve

Delete the additional RteEventRefandretainonly one RteEventRef.

9.41 ASW4X_RTTM_0058

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping should not have more thanone UsedOsEventRefcon-figured.

How to Solve

Delete the additional UsedOsEventRefandretainonly one UsedOsEventRef.

9.42 ASW4X_RTTM_0064

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping all the Containers shouldhave reference Definition-configured.

How to Solve

Configure the reference Definitionfor the subcontainers under teEventToTask-Mapping.

9.43 ASW4X_RTTM_0066

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping the attribute "RteActivationOffset"shouldhave a value greater thanzero.

How to Solve

Configure the RteActivationOffsetvalue tobe greater thanzero.

9.44 ASW4X_RTTM_0067

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping the attribute "RteActivationOffset"shouldhave a value greater thanzero.

How to Solve

Configure the RteActivationOffsetvalue tobe greater thanzero.

9.45 ASW4X_RTTM_0068

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping should have only one "RteActivationOffset"config-ured.

How to Solve

Delete the additional RteActivationOffsetina container tohave only 1RteActivationOffsetvalue.

9.46 ASW4X_RTTM_0069

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping the attribute "RtePositionInTask"shouldhave a value greater thanzero.

How to Solve

Reconfigure the RtePositionInTask value tobe greater thanzero.

9.47 ASW4X_RTTM_0070

Error Description

InRTEModule containingthe RteSwComponentInstance configuredwithRteEventToTaskMappingthe attribute "RtePositionInTask"shouldhave a value greater than zero.

How to Solve

Configure the RtePositionInTask value tobe greater thanzero.

9.48 ASW4X_RTTM_0071

Error Description

InRTEModule containingthe RteSwComponentInstance configured-withRteEventToTaskMappingthe attribute "RteActivationOffset"shouldhave a value configuredgreater thanzero.

How to Solve

Configure the RteActivationOffsetvalue toinRteEventToTaskMapping.

9.49 ASW4X_RTTM_0072

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping should have only one RtePositionInTask configured.

How to Solve

Delete the additional RtePositionInTask ina container tohave only 1RtePositionInTask value.

9.50 ASW4X_RTTM_0076

Error Description

InRTEModule containingthe RteSwComponentInstance shouldnothave duplicate RteEventToTaskMapping.

How to Solve

Delete the duplicate containers inthe RteEventToTaskMapping

9.51 ASW4X_RTTM_0078

Error Description

InRTEModule containingthe RteSwComponentInstance configured-withRteEventToTaskMappingthe Container RteEventRefconfiguredshouldbe fromthe same SwcInternalBehavior configuredfor RteSwComponentInstance.

How to Solve

Reconfigure the value ofthe RteEventReftoaneventwhichis fromthe same swcInternalBehavior configured for RteSwComponentInstance

9.52 ASW4X_RTTM_0079

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping the RteEventRefshouldnothave any proxy values anditshouldbe configured.

How to Solve

Configure the RteEventRefinthe RteEventToTaskMappingtoaneventfromthe SwcInternalBehavior configured for RteSwComponentInstance.

9.53 ASW4X_RTTM_0081

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping having RteEventRefshouldrefer a validRTEEvent.

How to Solve

Configure the RteEventRef in the RteEventToTaskMapping to an event from the SwcInternalBehavior configured for RteSwComponentInstance.

9.54 ASW4X_RTTM_0082

Error Description

InRteEventToTaskMappingConfiguredwithRteEventRefshouldrefer a RunnableEntity.

How to Solve

Reconfigure the value TimingEventtohave a validStartOnEvent.

9.55 ASW4X_RTTM_0083

Error Description

InRteEventToTaskMappingConfiguredwithRteEventRefshouldrefer a RunnableEntity

How to Solve

Reconfigure the value TimingEventtohave a validStartOnEvent.

9.56 ASW4X_RTTM_0084

Error Description

InRteEventToTaskMappingConfiguredwithRteMappedToTaskRefshouldrefer a OSModule whichis referedin any ofthe EcucModuleConfigurationValuesRef-Conditional.

How to Solve

Reconfigure the reference value ofRteMappedToTaskReftoa OSModule whichis referedinEcucModuleConfigurationValuesRefConditional.

9.57 ASW4X_RTTM_0085

Error Description

InRteEventToTaskMappingthe RteMappedToTaskRefshouldrefer a validOs Task.

How to Solve

Reconfigure the RteMappedToTaskReftorefer a validOStask.

9.58 ASW4X_RTTM_0086

Error Description

InRteEventToTaskMappingthe RteMappedToTaskRefshouldrefer a validOs Task.

How to Solve

Reconfigure the RteMappedToTaskReftorefer a validOStask.

9.59 ASW4X_RTTM_0087

Error Description

InRteEventToTaskMappingthe value ofthe reference RteUsedOsEven-tRefshouldrefer a OsEventwhichis configuredfor the OsTask referencedby the RteEventToTaskMappingContainer.

How to Solve

1.Reconfigure the OSEventfor the RteUsedOsEventReftohave same value as thatofOsTask referencedby the RteEventToTaskMapping.

9.60 ASW4X_RTTM_0088

Error Description

InRteEventToTaskMappingthe reference UsedOsEventRefshouldrefer toa OSE-vent.

How to Solve

Reconfigure the UsedOsEventReftorefer a validOSEvent.

9.61 ASW4X_RTTM_0089

Error Description

InRteEventToTaskMappingthe reference UsedOsEventRefshouldrefer toa OSEvent.

How to Solve

Reconfigure the UsedOsEventReftorefer a validOSEvent.

9.62 ASW4X_RTTM_0090

Error Description

InRteSwComponentInstance the SubContainers shouldhave a definationRefconfiguredor it's a proxy.

How to Solve

Configure the definitionfor the Subcontainer.

9.63 ASW4X_RTTM_0091

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteNVRamAllocation should have correspondingSwcInternalBehavior configured.

How to Solve

Configure the SwcInternalBehavior for the RteSwComponentInstance.

9.64 ASW4X_RTTM_0094

Error Description

There are more thanone RteSoftwareComponentInstanceRefconfiguredwithRteSoftwareComponent.

How to Solve

Delete the additional RteSoftwareComponentInstanceRefandretainonly one RteSoftwareComponentInstanceRef

9.65 ASW4X_RTTM_0095

Error Description

InRTE Module configured withRteSwComponentInstance, the reference RteSwComponentInstanceRefshould be configuredtoa validSwComponentPrototype.

How to Solve

Configure the RteSwComponentInstanceReftoa validSwComponentPrototype.

9.66 ASW4X_RTTM_0098

Error Description

InRTEModule configuredwithRteSwComponentInstance, the reference RteSwComponentInstanceRefshouldbe configuredtoa validSwComponentPrototype.

How to Solve

Configure the RteSwComponentInstanceReftoa validSwComponentPrototype.

9.67 ASW4X_RTTM_0099

Error Description

InRTEModule configuredwithRteSwComponentInstance, the reference RteSwComponentInstanceRefshouldbe configuredtoa validSwComponentPrototype.

How to Solve

Configure the RteSwComponentInstanceReftoa validSwComponentPrototype.

9.68 ASW4X_RTTM_0102

Error Description

In RTE Module configured with RteSwComponentInstance for the RteSwComponentInstanceRef the reference Definitionshouldbe configured.

How to Solve

Configure the reference Definitionfor RteSwComponentInstanceRef.

9.69 ASW4X_RTTM_0103

Error Description

In RTE Module configured with RteSwComponentInstance for the RteSwComponentInstanceRef the reference Definitionshouldbe configured

How to Solve

Configure the reference Definitionfor RteSwComponentInstanceRef.

9.70 ASW4X_RTTM_0105

Error Description

InRTEModule containingthe RteSwComponentInstance configured-withRteEventToTaskMappingthe Shortname is notconfigured.

How to Solve

Configure the shortname for the RteEventToTaskMapping.

9.71 ASW4X_RTTM_0106

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping, the ActivationOffsetshouldbe configuredwith-validvalue.

How to Solve

Configure the ActivationOffsetwitha validvalue.

9.72 ASW4X_RTTM_0110

Error Description

In RTE Module containing the RteSwComponentInstance configured with RteEventToTaskMapping, the RtePositionInTask configuredshouldnothave du-plicate values.

How to Solve

Configure the ActivationOffsetwitha validvalue.

9.73 ASW4X_RTTM_0112

Error Description

InRTEModule containingRteSwComponentInstance doesn'thave shortname configured.

How to Solve

Configure Shortname for the RteSwComponentInstance

9.74 ASW4X_RTTM_0113

Error Description

The Module Configuredinthe EcucModuleConfigurationValues shouldhave EcuExtractconfigured.

How to Solve

Configure the EcuExtractwithvalidvalue.

9.75 ASW4X_RTTM_0115

Error Description

The Module Configuredinthe EcucModuleConfigurationValues shouldhave Definitionconfigured.

How to Solve

Configure the Definitionwithvalidvalue.

9.76 ASW4X_RTTM_0119

Error Description

In RteSwComponentInstance containing the RteEventToTaskMapping all the runnables configured in the reference value ofRteEventRef, whichare mappedto the same modeDeclarationof modeGroup, should be mapped tosame OSTask inthe reference value ofRteMappedToTaskRef.

How to Solve

Reconfigure the value sothatthe OSTask inRteMappedToTaskRefsothatall the runnables configuredwithsame modeDeclarationofmodGroupis mappedtos-ame OSTask.

9.77 ASW4X_RTTM_0120

Error Description

In RteSwComponentInstance containing the RteEventToTaskMapping runnable triggered by ON- ENTRY SwcModeSwitchEvent should have highervalue for PositionInTask than the runnable triggered on ON-EXIT and ON-TRANSITION ModeSwitchEvents.

How to Solve

Reconfigure the RtePositionInTask in RteEventToTaskMapping so that runnable triggered by ON- ENTRY SwcModeSwitchEvent should have highervalue for PositionInTask than the runnable triggered on ON-EXIT and ON-TRANSITION ModeSwitchEvents.

9.78 ASW4X_RTTM_0122

Error Description

The EcucValueCollectionshouldhave a shortname configured.

How to Solve

Configure the Shortname for the EcucValueCollection.

9.79 ASW4X_RTTM_0124

Error Description

InSystemRootSwCompositionshouldbe configured.

How to Solve

Configure RootSwCompositioninEcucValueCollection.

9.80 ASW4X_RTTM_0125

Error Description

In EcucValueCollection the reference EcuExtract configured with System should contain RootSwComposition that has SoftwareCompositionconfiguredinit.

How to Solve

Configure RootSwCompositioninEcucValueCollection.

9.81 ASW4X_RTTM_0127

Error Description

InSystem, the category value shouldbe configuredtoa validvalue.

How to Solve

Reconfigure the category toa validvalue.

9.82 ASW4X_RTTM_0129

Error Description

InSystem, the category value shouldbe configuredtoa validvalue

How to Solve

Reconfigure the category toa validvalue.

9.83 ASW4X_RTTM_0132

Error Description

InOSTask the value configuredfor OsTaskPriority shouldbe oftype Integer.

How to Solve

Reconfigure the value ofOsTaskPriority withanInteger.

9.84 ASW4X_RTTM_0133

Error Description

InOSTask the value configuredfor OsTaskActivationshouldbe oftype Integer.

How to Solve

Reconfigure the value ofOsTaskActivationwithanInteger.

9.85 ASW4X_RTTM_0134

Error Description

The RteSwComponentInstance container shouldhave RteEventToTaskMap-pingContainer.

How to Solve

Configure the RteEventToTaskMappingfor the RteSwComponentInstance.

9.86 ASW4X_RTTM_0135

Error Description

InRteEventToTaskMappingthe reference RteComponentInstanceRefshouldbe configured.

How to Solve

Configrue validdefinitionfor the reference.

9.87 ASW4X_RTTM_0136

Error Description

In RteEventToTaskMapping the RteEventRef value refers to a TimingEvent whose StarOnEvent refers to a RunnableEntity whichis notfromthe same SwcInternalBehavior.

How to Solve

Reconfigure the RunnableEntity tobe fromthe same SwcInternalBehavior.

9.88 ASW4X_RTTM_0137

Error Description

In top level Composition the SwComponentPrototype has a reference type, this should refer a valid ComponentPrototype.

How to Solve

Reconfigure the reference type toa validSwComponentPrototype.

9.89 ASW4X_RTTM_0138

Error Description

In toplevel composition CompositionSwComponentType containing the SwComponentPrototype the reference type shouldbe configured.

How to Solve

Configure the reference type for SwComponentPrototype.

9.90 ASW4X_RTTM_0139

Error Description

IThe Virtual MappedOsTask has Eventtotask mappingwhichis invalidfor preemption.

9.91 ASW4X_RTTM_0140

Error Description

Runnable entity is missingor configuredas Proxy

9.92 ASW4X_RTTM_0141

Error Description

SwComponentPrototype is missingor configuredas Proxy.

9.93 ASW4X_RTTM_0142

Error Description

RTEEventis missingor configuredas Proxy

9.94 ASW4X_DT_0143

Error Description

InvalidCompuconfiguration, Inverse cannotbe computed

10. Variant Validations

This sectiondeals withthe error descriptionofVariantvalidations. VariantValidations are specifictoVarianttemplate inautosar.

10.1 ASW440_SYS_Constr_2594 [Constr_2594]

Error Description

Assigningvalues toSwSystemConstwhichdepends directly or indirectly onthis value assignment, is forbidden.

QuickFix

1. Modify swSystemValueconsttohave a numerical value or other systemconstants (VariationPointdialogshould popup).

10.2 ASW440_SYS_Constr_2507 [Constr_2507]

Error Description

Assigningvalues toSwSystemConstwhichdepends directly or indirectly onthis value assignment, is forbidden.

QuickFix

Remove the reference.

10.3 ASW440_SYS_Constr_2521 [Constr_2521]

Error Description shortLabel ofAttributeValueVariationPointshall be unique withinnextenclosingidentifiable.

QuickFix

1. Change the shortLabel ofthe AttributevalueVariationPoint.
2. Adda runningnumber for all repeatedshortLabel IntegerValueVariationPoint.

10.4 ASW440_SYS_Constr_2514 [Constr_2514]

Error Description

The combinationofshortName andshortLabel shall be unique withinthe nextenclosingIdentifiable.

QuickFix

1. Change the shortLabel ofthe VariationPoint.
2. 2. Runningnumber for shortLabel.

10.5 ASW4X_GEN_2504- constr_2504,constr_2518, constr_2577 and constr_ 2578 [Constr_2504]

Error Description

This is constraints are applicable only for Elements configuredwithvariationpoints.

Error Scenario: Ifthe BindingTime configuredinConditionByFormula ofa VariationPointis greater thanlatestbindingtime ofthe Elementtowhichitis configuredthenitis anerror.

QuickFix

1. Setthe allowedlatestbindingtime.

10.6 ASW4X_GEN_2519- constr_2519 [Constr_2519]

Error Description

This is constraints are applicable only for PreDefinedVariants.

Error Scenario: ifPredefinedVariantalongwithits includedVariants refers tomore thanone SwSystemconstantValueSet, thenany twoor more SwSystemConstantValue referringtosame SystemConstantshall have same value attributes else itwouldbe aninvalidconfiguration.

How to Solve.

1.Configure same value attribute for SwSystemConstantValue, IfPredefinedVariantalongwithits includedVariants refers tomore thanone

SwSystemconstantValueSetandany twoor more SwSystemCon-
stantValue referring tosame SystemConstant.

11. AutoSignalMapping Error Description

This section deals with the error description of AutoSignalMapping validations.

11.1 AutoSignalMapping_WARN_01

Error Description :

For SenderReceiverToSignalMapping, the Signal <SIG_NAM> and data element <DATA_ELE_NAME> are matched based on naming rules <NAMING RULE> but the generation of data mapping skipped due to the incompatibility of base type configuration from ISignal and Data element. Compatibility Error: <Error Message from Compatibility framework>.

How to Resolve :

Check the corresponding error message in the XML report to know the compatibility error and fix the error in base type elements configured in Isignal and the data element to generate the respective data mapping.

11.2 AutoSignalMapping_WARN_02

Error Description :

For SenderReceiverToSignalMapping, the Signal <SIG_NAM> and data element <DATA_ELE_NAME> are matched based on naming rules <NAMING RULE> but the generation of data mapping skipped due to the incompatibility of implementation data type from ISignal and provider PortComSpec or the provider PortComSpec does not contain the basetype configuration. Compatibility Error: <Error Message from Compatibility framework>.

How to Resolve :

Check the corresponding error message in the XML report to know the compatibility error and fix the error in implementation data type elements configured in Isignal and the data element to generate the respective data mapping.

11.3 AutoSignalMapping_WARN_03

Error Description :

For SenderRecRecordTypeMapping under SenderReceiverToSignalGroupMapping, the Signal <SIG_NAM> and data element

<DATA_ELE_NAME> are matched based on naming rules <SUB_ELEMENT_NAMING RULE> but the generation of data mapping skipped due to the incompatibility of base type configuration from ISignal and Data element. Compatibility Error: <Error Message from Compatibility framework>.

How to Resolve :

Check the corresponding error message in the XML report to know the compatibility error and fix the error in base type elements configured in Isignal and the data element to generate the respective data mapping.

11.4 AutoSignalMapping_WARN_04

Error Description :

For SenderRecRecordTypeMapping under SenderReceiverToSignalGroupMapping, the Signal <SIG_NAM> and data element <DATA_ELE_NAME> are matched based on naming rules < SUB_ELEMENT_NAMING RULE > but the generation of data mapping skipped due to the incompatibility of implementation data type from ISignal and Data element and "Consider Port ComSpec for compatibility (system template arr 4.31, table 6.9)" is selected in the preferences. Compatibility Error: <Error Message from Compatibility framework>.

How to Resolve :

Check the corresponding error message in the XML report to know the compatibility error and fix the error in implementation data type elements configured in Isignal and the data element to generate the respective data mapping.

11.5 AutoSignalMapping_WARN_05

Error Description :

For SenderRecArrayTypeMapping under SenderReceiverToSignalGroupMapping, the Signal <SIG_NAM> and data element <DATA_ELE_NAME> are matched based on naming rules < SUB_ELEMENT_NAMING RULE > but the generation of data mapping skipped due to the incompatibility of base type configuration from ISignal and Data element. Compatibility Error: <Error Message from Compatibility framework>.

How to Resolve :

Check the corresponding error message in the XML report to know the compatibility error and fix the error in base type elements configured in Isignal and the data element to generate the respective data mapping.

11.6 AutoSignalMapping_WARN_06

Error Description :

For SenderRecArrayTypeMapping under SenderReceiverToSignalGroup-Mapping, the Signal <SIG_NAM> and data element <DATA_ELE_NAME> are matched based on naming rules < SUB_ELEMENT_NAMING RULE > but the generation of data mapping skipped due to the incompatibility of implementation data type from ISignal and Data element and “Consider Port ComSpec for compatibility (system template arr 4.31, table 6.9)” is selected in the preferences. Compatibility Error: <Error Message from Compatibility framework>.

How to Resolve :

Check the corresponding error message in the XML report to know the compatibility error and fix the error in implementation data type elements configured in Isignal and the data element to generate the respective data mapping.

11.7 AutoSignalMapping_WARN_07

Error Description :

For the mapping type <mapping_type>, the Signal <SIG_NAM> and data element /ClientServerOperation/TriggerName<DATA_ELE_NAME> are matched based on the naming rules <NAMING RULE> but the generation of data mapping is skipped due to the preference “Consider only open ports for autosignal mapping” is selected but the port <PORT_NAME> considered is connected.

How to Resolve :

Check the corresponding error message in the XML report to know the error and fix the error by disabling Consider only open ports preference in preference page to generate the respective data mapping.

11.8 AutoSignalMapping_WARN_08

Error Description :

For the mapping type <mapping_type>, the Signal <SIG_NAM> and data element /ClientServerOperation/TriggerName<DATA_ELE_NAME> are matched based on the naming rules <NAMING RULE> but the generation of data mapping is skipped due to the preference “Consider only connected ports for autosignal mapping” is selected but the port <PORT_NAME> considered is open.

How to Resolve :

Check the corresponding error message in the XML report to know the error and fix the error by disabling Consider only connected ports preference in preference page to generate the respective data mapping.

11.9 AutoSignalMapping_WARN_09

Error Description :

For the mapping type <mapping_type>, the Signal <SIG_NAM> and data element /ClientServerOperation/TriggerName<DATA_ELE_NAME> are matched based on the naming rules <NAMING RULE> but the generation of data mapping is skipped due to the port type is not matched with the type specified in the naming rule.

How to Resolve :

Check the corresponding error message in the XML report to know the error and fix the error by selecting appropriate port type specified in the naming rule to generate the respective data mapping.

12. EOC Validations Error Description

This section deals with the error description of EOC Validations.

12.1 1652252 4-4.0-1 EOC Validation #1

Error description:

Event is present in SWCD/BSWMD files but not present in EOC's.

How To Resolve:

Schedule the events in EOC if exists.

12.2 1652254 4-4.0-2 EOC Validation #2

Error description:

Proxy events: EOC Event is proxy under EOC constraint.

How To Resolve:

Map the Event to EOC that are present in the AR package and not the proxy event.

12.3 1652270 4-4.0-3 EOC Validation #3

Error description:

The Events listed in EOC are not scheduled under the same OS task instead scheduled across different tasks.

How To Resolve:

All the events which are present under EOC must be scheduled under one particular OS task.

12.4 1652271 4-4.0-4 EOC Validation #4

Error description:

Init Event mapped to EOC, but it is scheduled in non-init task.

How To Resolve:

Schedule the init events under init tasks.

12.5 1652272 4-4.0-5 EOC Validation #5

Error description:

Event present in EOC but missing in OS task.

How To Resolve:

Schedule the EOC events in OS task.

12.6 1652273 4-4.0-6 EOC Validation #6

Error description:

Event not scheduled after/directly after the Event in OS task according to the order in the EOC constraint.

How To Resolve:

Schedule the Events in OS task in the same order as specified in EOC.

12.7 1652274 4-4.0-7 EOC Validation #7

Error description:

Proxy entities: Executable Entity is proxy under EOC constraint.

How To Resolve:

Map the Entity to EOC that are present in the AR package and not the proxy entity.

12.8 1652275 4-4.0-8 EOC Validation #8

Error description:

Entity of Event not scheduled after/directly after the Entity of Event in OS task according to the order in the EOC constraint.

How To Resolve:

Schedule the Entities in OS task in the same order as specified in EOC.

12.9 1658495 4-4.0-10 EOC Validation #10

Error description:

Order of Events/Runnables of EOC is in contradiction with order of Events/Runnables of EOC.

How To Resolve:

Schedule order of Events/Runnables of EOC with same order of Events/Runnables of other EOC's.

13 Contact Information

Technical Support

For details of your local sales office as well as your local technical support team and product hotlines, take a look at the website: www.etas.com/hotlines

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